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1. Introduction

Radar has been used for countless applications since its advent. A particularly useful implementation is one that utilizes the Doppler-effect, which tells us that there will be a frequency shift induced in a wave reflected from an object; if that object has motion relative to the source of the wave. Through measurement of this frequency shift the velocity of the object can be determined. Today, radar is utilized mainly for air traffic control, defence and weather monitoring. Increasingly, however, radar is finding more small scale applications, such as in the medical and security fields.

In the medical field a lot of attention has been given to Doppler radar monitoring of vital signs, such as the heart rate. The displacements of the heart during the cardiac cycle will induce Doppler shifts in the reflection of a wave transmitted towards it, which if captured and processed correctly will reveal detail about the functionality of the heart. The advantage of such a system is that it allows contactless monitoring to take place; which can, for example, aid those patients who are expected to stay in hospital for a significant tenure or monitor those suffering from third degree burns. Furthermore, heart abnormalities may cause variations in the normal Doppler signature, which when studied could help determine any functional problems that exist. In the security field such a system could, for example, be used to covertly monitor the heart rate of a subject in an airport, where the aim would be to track the heart rate of a subject to see how they respond to strenuous questioning, allowing for breaches in security to be uncovered.

Although this concept has been successfully implemented for the applications mentioned, the systems are often large and obtrusive. Therefore, it is desirable to develop a compact unit that can be discretely hidden or moved around easily. This report follows the development of such a device, with the intention to provide a comprehensive basis for future work. It covers two main areas: firstly, the hardware associated with generating and capturing the Doppler shifts; and secondly, the software used to process them.

2. Project Aim

The main aim of this project was to develop a portable device, utilizing the Doppler-effect, which could be used to monitor the heartbeat of a human being. Although this goal remained the same throughout the project, there was some deviation from the project plan set out in the project proposal. Originally, it was desirable to be able to use the portable device to identify functional problems with the heart as well as establishing the heart rate; however, as the project evolved this seemed to fall more and more out of scope, due to time constraints. Taking this into consideration the overall project goal was to develop a portable device which could accurately determine the heart rate of a subject through contactless means.

3. Theory

3.1. Heart

The human heart is the most vital organ in the human body. It provides the brain and muscles with a continuous supply of oxygen, allowing them to function correctly. It is important to discuss the operation of the heart as it has influence on the development of the heart rate monitoring device.

3.1.1. Mechanics of the Heart

The heart is divided into four chambers, as shown in Figure 1. The contraction and relaxation of these chambers results in a flow of blood to the lungs and around the body. They are controlled by the heart's conduction system, which contains nodal tissues that produce and deliver electrical impulses. It is these impulses which induce the signal variations in a traditional Electrocardiogram (ECG), and is known as automaticity. [1]

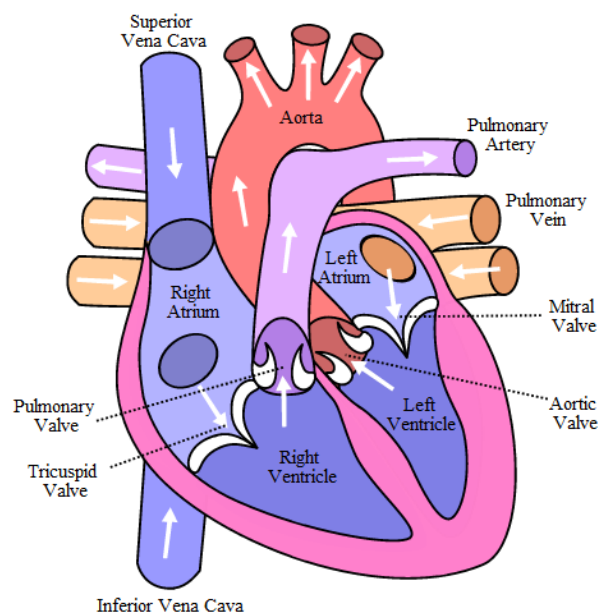


Fig 1 The human heart [2]

The de-oxygenated blood from the body enters the right atrium, where it is pumped through the tricuspid valve into the right ventricle. The de-oxygenated blood is then pumped through the pulmonary valve and towards the lungs. When passing through the lungs the blood becomes oxygenated and then enters the heart again through the left atrium, where it is pumped through the mitral valve into the left ventricle. Finally, the oxygenated blood is pumped through the aortic valve into the aorta where it is then transported around the body. This is shown in Figure 2.

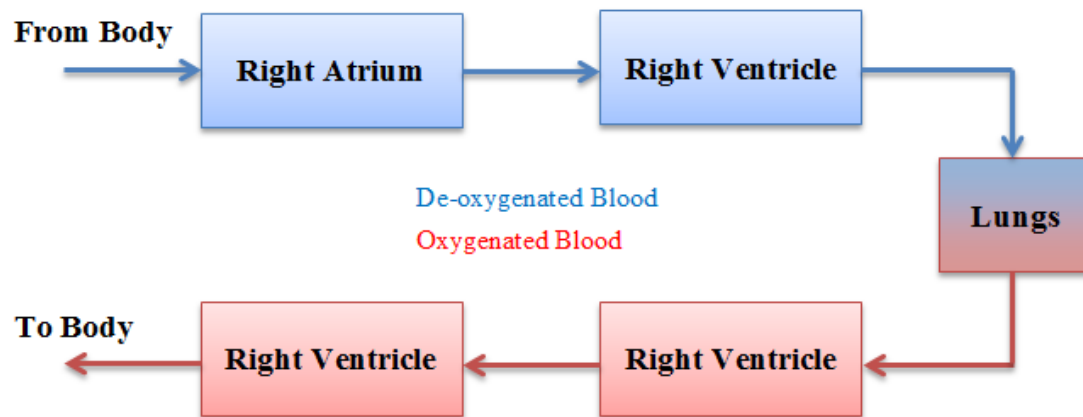


Fig 2 The blood flow process

3.1.2. Cardiac Cycle

A single heartbeat can be seen as a series of events known as the cardiac cycle. This cycle is comprised of six phases, as shown in Figure 3.

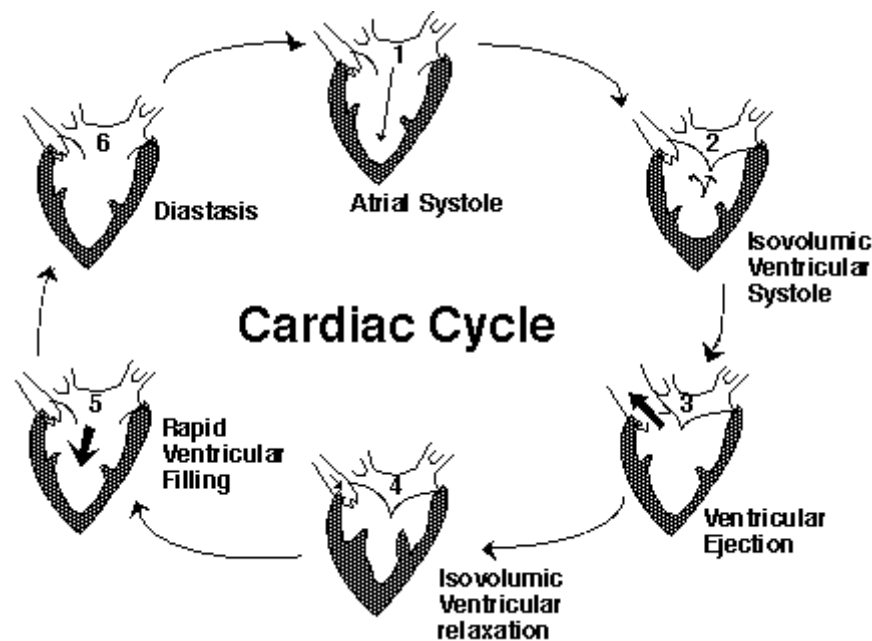


Fig 3 Phases of the cardiac cycle

Phase 1: Atrial Systole (Duration = 0.11s)

This is when both the left and right atria contract, causing an increase in pressure and the flow of blood into the ventricles. Only 20% of the ventricles volume is filled in this phase. The atrioventricular valves are shut, preventing backflow as ventricle pressure surpasses that of the atriums. [3]

Phase 2: Ventricular Isometric Contraction (Duration = 0.5s)

The first thing to happen in this stage is the closure of all the heart valves: aortic, pulmonary and atrioventricular, and results in the first heart sound (S1). In this period the ventricles contract while the volume of blood remains constant, causing a rapid increase in ventricular pressure.

Phase 3: Ventricular Ejection (Duration = 0.09s + 0.13s)

This phase is split into two separate events. The first event is the opening of the semilunar valves (aortic and pulmonary) valves. These valves open when the pressure in both ventricles exceeds the pressure in both the aortic and pulmonary arteries. Typically this pressure is around 120mmHg (16kPa). [4]

The second phase is the decreased ejection period, where the blood flow decreases. The ventricular pressure becomes lower than that of the aortic and pulmonary artery pressure, subsequently shutting the semilunar valves and producing the second heart sound (S2).

Phase 4: Isometric Ventricular Diastole (Duration 0.08s)

The ventricles relax and their pressure falls, whereas the atrial pressure increases due to the atria filling with blood. This causes the atrioventricular and semilunar valves to close, preventing blood being pumped into the ventricles.

Phase 5: Ventricular Filling (Duration = 0.11s)

As the ventricular pressure drops the atrioventricular valves open and blood is pumped rapidly into the ventricles, until the heart reaches a state of diastasis.

Phase 5: Diastasis (Duration is dependent on heart rate)

This is the resting period of the heart and can be seen as the first or last phase of the cardiac cycle. Most of the ventricular blood filling occurs in this phase (i.e. before atrial systole).

Figure 4 shows a graphical representation of the cardiac cycle. Notice that the heart sounds are produced when the atrioventricular valves are either opening or closing.

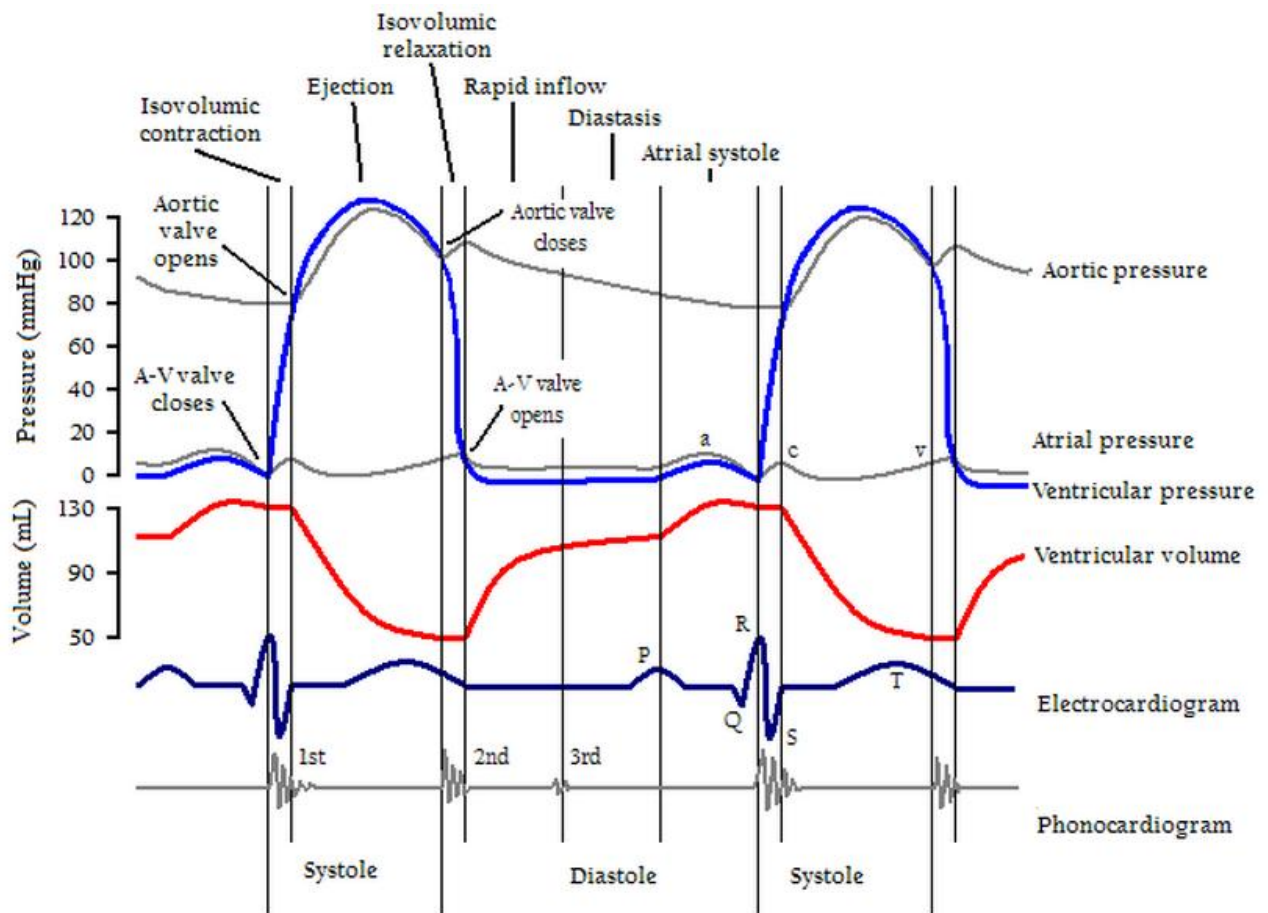


Fig 4 Graphical representation of the cardiac cycle

3.1.3. Arrhythmia

Arrhythmia is a type of heart disease which is defined as the irregularity of the heartbeat. It is a disorder in the contraction and relaxation patterns of the chambers of the heart. Arrhythmias can be classified in three ways:

- Tachycardia – heartbeats that are too fast.
- Bradycardia – heartbeats that are too slow.
- Fibrillation – heartbeats that are irregular.

They can also be classed by whether they start in the atria (supraventricular) or in the ventricles (ventricular), where the latter is the most serious. [2]

Although it is beyond the scope of this project for the device to be able to identify arrhythmias, the following work is included as a basis of future work. The next step in the development of this device should be the development of an algorithm to identify and alert the user of arrhythmia in the subject's heart.

The most common type of arrhythmia is atrial fibrillation which causes the atria to beat in an irregular manner and subsequently reduce the efficiency of the heart. Figure 5 shows the effect of atrial fibrillation on an ECG.

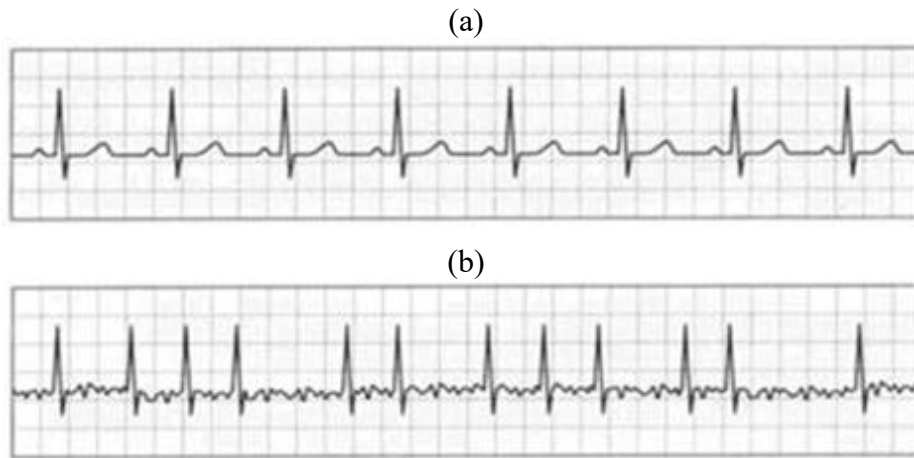


Fig 5 (a) ECG of normal heart rhythm (b) ECG of atrial fibrillation heart rhythm

The signals in Figure 5 are obtained from an ECG, which works on an entirely different principle as the technology proposed in this project. It will therefore be necessary to develop a number of custom heart signals, based on the expected Doppler shifts, which exhibit different types of arrhythmias and develop a processing algorithm based on them. The same will go for any other heart abnormalities the device is expected to identify.

3.2. The Doppler-Effect

The Doppler-effect has become the fundamental building block of almost all global radar systems. It is also the fundamental building block of the portable heart rate monitoring device developed in this project. The Doppler-effect is the increase (or decrease) in observed frequency of a reflected wave at a receiver due to the motion of a target towards (or away from) it. Figure 6 shows a simple illustration of this; notice how the movement of the heart causes a phase shift in the reflected signal.

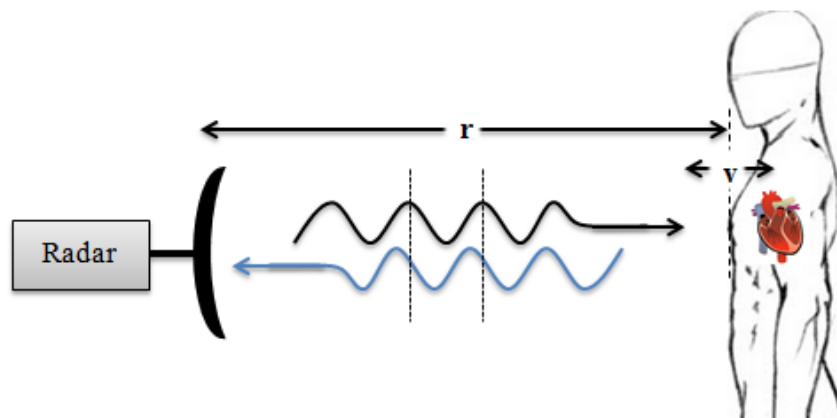


Fig 6 The Doppler effect

In this case r represents the distance from the radar antenna to the heart and v represents the relative velocity of the heart during expansion and contraction. The phase represented by the two-way path from the radar to the heart is given by Equation 1.

$$\phi = 2\pi \frac{2r}{\lambda} \quad [\text{Eqn 1}]$$

The Doppler shift is given by the rate of change of phase, given by Equation 2. If the heart is contracting (i.e. moving away from the radar) a phase lag is introduced and the Doppler frequency is negative. [5]

$$f_D = \frac{1}{2\pi} \frac{d\phi}{dt} = \frac{1}{2\pi} \frac{d}{dt} \left(\frac{4\pi r}{\lambda} \right) = \frac{2}{\lambda} \frac{dr}{dt} = \frac{2vf_0}{c} \quad [\text{Eqn 2}]$$

This theory can be expanded by introducing the micro-Doppler effect. This effect is caused by the modulation of the base Doppler shift by the moving parts of the target, resulting in sidebands around the target's Doppler frequency. For example, a propeller plane may be moving at a constant velocity towards a Doppler radar, resulting in a constant Doppler shift; the propellers on the plane may introduce further Doppler modulation of the base Doppler shift. This phenomenon is known as the micro-Doppler effect.

In this application a moving body (the bulk translation) will produce a base Doppler shift at the receiver and the heart will cause a micro-Doppler shift. Furthermore, if the body is kept still, then the heart (the bulk translation) will cause the base Doppler shift and, quite possibly, movements of the valves and chambers in the heart will cause micro-Doppler shifts.

From the derivation provided in the project proposal, the micro-Doppler shift of a vibrating part situated on or within the main target, such as the heart in a moving body is given by Equation 3.

$$f_{\text{microDoppler}} = \frac{2f}{c} \left[\frac{d}{dt} \mathbf{r}(t) \right]^T \cdot \mathbf{n} = \frac{4\pi f f_v D_v}{c} \cos(2\pi f_v t) \mathbf{n}_v \cdot \mathbf{n} \quad [\text{Eqn 3}]$$

This is a time-varying sinusoidal function, where the part is exhibiting a vibration characterised by frequency f_v and maximum amplitude D_v . The unit vector of the vibration direction is given by Equation 4. [6]

$$\mathbf{n}_v = [\cos \alpha_p \cos \beta_p, \sin \alpha_p \cos \beta_p, \sin \beta_p]^T \quad [\text{Eqn 4}]$$

As mentioned before the original scope of the project included the capability of the device to identify particular heart abnormalities as well as measuring the heart rate. This would most likely be achievable by ensuring the body is kept still, registering the micro-Doppler shifts induced by the internal mechanics of the heart and cross referencing with a healthy heart. However, as the project evolved the scope was reconsidered, due to the complexity of the original problem, and was limited to measuring the heart rate only. We can use the induced frequency shifts to determine the periodic movement of the heart, and therefore determine the heart rate.

From Equation 2 it is clear to see that using a higher frequency is more desirable, as it allows for a larger Doppler-shift to be observed. The reflected wave in this application is reflected from the surface of the chest and not directly from the heart. Therefore, the Doppler shift is

induced by the displacement of the chest and not the displacement of the heart. The heartbeat causes a displacement in the chest of around 0.3mm peak-to-peak. [7]

Some expected results are shown in Table 1.

Frequency, f	Wavelength, λ (m)	Phase Shift, ϕ (deg)
2.4 GHz	0.125	1.73
24 GHz	0.0125	17.3
40 GHz	0.0075	28.8

Table 1 Expected phase shifts using different frequencies

3.3. Doppler Radar

In order to exploit the Doppler-effect it is necessary to develop a Doppler radar. Although the actual Doppler module was not designed in this project, its principles will be described here. There are three main ways of producing the Doppler-effect and all have their relative advantages and disadvantages.

The first way is to implement the Doppler radar using the Continuous Wave (CW) principle. This involves continuously transmitting a high frequency signal and processing the reflected signal in a continuous manner. The advantage of such a system is that it is easier to operate than other Doppler radars. CW Doppler radars suffer from the inability to determine the range of the target. There can also be high levels feedback between the transmitter and receiver, leading to degradation in received signal quality. [8]

An extension of CW Doppler radar can be made by modulating the standard CW transmitted signal; the amplitude is kept constant, but the signal is Frequency Modulated (FM). The advantage of using FM is that now the range of the target can be determined. This is achieved by looking at the beat frequency which is produced when the echo signal is heterodyned with part of the transmitted signal. This principle stands for that of a stationary target; however, a moving target also overlays its own Doppler shift on the beat frequency. This can lead to incorrect range deduction and therefore must be corrected through averaging the beat frequency over multiple cycles of the FM waveform. [9]

Another technique which can be used to overcome the limitation of CW Doppler radar is pulse-Doppler radar, which can be used to determine both target velocity and range. Instead of transmitting continuously, the radar transmits intermittently and determines the range by the time difference between transmitted and received signals. The relative velocity of the target is simply determined from the Doppler-shifts apparent in the pulses. [10]

In the case of heart rate monitoring it is not a requirement to be able to measure the range of the heart from the radar and therefore a simple CW Doppler-radar can be used. Figure 7 shows an illustration of a CW Doppler-radar with a single channel.

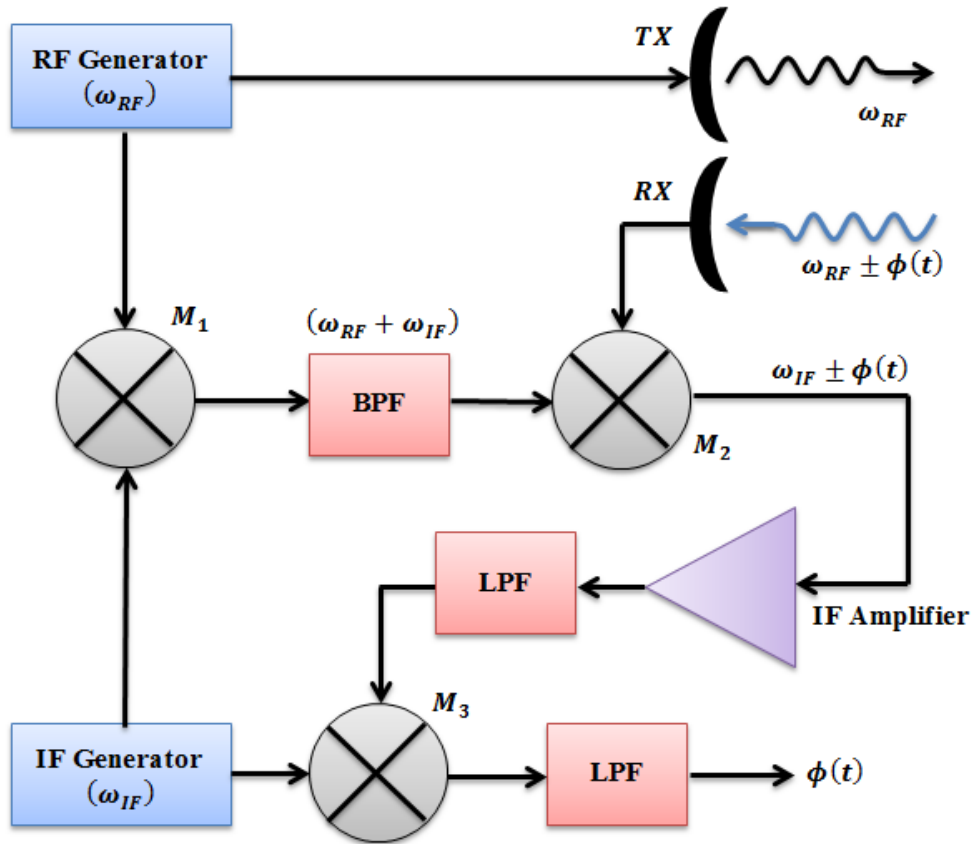


Fig 7 Single channel CW Doppler radar

The first operation of this radar is the mixing of the RF and IF signals in Mixer 1, which provides a stable Local Oscillator (LO); any unwanted frequencies are attenuated by the band pass filter. The output of this mixing stage is given by Equation 5.

$$\cos(\omega_{IF}t) \times \cos(\omega_{RF}t) = \frac{\cos((\omega_{IF}+\omega_{RF})t) + \cos((\omega_{IF}-\omega_{RF})t)}{2} \quad [\text{Eqn 5}]$$

Due to the Doppler-effect, the reflected signal acquires a time varying phase $\phi(t)$ relative to the movement of the heart. As part of the super-heterodyne topology, the reflected signal is received by RX and down-converted from RF to IF in Mixer 2. The output of this down-conversion is given by Equation 6.

$$\left(\frac{\cos((\omega_{IF}+\omega_{RF})t) + \cos((\omega_{IF}-\omega_{RF})t)}{2} \right) \times \cos(\omega_{RF}t + \phi(t)) \quad [\text{Eqn 6}]$$

This simplifies to Equation 7.

$$\frac{\cos(\omega_{IF}t + \phi(t))}{2} + \frac{\cos(\omega_{IF}t + 2\omega_{RF}t + \phi(t))}{4} + \frac{\cos(\omega_{IF}t - 2\omega_{RF}t + \phi(t))}{4} \quad [\text{Eqn 7}]$$

The reflected signal in the case of heart rate monitoring will be very weak and therefore the signal must be amplified. Furthermore, we are only interested in the low frequency term and consequently the signal is passed through a low pass filter before finally being demodulated in Mixer 3. The output of Mixer 3 is given by Equation 8.

$$\left(\frac{\cos(\omega_{IF}t + \phi(t))}{2} \right) \times \cos(\omega_{IF}t) = \frac{\cos(2\omega_{IF}t + \phi(t))}{4} + \frac{\cos(\phi(t))}{4} \quad [\text{Eqn 8}]$$

This leaves a term which is dependent only on the phase shift of the signal and can be isolated by passing the signal through a low pass filter stage. This signal is the Doppler-shift introduced by the movement of the heart and can be used to calculate the heart rate.

A typical resting heart rate ranges between 60-100 beats per minute, but can be as low as 30 beats per minute and as high as 180 beats per minute. A typical respiration rate ranges between 12-20 breaths per minute, but can go up to 80 breaths per minute during heavy exercise. The actual Doppler information acquired by the reflected signal is of little interest to us in this application; however, the periodicity of the Doppler variations is.

Since both the respirational and cardiac cycles will induce Doppler shifts in the reflected signal its frequency spectrum will contain two peaks, relating to the periodicity of the Doppler shifts. For a respiration rate of 15 breaths per minute and heart rate of 70 beats per minute, the frequency spectrum will exhibit peaks at 0.25 Hz and 1.17 Hz respectively. This is shown in Figure 8. [11]

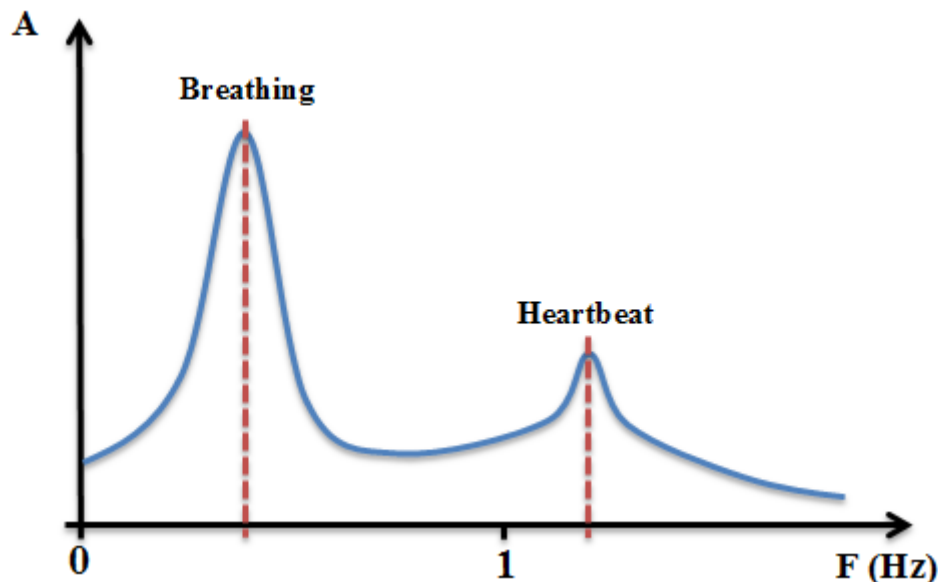


Fig 8 Expected frequency spectrum of reflected signal

From the maximum range of heart rates described above it is likely that the heartbeat peak will fall within the range of 0.5 Hz – 3 Hz. It is therefore appropriate to isolate this band of frequencies using a bandpass filter. Under periods of intense activity the respiration rate may be high enough for its peak to move into this frequency band, otherwise under normal conditions it will be under the lower frequency cut-off.

The heart has two main movements when operating: the first movement occurs when the heart is filling with blood and is called diastole; the second movement occurs when the heart contracts and is known as systole. Although it is beyond the scope of this project for the device to be able to identify abnormalities of the heart, it is important to provide a basis for future work.

If there was a discrepancy in the heart operation, it would be important for the user to be able to determine whether this discrepancy is occurring during diastole or systole. This can be achieved by introducing quadrature demodulation to the CW Doppler radar allowing for the sign of the Doppler frequency to be determined. [12]

Figure 9 shows how a quadrature demodulator is implemented.

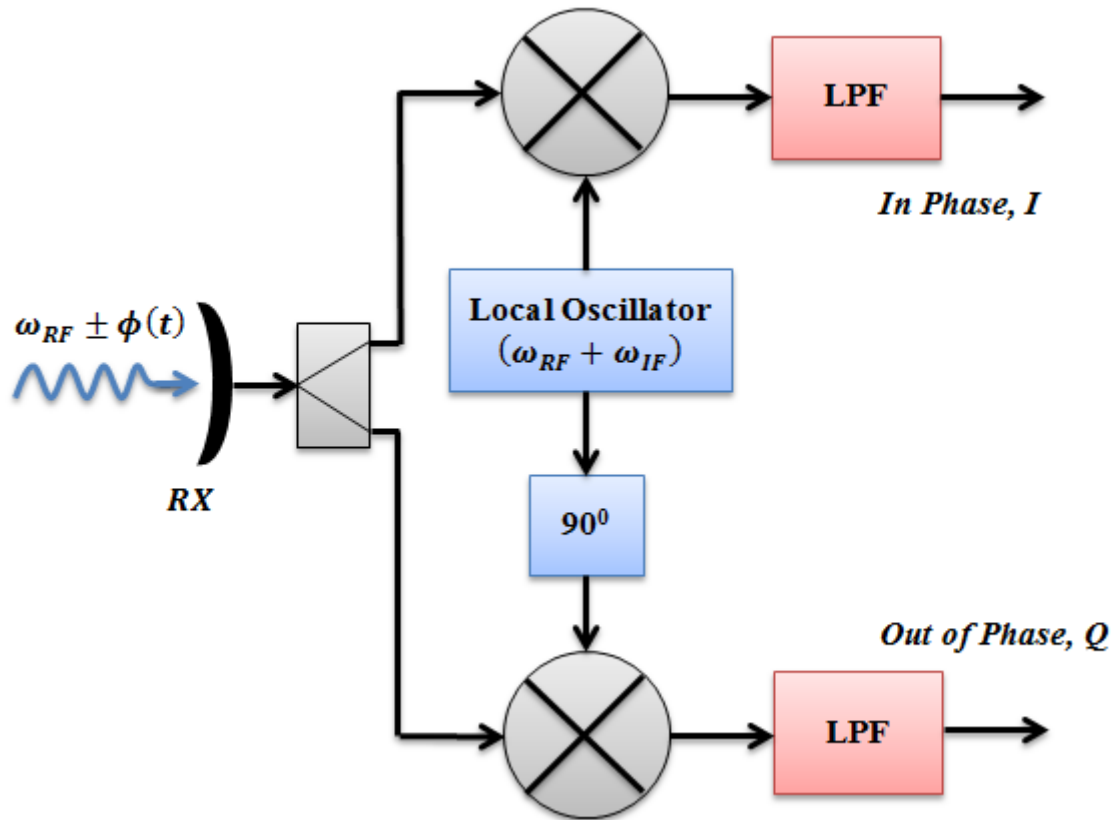


Fig 9 Quadrature demodulator

Equations 9 & 10 now apply. If the heart is in systole (i.e. moving away from the radar) the Doppler frequency is negative, and if the heart is in diastole the Doppler frequency is positive. This would be useful when the user is trying to determine where and when abnormalities are occurring. [13]

$$\phi = \tan^{-1} \left(\frac{Q}{I} \right) \quad [\text{Eqn 9}]$$

$$f_D = \frac{1}{2\pi} \frac{d\phi}{dt} \quad [\text{Eqn 10}]$$

3.4. Antenna

The accuracy of a Doppler radar system can be dramatically improved through careful antenna design. It is important to discuss the relevant issues associated with choosing and designing an antenna for this application. The main design decisions in this application are related to the frequency of operation and the desired compactness of the portable device.

Inherently, high frequency antennas are smaller in size than lower frequency ones and they are generally easier to design and manufacture. The only drawback is in the radar module itself, which becomes more complex and more expensive to implement as the operating frequency increases.

Realistically, there are two options available for this application. The first option is to use a planar antenna and the second option is to use some type of waveguide antenna, such as a horn antenna. The benefit of using a planar antenna is that it is significantly more compact than a horn antenna, although its maximum gain is limited.

Using Equations 11 & 12, the typical dimensions of a 24 GHz single patch antenna can be calculated. [14]

$$k_{nm}^2 = \left(\frac{n\pi}{L} \right)^2 + \left(\frac{m\pi}{W} \right)^2 \quad [\text{Eqn 11}]$$

$$f_{nm} = \frac{k_{nm}c}{2\pi\sqrt{\epsilon_r}} \quad [\text{Eqn 12}]$$

Where m and n determine the magnetic surface current mode and ϵ_r is the relative permittivity of the dielectric medium the patch is mounted on (FR-4 $\approx$ 4.3 F/m). For the fundamental mode of operation TM_{10} we get Equation 13:

$$L = \frac{c}{2f_{10}\sqrt{\epsilon_r}} \quad [\text{Eqn 13}]$$

This yields a resonant length of $L \approx 3$ mm, which is very small relative to the size of the portable device. The directivity achieved by a single patch of this size is around 7 dBi. The directivity of the patch can be used to estimate the link budget between the radar module and the target. The link budget is governed by the radar equation, given by Equation 14. [15]

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^2 r^4} \quad [\text{Eqn 14}]$$

In this case $G_t = G_r$ as the same antenna is used for both transmitting and receiving. The Radar Cross Section (RCS) of human cardio-respiratory motion when in the supine position is 0.326 m^2 . In the prone position this increases to 2.9 m^2 . If the distance r between the antenna and the chest is held at 0.5m and the transmitted power P_t is 10 dBm , then the power received by the radar for a subject in the supine position is -28.9 dBm . [16]

Realistically, the subject will be in the sitting position when being monitored. This will act to lower the RCS of the cardio-respiratory motion and subsequently lower the received power. By increasing the number of patches on the antenna its gain can be significantly improved, increasing the power received by the radar and improving signal clarity. At 24 GHz a five element array can provide a gain of 12 dBi , which will increase the power received by the radar to -18.8 dBm . [17]

Horn antennas are generally large structures that work by transforming guided waves into radiated waves through the use of a flared horn. They normally have broadband characteristics and the opening of the horn may be regarded as a true aperture, for analysis by aperture theory. The typical dimensions of a horn antenna operating at 24GHz are as follows: Height = 38 mm ; Width = 48 mm ; Length = 48 mm . A typical gain achieved by an antenna of this size operating at this frequency is 17 dBi , which would increase the power received at the radar to -2.9 dBm . [18]

Table 2 shows the various received power levels for different antenna types for both supine and prone positions.

Antenna Type	Gain (dBi)	Received Power (dBm)	
		Supine	Prone
Single Patch	7	-28.9	-13.4
Planar Array	12	-18.8	-9.4
Horn Antenna	17	-2.9	0.62

Table 2 Expected received power levels for different antennas

4. Prototype Design

4.1. Universal Radio Software Peripheral (URSP)

It was first thought that the proposed ‘Heart Rate Monitor’ device could be prototyped using the URSP N210 devices. These devices interface with LABVIEW, where signal processing techniques can be used to extract useful information. However, after extensive testing it became apparent that the URSP devices were unsuitable for this application for a number of reasons. The first reason was that the 2.4 GHz operating frequency was far too low to obtain a reasonable Doppler shift from the heart. Secondly, the useful part of the signal was indistinguishable from the high levels of noise. Finally, the processing power of the computer fell short of the device requirements, introducing latency and inaccuracy to the results.

4.2. Breadboard Design

4.2.1. Doppler Module

The main reason for using the URSP devices in the first place was to utilize their built in CW Doppler radar systems. As mentioned the operating frequency was too low for this application. It therefore became desirable to either find or build a higher frequency Doppler radar module. It was decided that the K-band Quadrature Doppler module NJR4261J would be used for the remainder of this project. This decision was based on the fact that the module had proved successful in similar applications and had a desirable operating frequency.

Its main operational characteristics are as follows:

- Operation frequency: 24 GHz.
- Antenna: Patch antenna (10 dBi), WR-42 Horn antenna (17 dBi).
- I & Q baseband outputs.
- Compact size (44 mm × 22 mm × 9 mm).
- Power consumption: 50 mA at 5 V supply.
- Output power: 8.2 dBm (typical).

The breadboard prototype consisted of the Doppler module, a filter stage and a gain stage.

4.2.2. Filter Stage

The low-pass filter has a cut-off frequency of 23 Hz and is required to filter out interfering signals, such as mains noise. It is implemented as an active Sallen-Key second order filter. The advantage of this 2-pole filter is that the frequency response is a lot sharper, obtaining a 40dB/decade roll-off.

The Sallen-Key filter topology is shown in Figure 10.

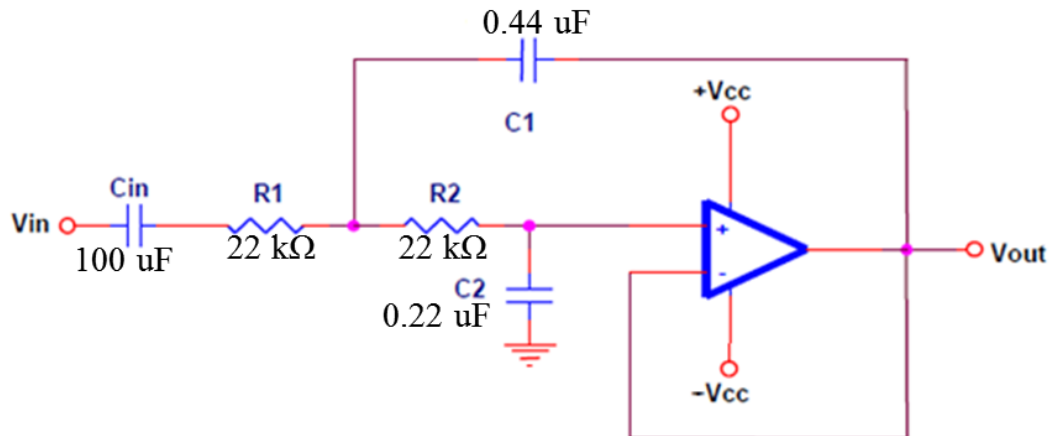


Fig 10 Dual-supply Sallen-Key low-pass filter topology

The cut-off frequency of this filter is given by Equation 15. [19]

$$f_o = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}} \quad [\text{Eqn 15}]$$

4.2.3. Gain Stage

The output voltage of the I & Q channels, due to typical subject movements, was around 20mV peak-to-peak and it was therefore necessary to implement a gain stage. The gain stage is implemented using a simple inverting op-amp topology, shown in Figure 11, and follows the filtering stage.

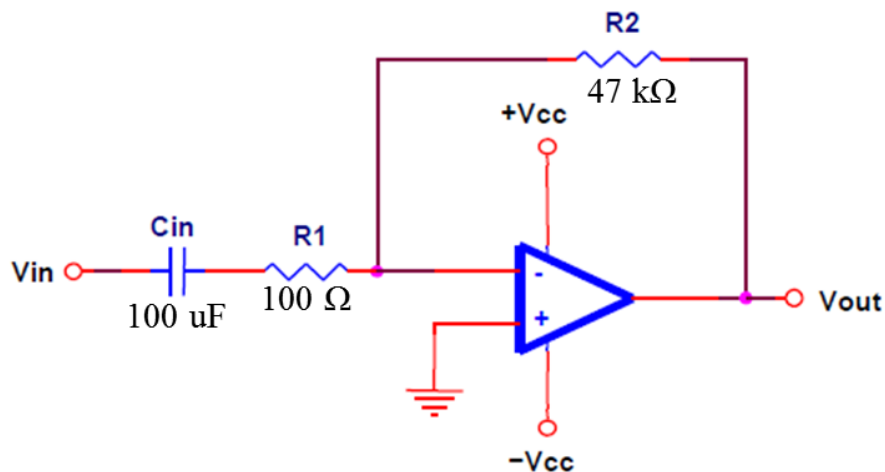


Fig 11 Inverting op-amp topology

The gain in this case was 470 and was set using Equation 16. [20]

$$G = \frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1} \quad [\text{Eqn 16}]$$

4.2.4. Prototype Schematic

The analogue-to-digital conversion (ADC) was performed by a NI-DAQ; a USB peripheral with 12 bit resolution and up to 5 kHz sampling rate. The circuit was supplied by a dual-rail supply of $\pm 9\text{ V}$ in order to make use of the dynamic range of the DAQ. The Doppler module voltage was limited to 5 V and therefore a voltage regulator was used to step down the supply. The I & Q channels had identical but separate filtering and amplification circuitry and were sampled by the DAQ. The prototype schematic is shown in Figure 12.

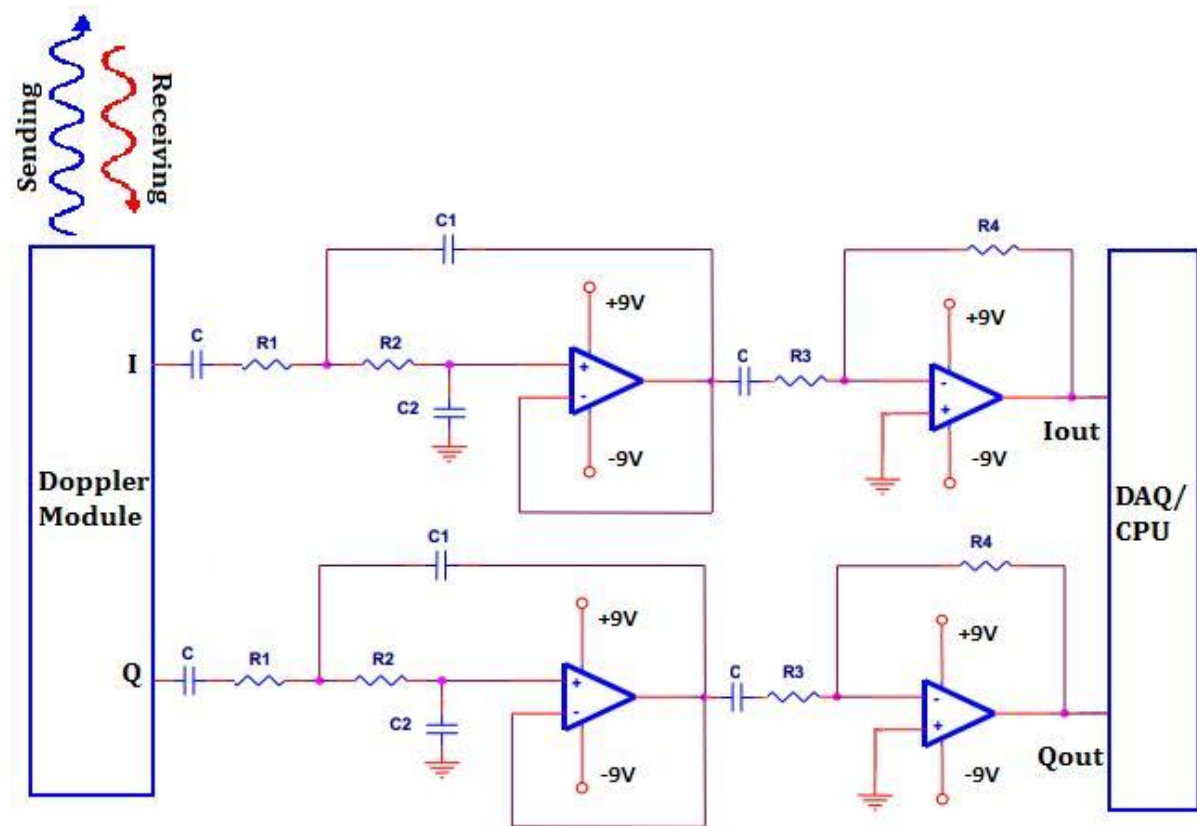


Fig 12 Prototype circuit design

The component values used in the prototype are shown in Table 3.

Component	Value
C	100 μF
R ₁ , R ₂	22 k Ω
C ₁	0.44 μF
C ₂	0.22 μF
R ₃	100 Ω
R ₄	47 k Ω

Table 3 Component values used in prototype

4.3. Prototype PCB Design

The prototype that was built in Section 4.2 was tested and the initial results were promising. The Doppler shifts associated with the respirational and cardiac cycles were clearly distinguishable in the signal. As a crucial part of this project was to develop an algorithm which could determine the heart rate of the subject, it became desirable to design a more robust and compact circuit which could be used for extensive testing. This was best achieved through the development of a Printed Circuit Board (PCB). In addition, the PCB made it easier to spot design faults and also prevent parasitic capacitance altering the electronic characteristics of the circuit.

The first PCB designed in this project was based on the prototype developed in Section 4.2 and is shown in Figure A1 and Figure A2 (Appendix A).

The regulator LM7805 used in the PCB to supply the Doppler module with 5 V from the raw 9 V supply has the following operational characteristics:

- Dropout voltage (required headroom): 2 V
- Power Dissipation: $(9\text{V} - 5\text{V}) \times 50\text{ mA} = 0.15\text{ W}$

In Figure A1, C_{10} and C_{14} are electrolytic decoupling capacitors, which not only improve the transient time response but also aid in noise rejection. In addition, they decouple the circuit from the power supply and act to stabilize the regulator internal loop. The filter and gain stages of the circuit were implemented using a quad op-amp and ADC was carried out by the NI-DAQ.

5. Final Product Design

The overall objective of this project was to develop a portable heart rate monitoring device. The prototype PCB designed in Section 4.3 was used extensively for testing and development; however, for it to become stand-alone additional functionality was required. It was therefore necessary to redesign the device. Again, the K-band Quadrature Doppler module NJR4261J was used as the CW radar system. The additional functionality consisted of on-board ADC, a power supply, an infra-red reference finger sensor, a microcontroller and an embedded display.

5.1. Analog-to-Digital Converter (ADC)

Through observation of the results obtained from the PCB design, it was decided that more resolution was required in the sampled signal. The ADC solution also had to be embedded on the final PCB design. After an extensive product search, the ADS1147 from Texas Instruments was decided on.

The ADS1147 has the following operational characteristics:

- 16 bit resolution with no missing codes, 4 inputs.
- Up to 2 kHz sampling rate for a single channel. To maintain 16 bit effective resolution, the sampling rate should not exceed 320 samples per second (SPS).
- Fast settling time when changing input channel. Multiple channels can therefore be sampled without significant overhead.
- Relatively low cost and low power consumption.

A minimum of two channels are required to sample I & Q channels. The reference IR finger sensor does not require such high resolution and can therefore be sampled by the microcontroller's built-in ADC. The input voltage range of this ADC is 5V, which sets the limit on the maximum signal swing. The analogue processing could still be carried out on higher-voltage rails ($\pm 9V$), but this would have to be divided down to a final range of 5V.

5.2. Microcontroller

The embedded microcontroller used to process the signal had to meet a number of stringent processing power and memory requirements. It must be able to hold all the sampled data in RAM memory and have enough flash memory to store the code required for the algorithm implementation.

After an extensive product search, a microcontroller based on the ARM Cortex-M3 architecture was selected. This particular microcontroller is called the LPC1769, shown in Figure 13, and is manufactured by NXP. The reasons for choosing this particular part were its processing power and the availability of a very low-cost development board.

The microcontroller was shipped on a PCB that consisted of two parts: the programmer and the microcontroller part.

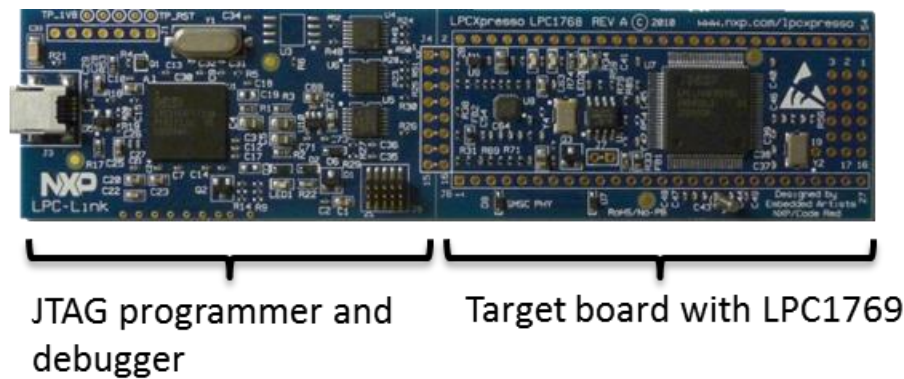


Fig 13 LPC1769 microcontroller and development board

The LPC1769 has the following operational characteristics:

- 64 kbytes of flash program memory.
- 32 kbytes of RAM memory.
- Up to 120 MHz operation, one instruction per clock.
- Power efficient: 30 mA current consumption at 40 MHz.
- Built-in 12 bit ADC with up to 200 kSPS sampling rate.
- Full debugging support.
- Extensive software library, free development tools.

During the development process the two boards were left joined together. Once the code was ready and ported to the microcontroller the boards were separated. A pin connector was attached so that the target board can be re-programmed.

Compared to simple AVR and PIC microcontrollers, ARM Cortex has much more complex clocking, interrupt and execution error handling. The software has to initialise the part correctly, otherwise it will not run, or run at the wrong frequency. Fortunately, a software library called Cortex Microcontroller Software Interface Standard (CMSIS) is provided for free, and introduces a way of accessing special features of the microcontroller with the standard C-language syntax.

Also, a DSP library is provided for this family of microcontrollers, including functions for all the common digital signal processing tasks such as:

- Basic real and complex number arithmetic.
- Complex FFT transforms.
- Convolution.
- Trigonometry functions.
- FIR and IIR filters.
- Matrix operation functions.

All these functions are available for both floating-point and fixed-point formats.

5.3. Power Supply

5.3.1. Considerations

For the circuit to be portable, it must be powered from a battery. The battery must satisfy the current requirements, shown in Table 4, for a desired operational period.

Component	Maximum Current Consumption (mA)	Essential Current Consumption (mA)	Regulator Number
Doppler Module	50	50	1 – Dedicated Regulator
LCD Display	1.3	1.3	2 – Main Vcc
LCD Backlight	70	-	
ADS1147 (ADC)	0.25	0.25	
LPC1769 (μ C)	30	30	3 – 3.3 Volts
Finger Sensor LED	100	-	Directly from Battery
Total	250	80	

Table 4 Power consumption of portable device

The maximum current of 250mA that is expected to be drawn by the device is too high for the batteries to provide a sustainable supply and achieve their nominal capacity. However, when the LCD backlight and IR finger are not in use the current consumption is brought down to 80mA. Furthermore, the voltage regulators introduce a level of inefficiency to the device. From the datasheet of the switching boost converter MCP1623, its maximum output current for an input voltage of 2.2 V (1.1 V for each battery cell when drained) is around 115 mA. It can also be noted that both the LCD backlight and IR finger sensor LED can be powered directly from the battery, by-passing any voltage regulators and improving efficiency.

Another way to minimise power consumption is by putting the microcontroller in sleep mode. Continuous sampling operations occupy very little processor time, and the algorithm calculations take around 150ms per one-second cycle. The power consumption in sleep mode is almost negligible in context of this application and it would therefore be possible to reduce microcontroller power consumption by a factor of 5.

Power supply design is important for this device as ADC requires a clean stable power supply. The DC-DC boost converters produce a lot of switching noise (ripple). To smooth it effectively, Low Drop-Out (LDO) regulators with a high Power Supply Rejection Ratio (PSRR) are used after the DC-DC converter to smooth the supply voltage.

5.3.2. Battery Options

The following is a list of battery types that can be used to power the portable device:

- Lithium: capacities of around 1000-1500 mAh, voltage $\approx$ 3.7 volts.
- Alkaline: each cell gives 1.5V, around 2200 mAh capacity.
 - Actual capacity is less if discharged at high current.

- Rechargeable AA/AAA: each cell gives $\approx 1.2\text{V}$, less as its being discharged.
- 9V Cell: consists of 6 individual cells, quite low capacity of around 500 mAh.

Both single-supply and dual-supply power configurations were considered for the portable device. The advantage of using a dual-supply ($\pm\text{V}$) is that it gives a greater voltage swing and most op-amps are optimised for dual-supply. However, it is more difficult to generate a dual-supply from a battery as additional components are required, such as a voltage inverter, to generate the negative voltage, and a negative voltage regulator.

After considering the options outlined above, it was decided a single-supply alkaline power supply would be used. The power supply arrangement is shown in Figure 14.

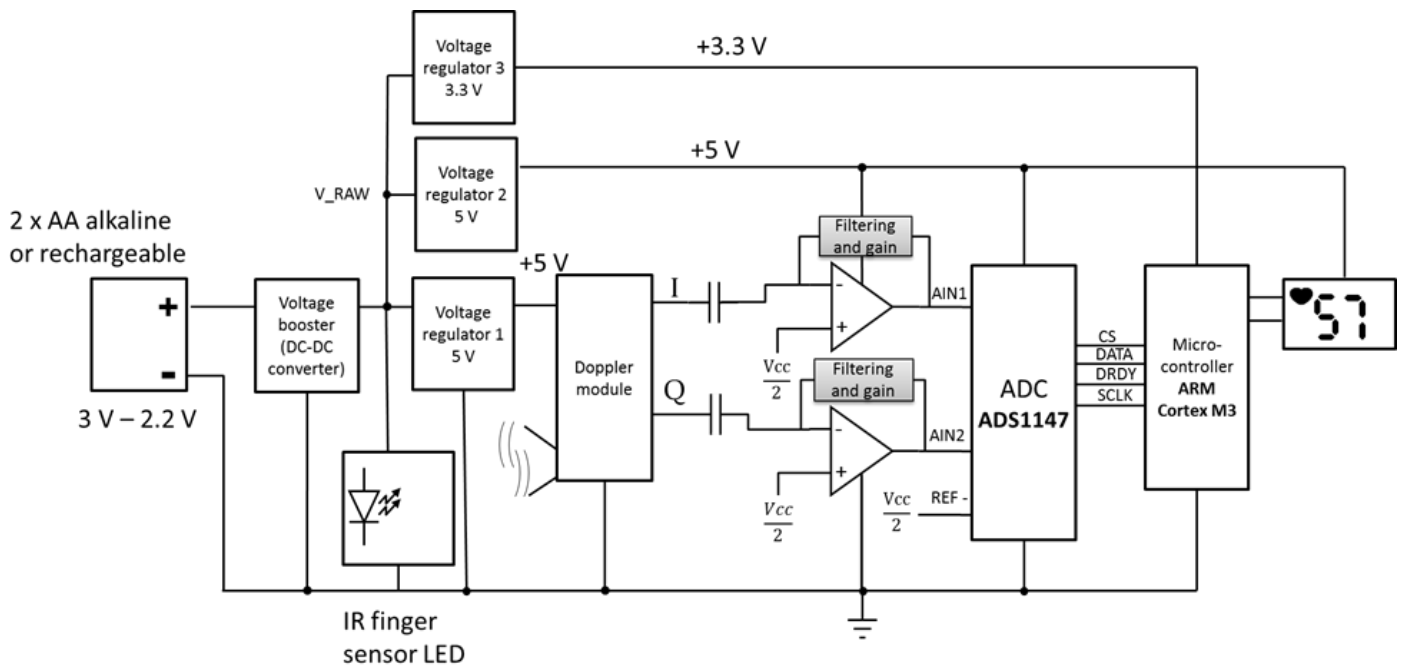


Fig 14 Power supply arrangement

Through previous tests, the Doppler module was revealed to be very sensitive to power supply ripple and it therefore has its own dedicated voltage regulator.

The only advantage of using a dual-supply in this case would be a larger signal swing, which could lead to a greater SNR. However, the noise embedded in the output of the Doppler module is a lot higher than the floor noise of the system, and gets amplified together with the signal. Therefore the signal and the noise get amplified equally and a higher supply voltage will not necessarily result in a greater SNR. The raw output voltage V_RAW can be adjusted to $\approx 5.5\text{V}$ to give the LDOs enough headroom.

5.4. Finger Sensor

In order to test the accuracy of the device an infra-red finger sensor was built into the final product design. Unlike the Doppler radar, the finger sensor requires contact between the subject and the device.

The finger heart beat sensor works by shining infra-red light into body tissue that is rich in blood vessels. Depending on the oxygenation level in the blood, different amounts of IR light are reflected and absorbed. Since changes in oxygenation are synchronous with the cardiac cycle, it is possible to detect the heart beat by tracking variations in the reflected IR light.

The circuit schematic of the finger sensor is shown in Figure 15. The infra-red LED and sensor diode are placed next to each other and tilted towards each other. The tip of the finger is placed on top of them, as shown in Figure 16.

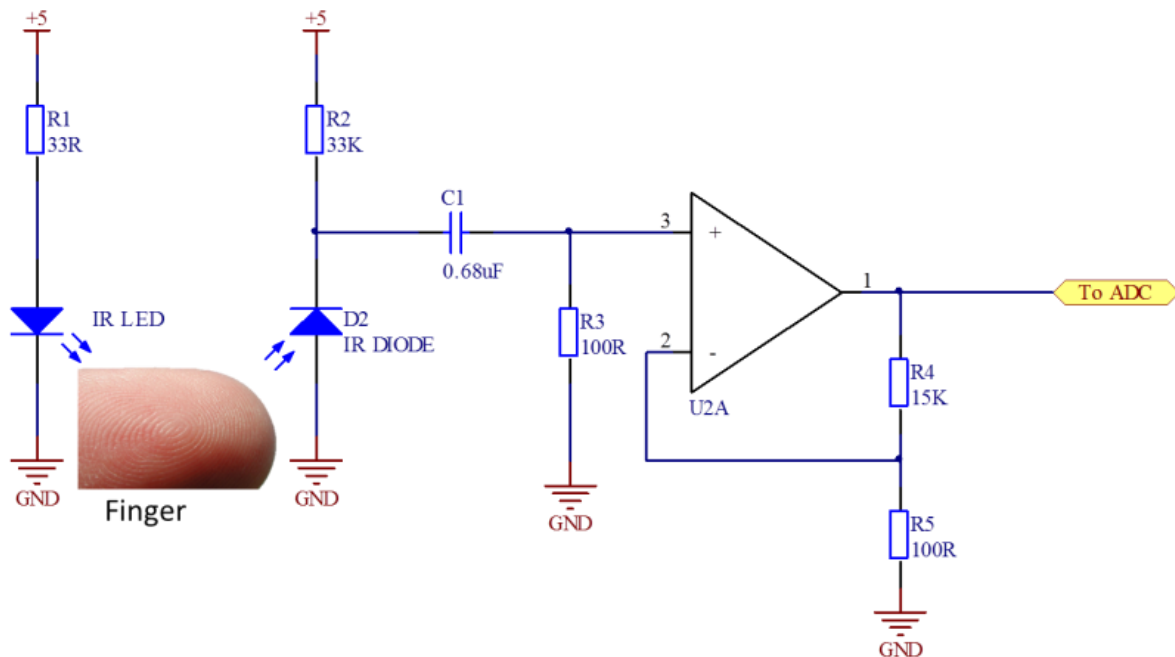


Fig 15 IR finger sensor circuit schematic

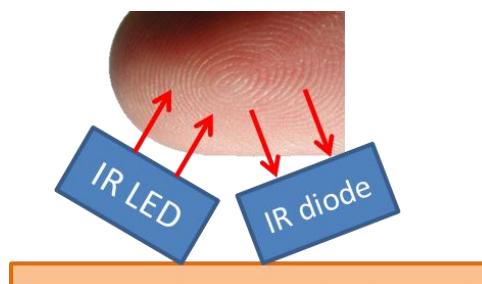


Fig 16 Finger sensor operation

5.5. PCB Design

Similarly to the prototype design it was desirable to design a PCB for the final product as well. It was decided, for compactness and durability, that surface mount technology would be used. In addition to the extra functionality that was added, some alterations were made to the original prototype design. In the original prototype design there was a single gain stage of 470. In the final product design this was increased to 1000 and was split into two stages, which allowed for better noise rejection and stability.

The two stage gain topology is shown in Figure 17.

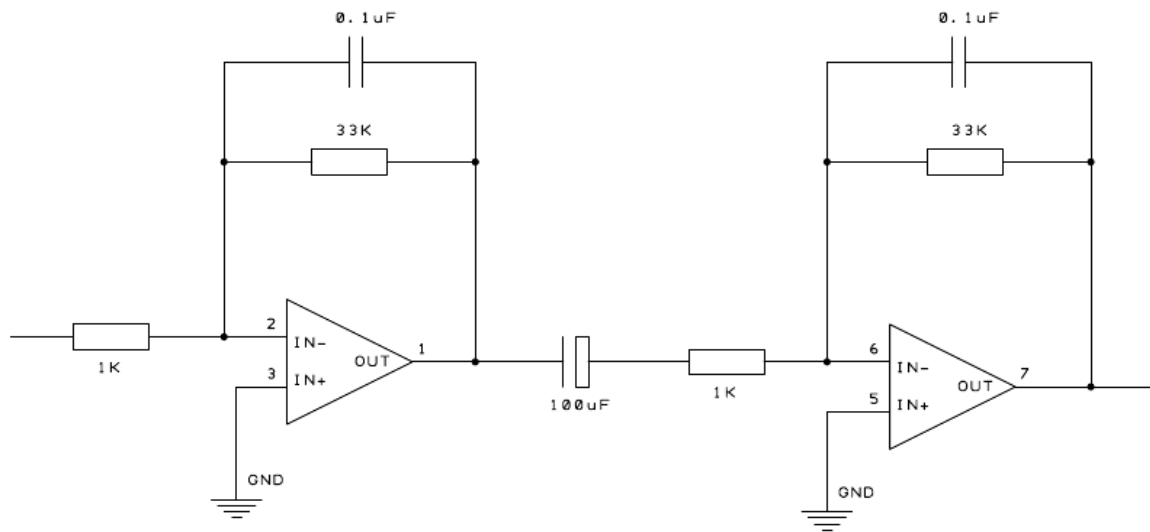


Fig 17 Two stage gain topology

In this case both stages had a gain of 33, giving an approximate overall gain of 1000. Although it was intended that this change was made before the fabrication of the PCB, it was not and because of the limited space on the final PCB it was difficult to simply add an extra gain stage. Instead, the required gain was implemented by setting the gain of the first stage to 37 and the setting internal gain of the ADS1147 to 32. With this approach there is minimal interference with the PCB, but at the cost of losing a pole in the low-pass filter.

The final PCB circuit schematic is shown in Figure A3 and the corresponding layout is shown in Figure A4. Included in Appendix A is a brief guide to the pin-outs of the final PCB and details on the connections between the PCB and microcontroller. This is intended for debugging purposes and is shown in Table A1.

6. Algorithm

A significant part of this project consisted of establishing a robust algorithm which would take the raw signal data, obtained from Data Acquisition (DAQ), process it and display a meaningful result. The result in this case is the heart rate. A number of Digital Signal Processing (DSP) techniques have been tried and the following sections will outline the development of this algorithm. Figure 18 & 19 show the raw signal data without breathing and with breathing respectively, although both signals have been subjected to oversampling and averaging which is discussed in the next section.

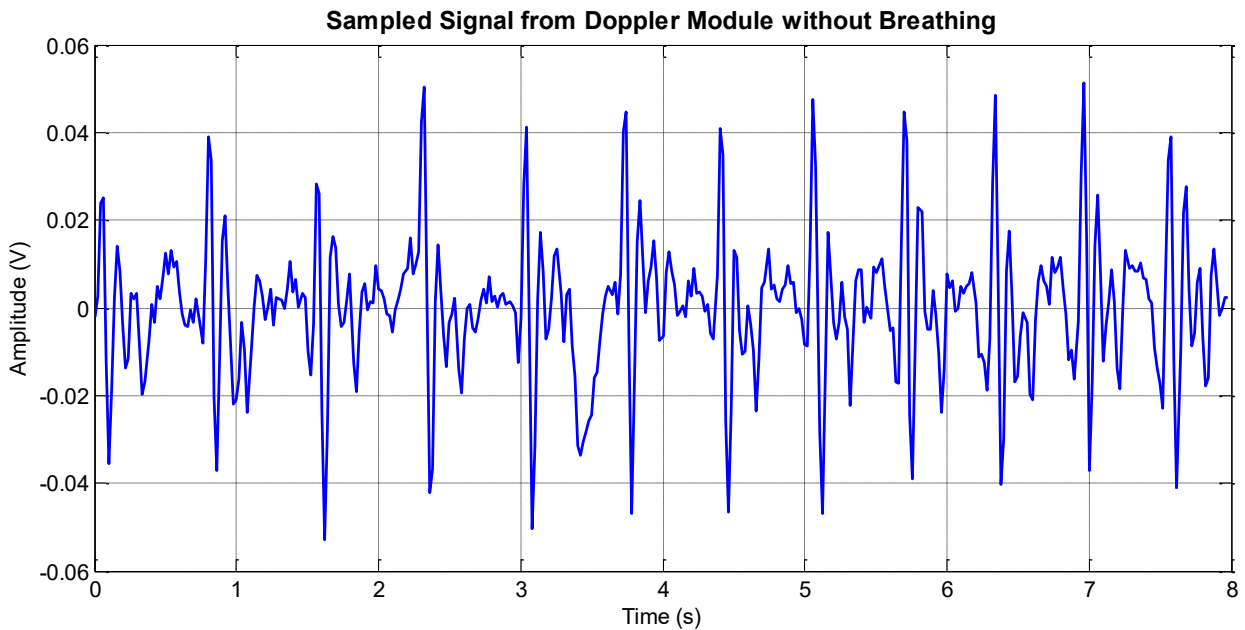


Fig 18 Raw signal data without breathing

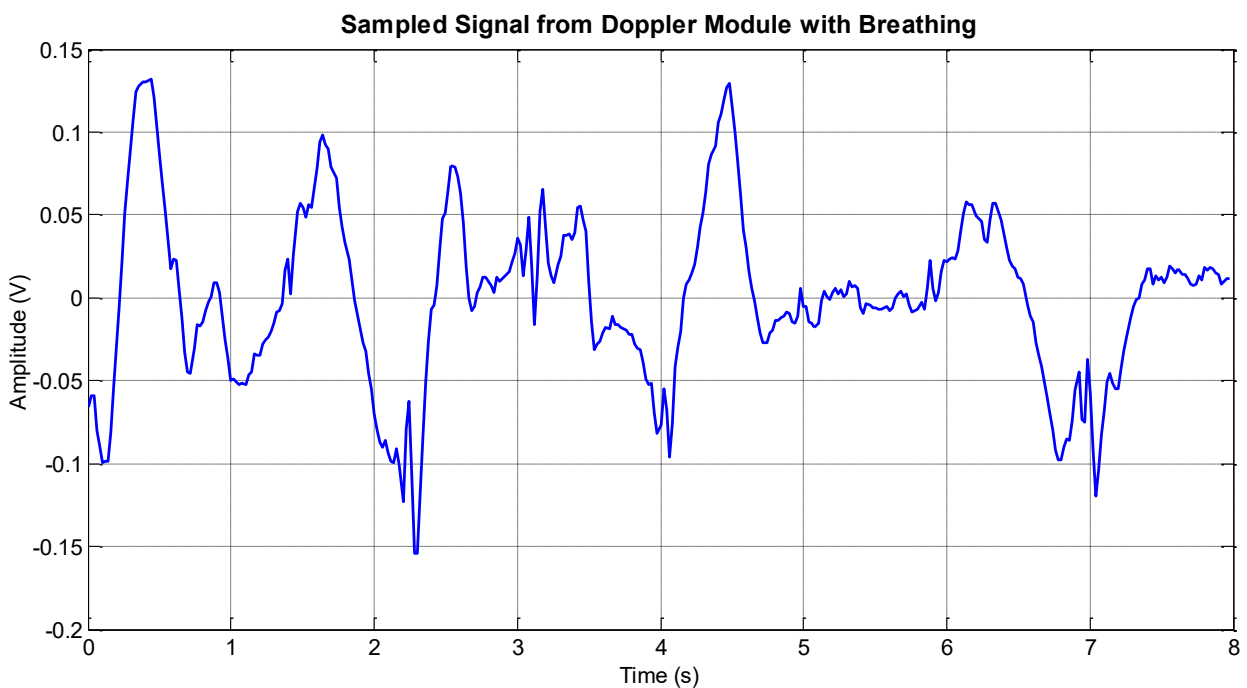


Fig 19 Raw signal data with breathing

It is clear to see the periodicity of the signal introduced by the heartbeat in Figure 18. In Figure 19 however, the noise introduced by breathing is much larger than that of the heartbeat and therefore a more complex signal processing problem is presented. An initial algorithm was established by concentrating primarily on the raw signal data without breathing.

There are two stages to the development of this algorithm. The first stage concentrates on its development in MATLAB and the second stage consists of the implementation of the algorithm on the microcontroller.

6.1. Development – No Breathing

6.1.1. Oversampling & Averaging

Oversampling & Averaging is a common technique used to enhance ADC resolution. As the signal from the Doppler module is inherently noisy, resolving a series of bits into a signal bit can improve its resolution and eliminate its random noise.

This technique requires a higher number of samples, which is achieved by oversampling the signal. Equation 17 gives the oversampling frequency: [19]

$$f_{oversampling} = 4^n \times f_{nyquist} \quad [\text{Eqn 17}]$$

Where n is the required number of additional resolution bits. There are a number of criteria which must be met for effective oversampling.

- The signal component of interest should not vary significantly during a conversion.
- There should be some noise present in the signal (small variations).
- The amplitude of the noise should be at least 1 Least Significant Bit (LSB).

After the samples have been acquired averaging occurs. However, this is not averaging in a conventional sense, where all m samples are added and then divided by m . Instead a moving average is implemented, where m readings are placed in a group and then an average of this group is taken. Figure 20 shows a series of moving averages on a set of data.

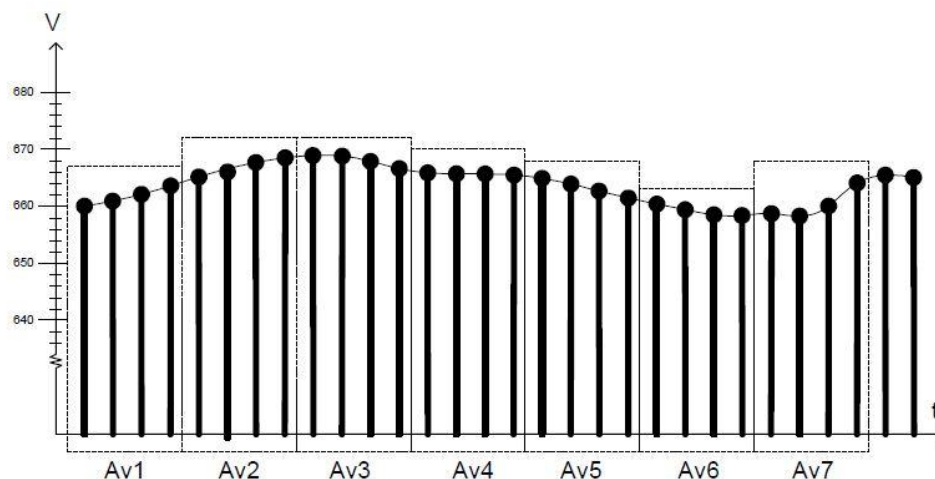


Fig 20 The moving average principle

If this principle is applied to the raw signal data then a significant improvement can be made to its resolution. Figure 21 shows a heartbeat signal, sampled at 50 Hz for a period of 10 seconds, without the implementation of this technique.

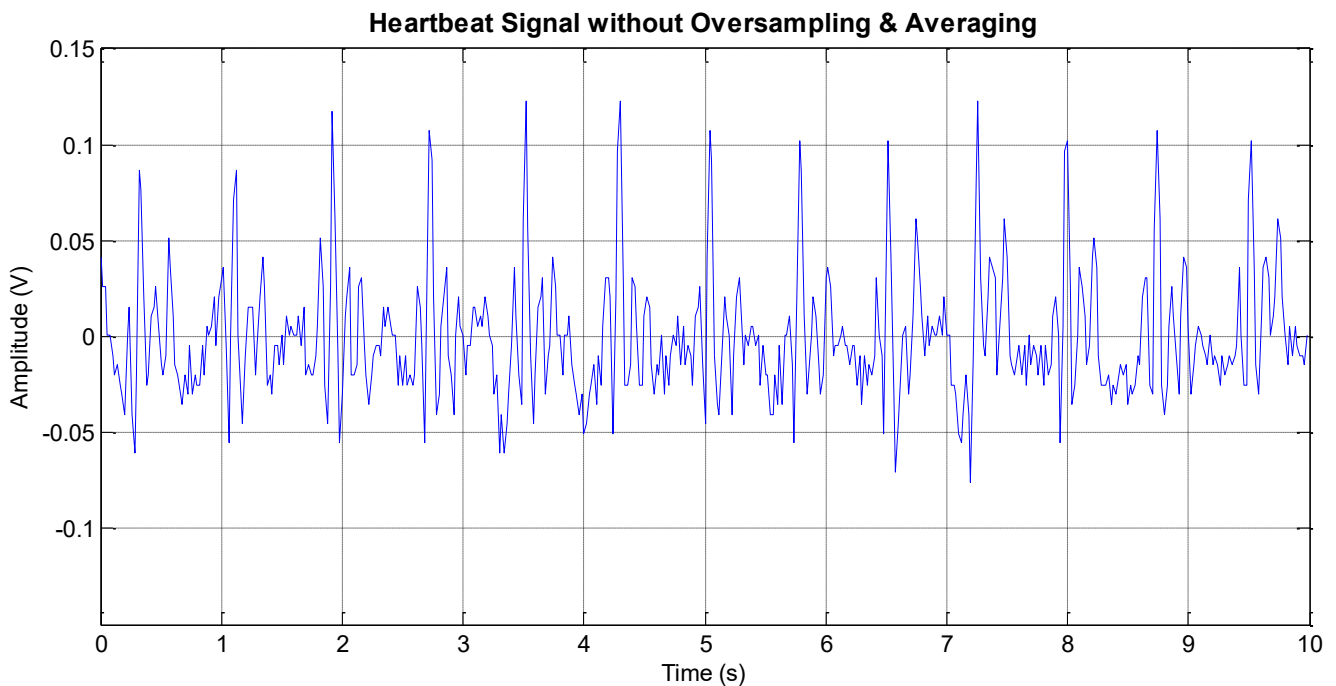


Fig 21 Heartbeat signal without Oversampling & Averaging

Figure 22 shows a heartbeat signal sampled at 3000 Hz, where the averaging factor m is 60. This produces a pseudo sampled digital signal at 50 Hz. It is clear to see the improved resolution.

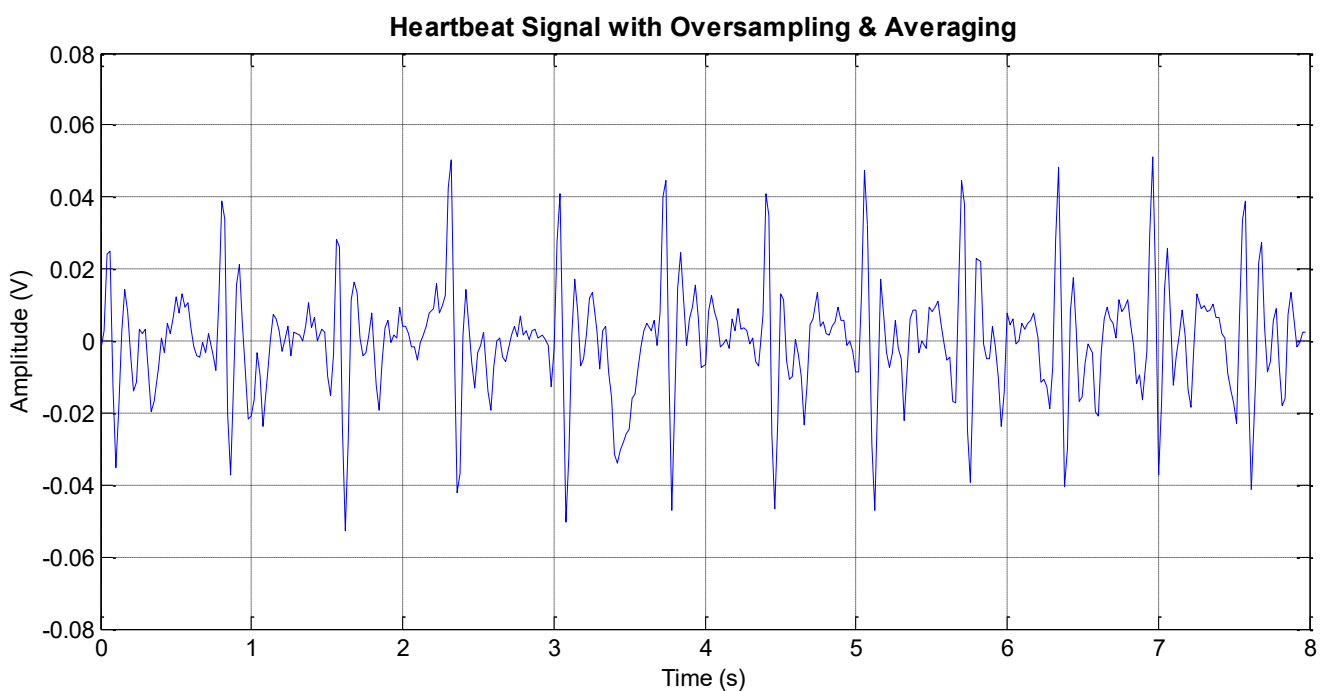


Fig 22 Heartbeat signal with Oversampling & Averaging

Oversampling & Averaging can be implemented in MATLAB by creating a vector of zeros, m times smaller than the number of samples taken during the oversampling period, and then averaging each group of m samples and assigning to a non-arbitrary index in this vector.

The following MATLAB script carries out this operation:

```
TIME = 10;
                                     % Sampling period
SAMPLING_RATE = 3000;
M = 60;                             % Averaging factor

% DAQ function
[data,time] = DAQ_signal_in(TIME,10,SAMPLING_RATE,M);

% Extraction of in-phase and quadrature components
N = TIME*SAMPLING_RATE;
data_I = data(1:N,1);
data_Q = data(1:N,2);

% Creation of zero vectors
average_I = zeros(N/M,1);
average_Q = zeros(N/M,1);

% for loop which averages and assigns samples to a vector index
for k=1:(N/M)-1
    index=k*M;
    average_I(k)=sum(data_I(index:index+M))/M;
    average_Q(k)=sum(data_Q(index:index+M))/M;
end

% Re-assigning the data vectors
data_I = average_I;
data_Q = average_Q;

% Sizing the vectors to avoid drop-off
data_I = data_I(1:end-(2*SAMPLING_RATE/M));
data_Q = data_Q(1:end-(2*SAMPLING_RATE/M));
data_I(end) = data_I(end-1);
data_Q(end) = data_Q(end-1);

% Mean removal
data_I = data_I-mean(data_I);
data_Q = data_Q-mean(data_Q);

% Ensures time vector is the same length as new data vectors
time = time(1:max(size(data_I)));
```

6.1.2. Spectrum Analysis

As the signal demonstrates periodic activity, the Fourier Transform can be used to try and identify the frequency of this periodic activity. The spectrum of the recorded signals can be calculated directly using the Fast Fourier Transform (FFT). Figure 23 shows the I-Channel and its respective frequency spectrum.

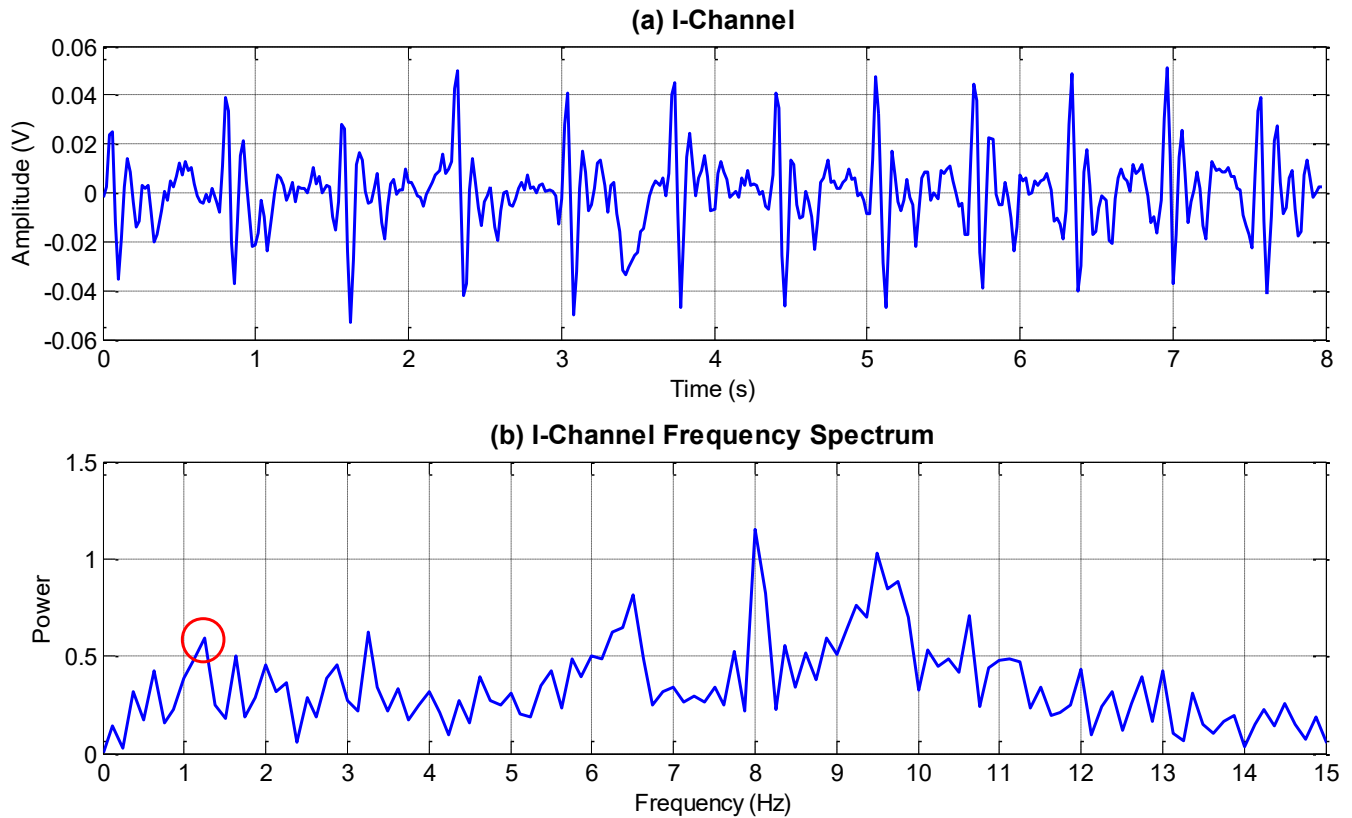


Fig 23 (a) I-Channel Signal (b) I-Channel Frequency Spectrum

The I-Channel signal in Figure 23(a) was taken from a subject at rest. Average resting heart rates fall between 60 and 100 beats per minute, and this will correspond to a peak in the frequency spectrum between 1 Hz and 1.7 Hz. In this range the most prominent peak falls at 1.25 Hz, corresponding to a heart rate of 75 beats per minute. [20]

It is prudent to apply the same theory to the Q-Channel. Figure 24 shows the Q-Channel and its respective frequency spectrum. Again, the prominent peak in the 1 to 1.7 Hz range is expected.

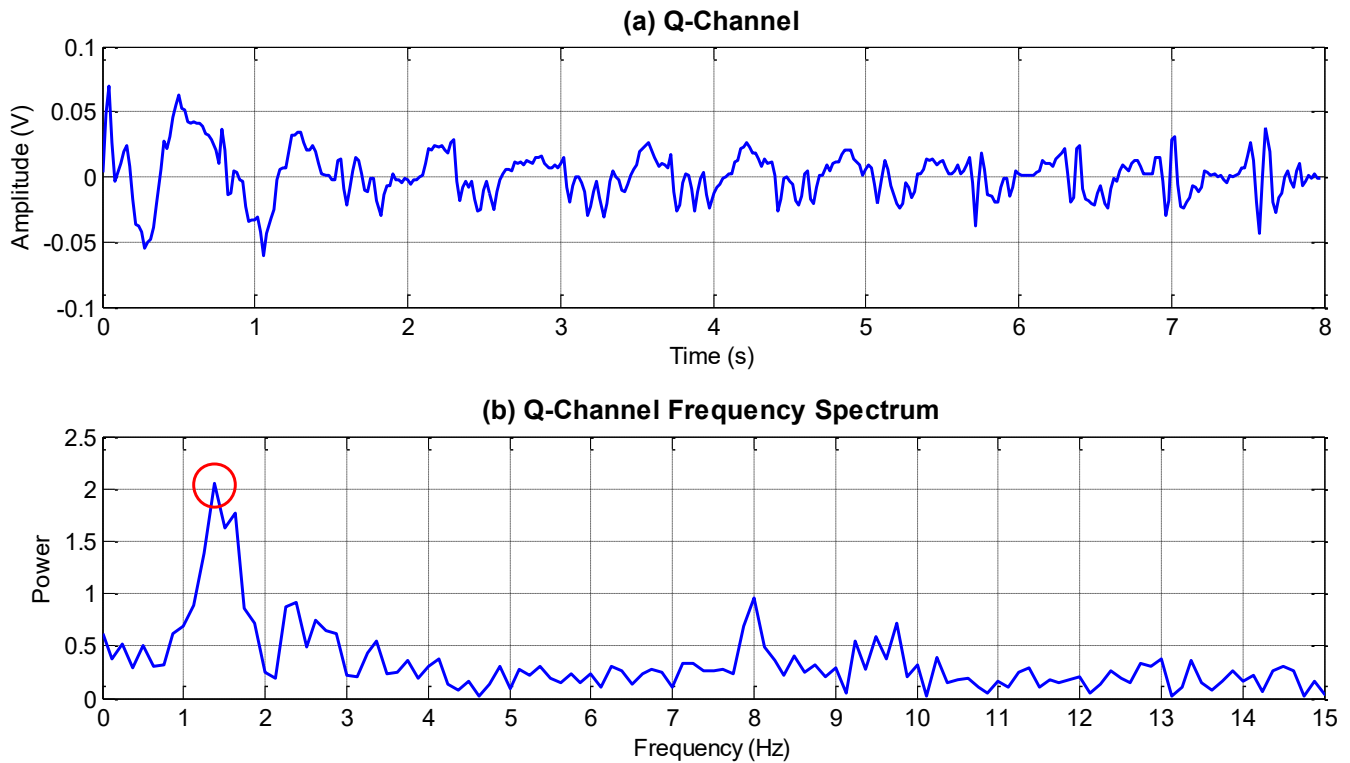


Fig 24 (a) Q-Channel Signal (b) Q-Channel Frequency Spectrum

In Figure 24(b) the prominent peak falls at 1.375Hz, corresponding to a heart rate of 83 beats per minute. There is obviously some discrepancy between these results, indicating that this method may not be as robust as necessary. One could argue this discrepancy could be avoided by using just one of the channels; however, there is no certainty as to which channel will provide the most accurate result.

This method is implemented in MATLAB using the following code:

```
% The frequency range and resolution is calculated
dF = (SAMPLING_RATE/M)/max(size(time));
f = 0:dF:SAMPLING_RATE/M-dF;
f = transpose(f);
f = f(1:size(f)/2);

% The I-Channel spectrum is calculated
spectrumI = (abs(fft(data_I)));
spectrumI = spectrumI(1:size(spectrum)/2);

% The Q-Channel spectrum is calculated
spectrumQ = (abs(fft(data_Q)));
```

6.1.3. Matched Filtering

Matched filtering is an effective tool in augmenting the presence of a known signal pattern. The matched filter is the optimal linear filter for maximising the Signal-to-Noise Ratio (SNR) of a signal. To implement a matched filter, a template must first be established; in this case the template is the signal variation that is introduced by the heartbeat. Figure 25 shows the heartbeat template, which will be referred to as the heartbeat kernel. [21]

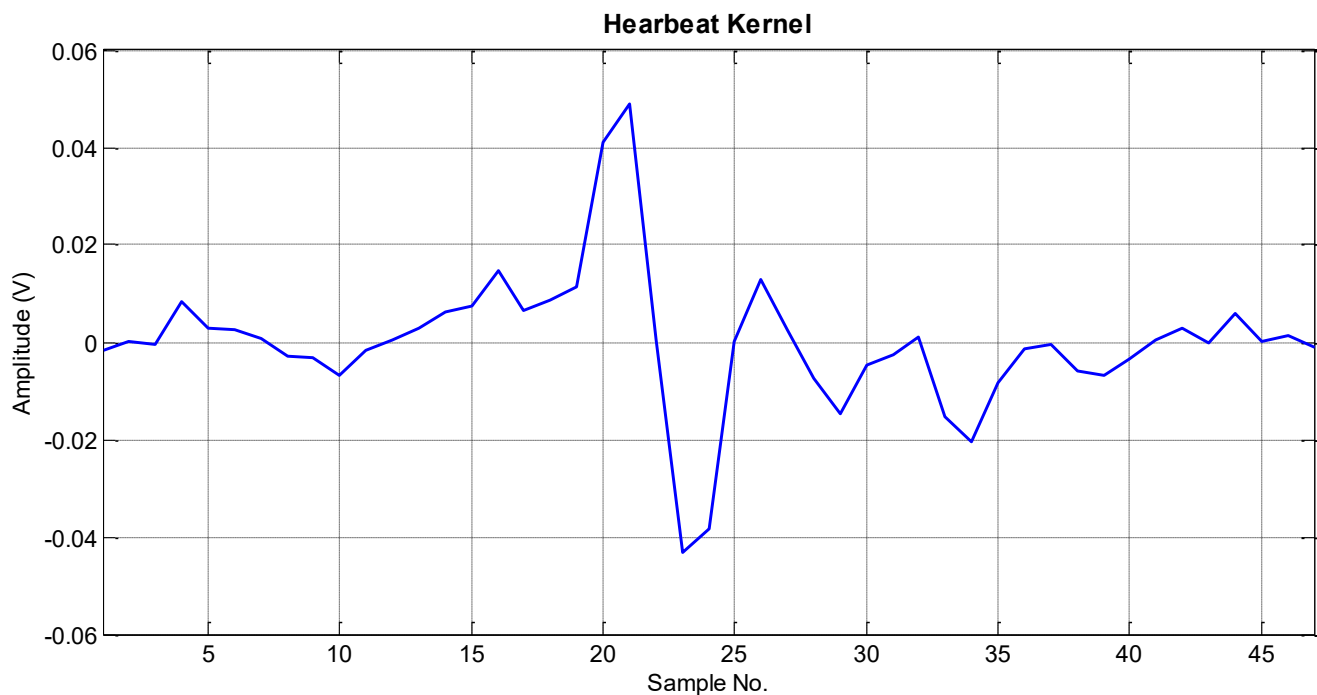


Fig 25 The heartbeat kernel used for Matched Filtering

The whole signal is then convolved with the time-reversed conjugated version of the heartbeat kernel. As the Doppler module outputs both In-Phase (I) and Quadrature (Q) channels matched filtering must be applied to both signals. The following code will implement this in MATLAB:

```
% I & Q channels are convolved with time-reversed conjugated hb_kernel
match_filteredI = conv(data_I,hb_kernel(end:-1:1));
match_filteredQ = conv(data_Q,hb_kernel(end:-1:1));

% Ensures vector lengths are the same as the time vector
match_filteredI = match_filtered(1:max(size(time)));
match_filteredQ = match_filtered(1:max(size(time)));
```

Figure 26 shows the I-channel before and after matched filtering. Although the signal after matched filtering has been suppressed in terms of its absolute magnitude, it is clear to see that the pattern has become clearer.

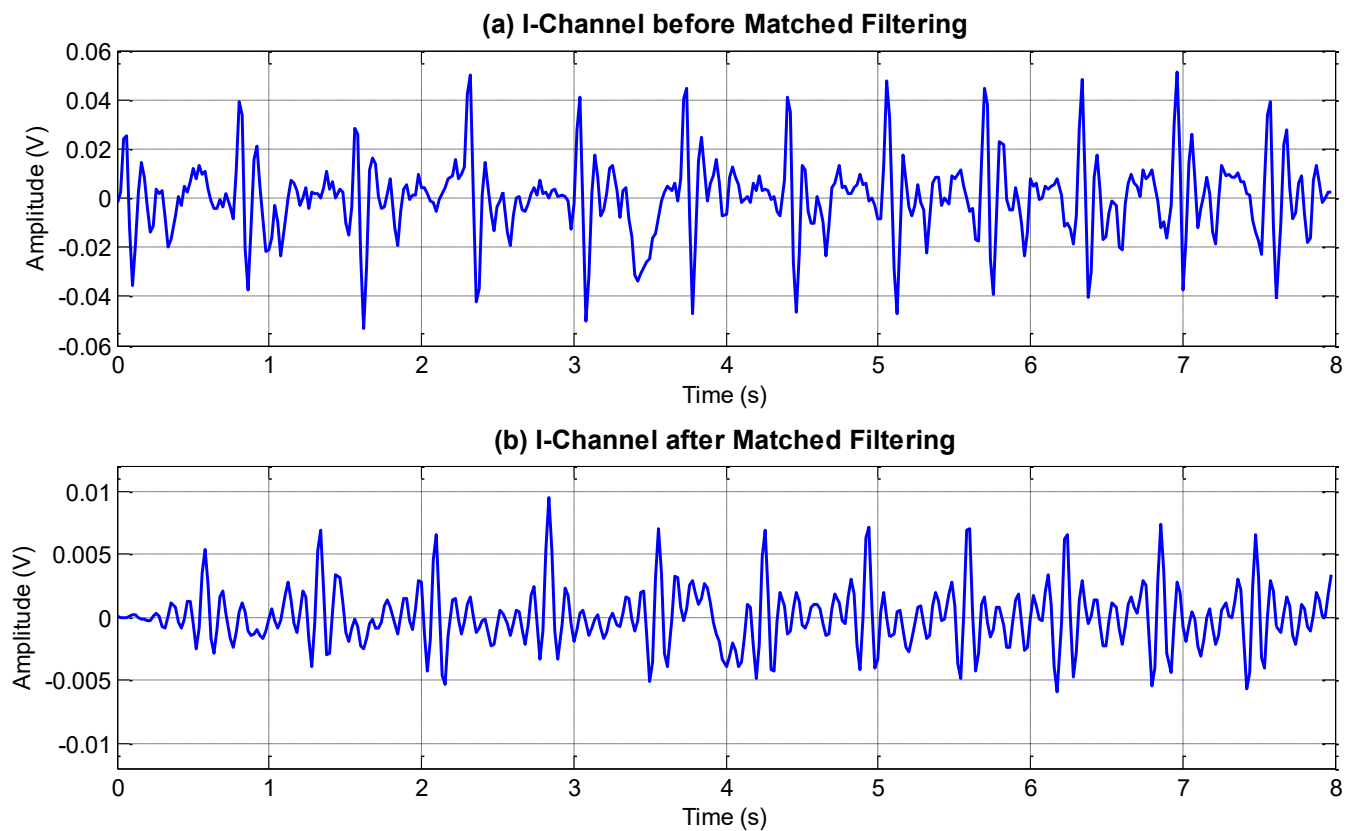


Fig 26 (a) I-Channel without Matched Filtering (b) I-Channel with Matched Filtering

One method of obtaining the heart rate from both these signals is by establishing a reference point in each heartbeat variation, such as the positive peak, and determining their relative period; from this the heart rate can be calculated. In Figure 26(a) the heart rate is worked out to be 88 beats per minute and in Figure 26(b) the heart rate is worked out to be 87 beats per minute.

The merit of using a matched filter is that it will detect the presence of the kernel, making patterns in the signal more prominent. In this case, the accuracy of the heart rate is retained. The fundamental flaw, in the use of matched filters, lies when the kernel does not match the pattern in the signal. For example, if the same process is applied to the Q-Channel, where the variation of the signal is somewhat different from that in the I-Channel, the matched filter is not as effective. This is clearly shown in Figure 27.

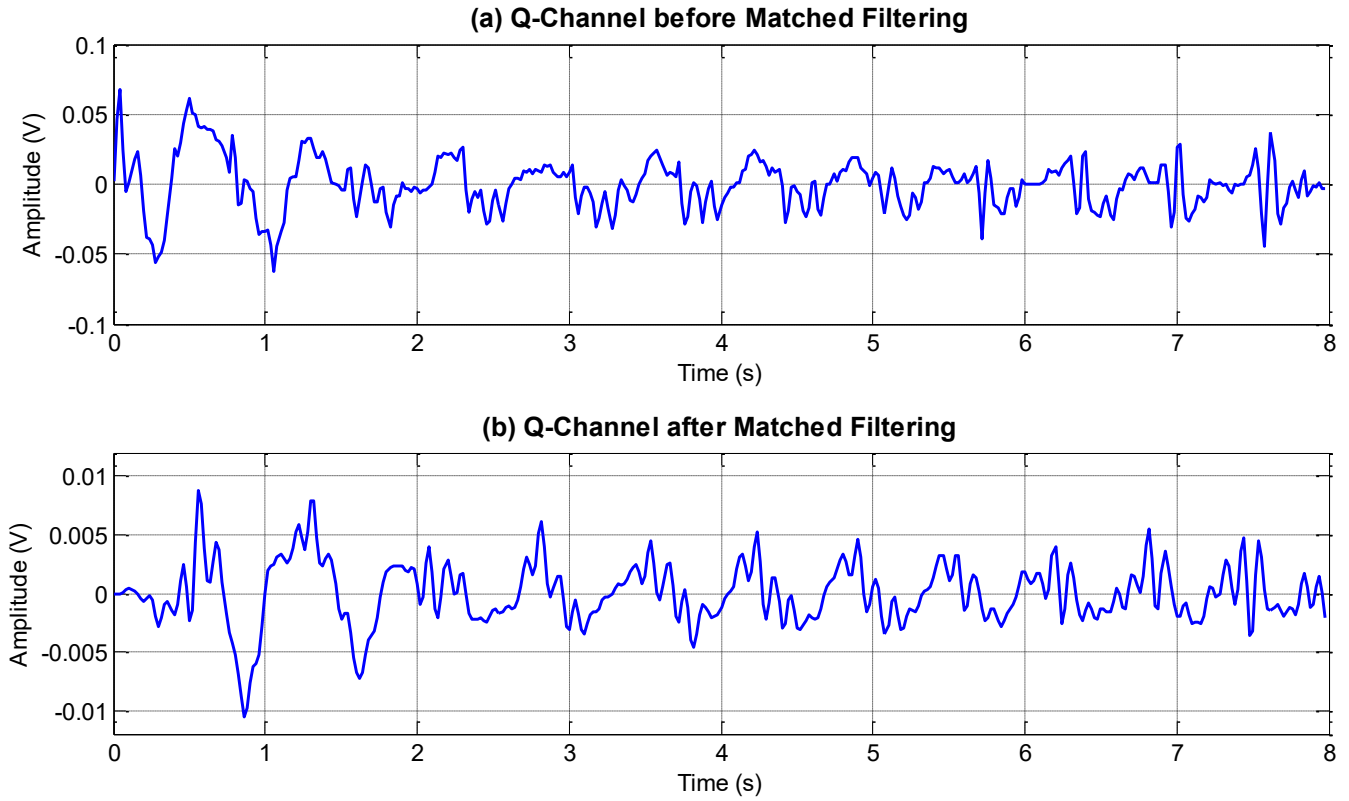


Fig 27 (a) Q-Channel without Matched Filtering (b) Q-Channel with Matched Filtering

This drawback is the main reason why matched filtering cannot be used for this algorithm. The pattern that is introduced by the heartbeat varies from use to use. As the heartbeat kernel has a fixed shape, it is rendered useless when the signal pattern varies. A dynamic kernel cannot be used as it will be impossible to extract the right portion of the signal; as measurements are not always started at the same stage of the cardiac cycle.

6.1.4. Continuous Wavelet Transform (CWT)

The Fourier Transform (FT) lends itself to the analysis of stationary periodic signals. The signal obtained from the Doppler module is periodic, but it does not have the characteristics of a stationary signal and it is therefore necessary to apply a different analysis technique. In this case, we are not only interested in the frequency distribution in spectrum of the obtained signal, but also its frequency distribution in time. [22]

The Continuous Wavelet Transform (CWT) has the attractive property of decomposing a signal into the time/frequency domain, and therefore allowing the analysis of specific frequency components over a time distribution. Equation 18 defines the CWT: [23]

$$S(a, \tau) = \frac{1}{\sqrt{a}} \int s(t) \psi^* \left(\frac{t-\tau}{a} \right) dt \quad [\text{Eqn 18}]$$

Where a is the scale of the transforming wavelet along the frequency axis and τ is its translation along the time axis. The integral in Equation 1 is seen as filtering the signal $s(t)$ through a series of wavelet mother functions, translated by τ and dilated by a .

There are a number of mother wavelet functions which can be used in the CWT and one of the most commonly used is the Morlet mother wavelet. The complex Morlet is given by Equation 19: [24]

$$\psi(x) = \pi^{-\frac{1}{4}} \left(e^{j\omega_0 t} - e^{-\frac{1}{2}\omega_0^2} \right) e^{-\frac{1}{2}x^2} \quad [\text{Eqn 19}]$$

Where ω_0 is the centre frequency of the wavelet and x is its associated bandwidth. A more robust method is to use the real part Morlet wavelet as the mother wavelet and is given in Equation 20: [25]

$$\psi(x) = e^{-\frac{1}{2}x^2} \cos(5x) \quad [\text{Eqn 20}]$$

Figure 28 shows the real part Morlet wavelet:

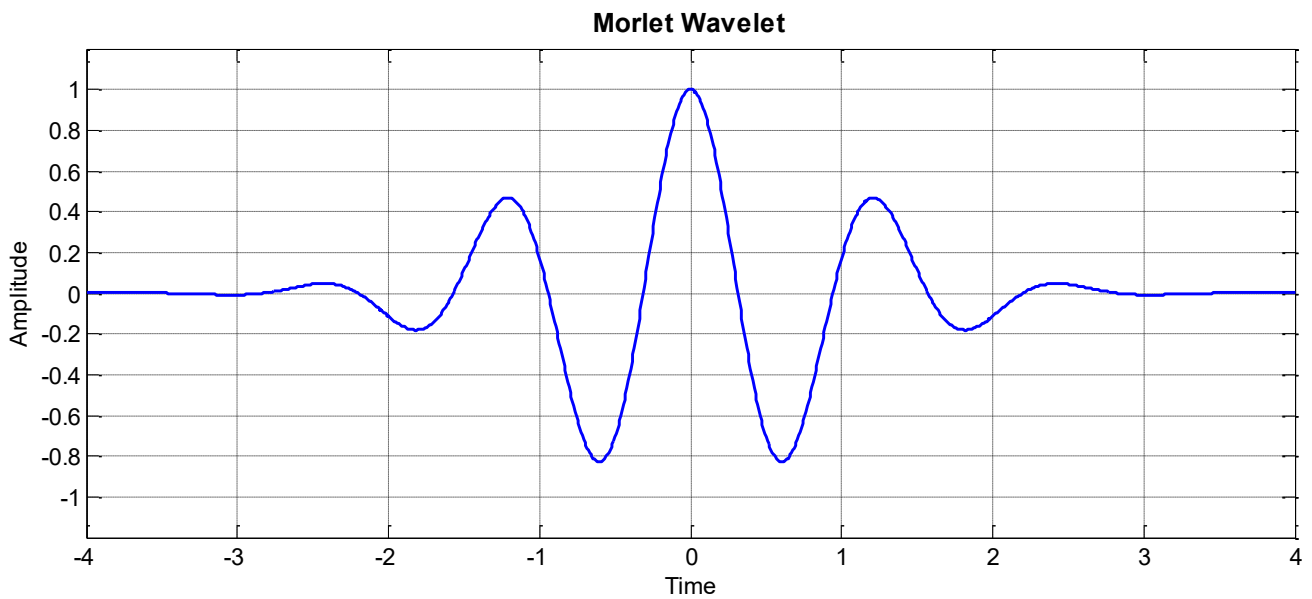


Fig 28 Real part Morlet wavelet

6.1.4.1. Example Waveform

The best way to demonstrate the usefulness of the CWT is through an example. Figure 29 shows a non-stationary periodic signal which will be used in this example. It consists of three sine waves of frequency 40 Hz, 20 Hz and 10 Hz, concatenated to form one signal waveform.

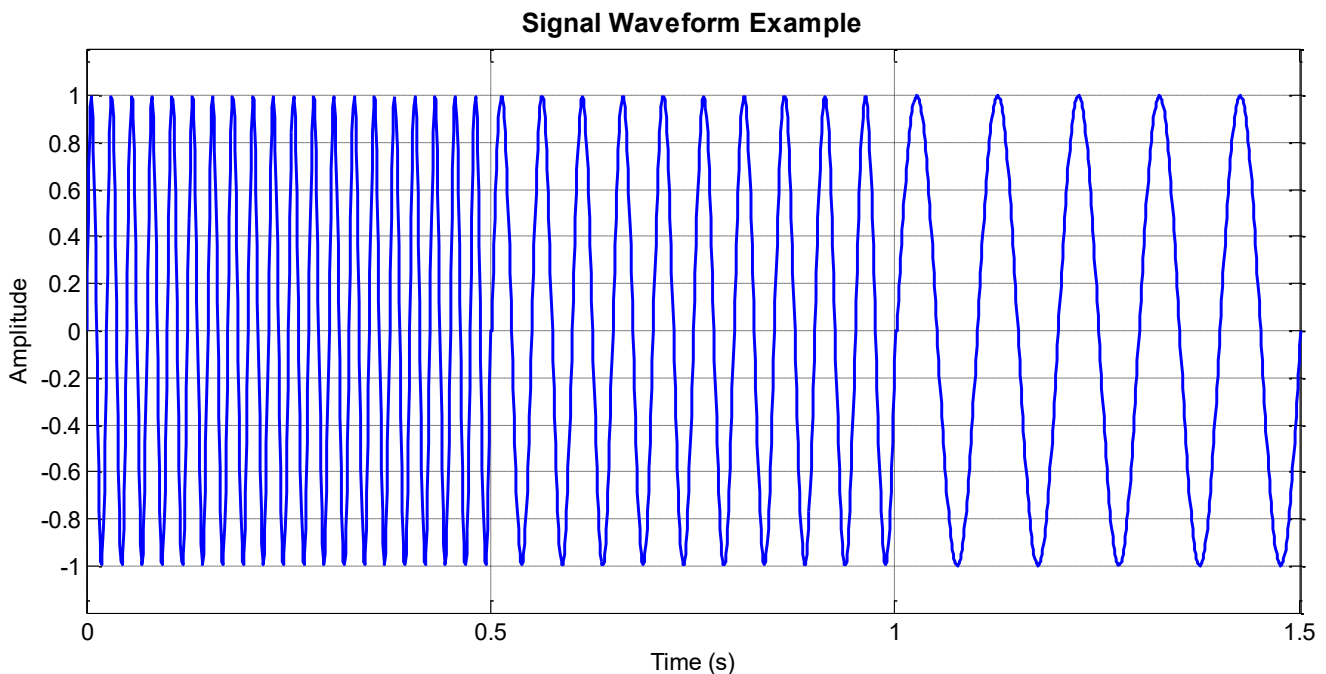


Fig 29 Non-stationary periodic signal

By performing the CWT on this signal, using the real part Morlet wavelet as the mother wavelet, the frequency/time distribution can be obtained and specific frequency coefficients can be isolated.

The following MATLAB code can be used to implement this example:

```
% Declaring the signal waveform
t=0:0.001:0.5;
y = [sin(2*pi*40*t) sin(2*pi*20*t) sin(2*pi*10*t)];
time=0:0.00098:1.5;

% Declaring frequency range of CWT
scale = (1:1:120);
% The scale values are converted to their pseudo-frequencies
F = scal2frq(scale,'morl',0.001);

% CWT
[coeffs] = cwt(y,scale,'morl', '3Dplot');
```

Figure 30 shows the 3D CWT plot of the example signal, where it is clear to see the three different frequencies of the signal and their subsequent positions in time. In this case we see both the positive and negative coefficient values. Positive coefficients correspond to peaks in the waveform and negative coefficients correspond to the troughs.

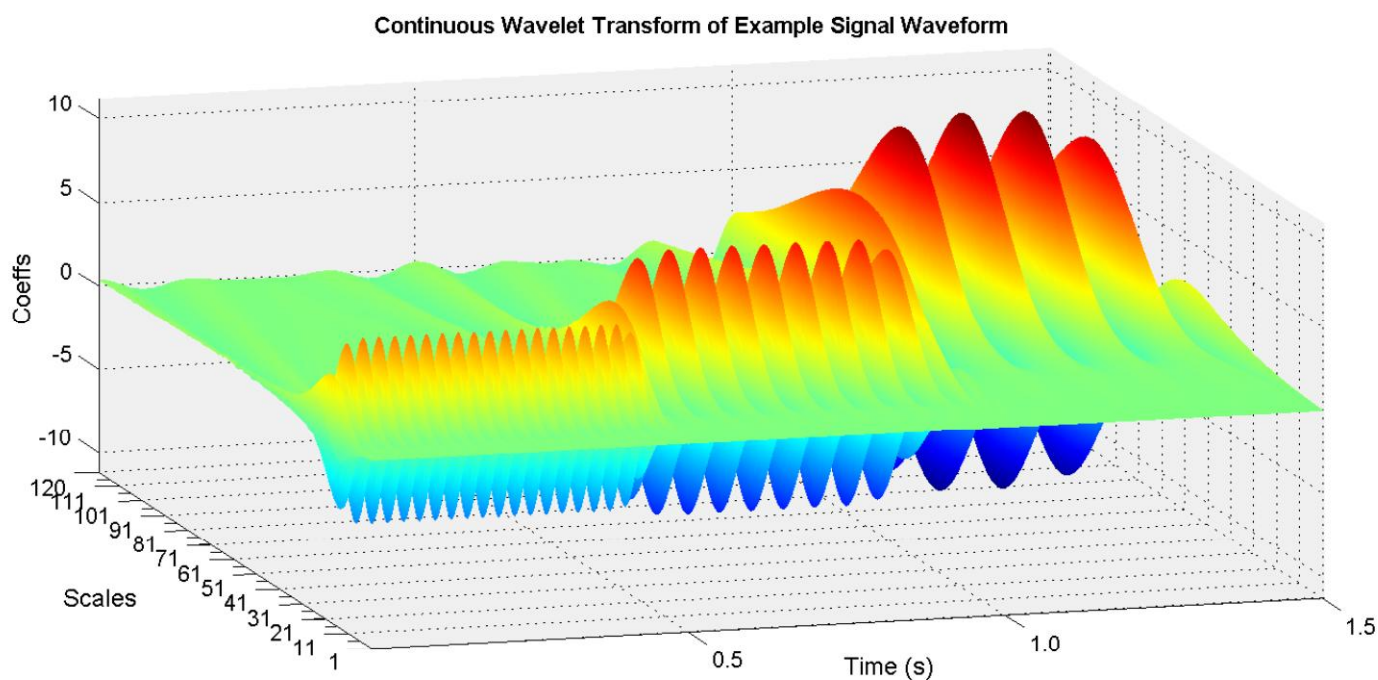


Fig 30 The CWT of the example signal waveform (3D plot)

A more informative view of this result can be obtained by plotting the 2D CWT of the example signal and is shown in Figure 31. The red areas correspond to positive coefficients, whereas the blue areas correspond to negative coefficients.

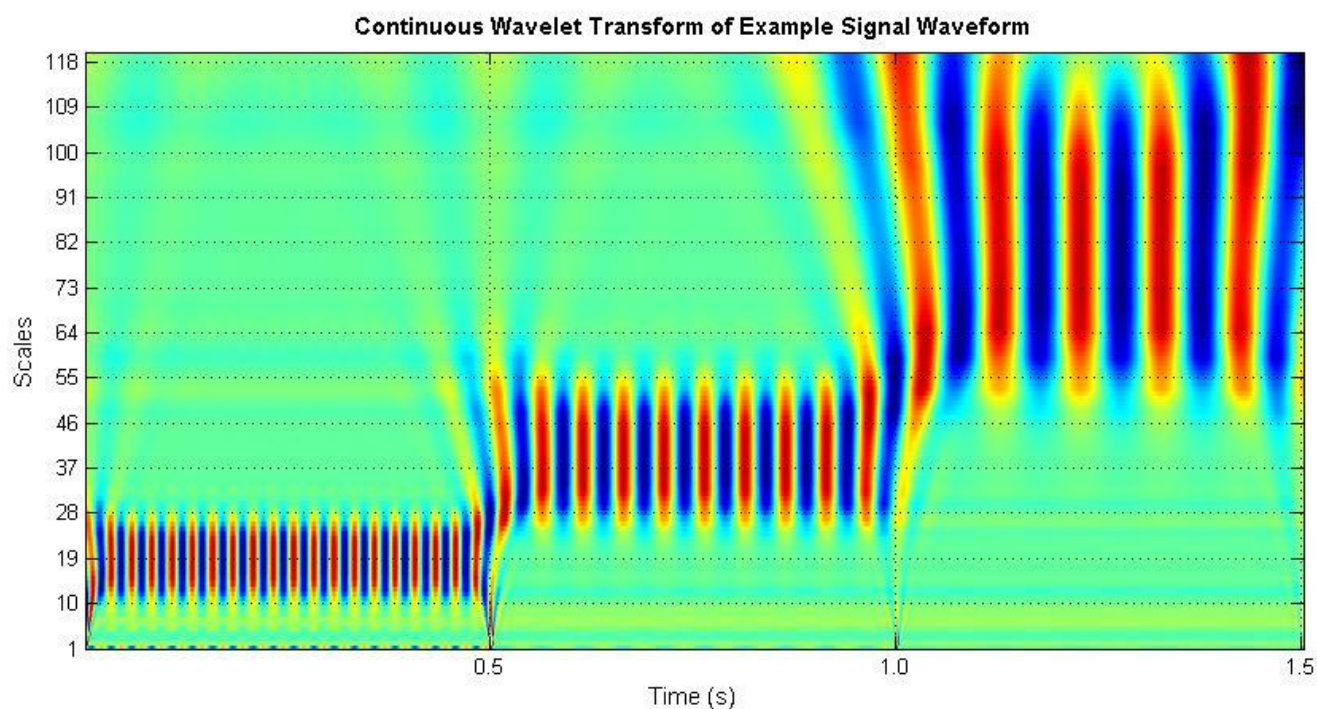


Fig 31 The CWT of the example signal waveform (2D plot)

By looking at the pseudo-frequencies corresponding to the scale values of the CWT, certain frequencies can be isolated. This is a useful tool in the analysis of the Doppler module signal as the frequency deviation induced by each heartbeat can be isolated and its periodic behaviour analysed. Figure 32 shows the isolation of each of the frequencies in the waveform, which is achieved by taking a tranche of the CWT relating to a specific scale/frequency.

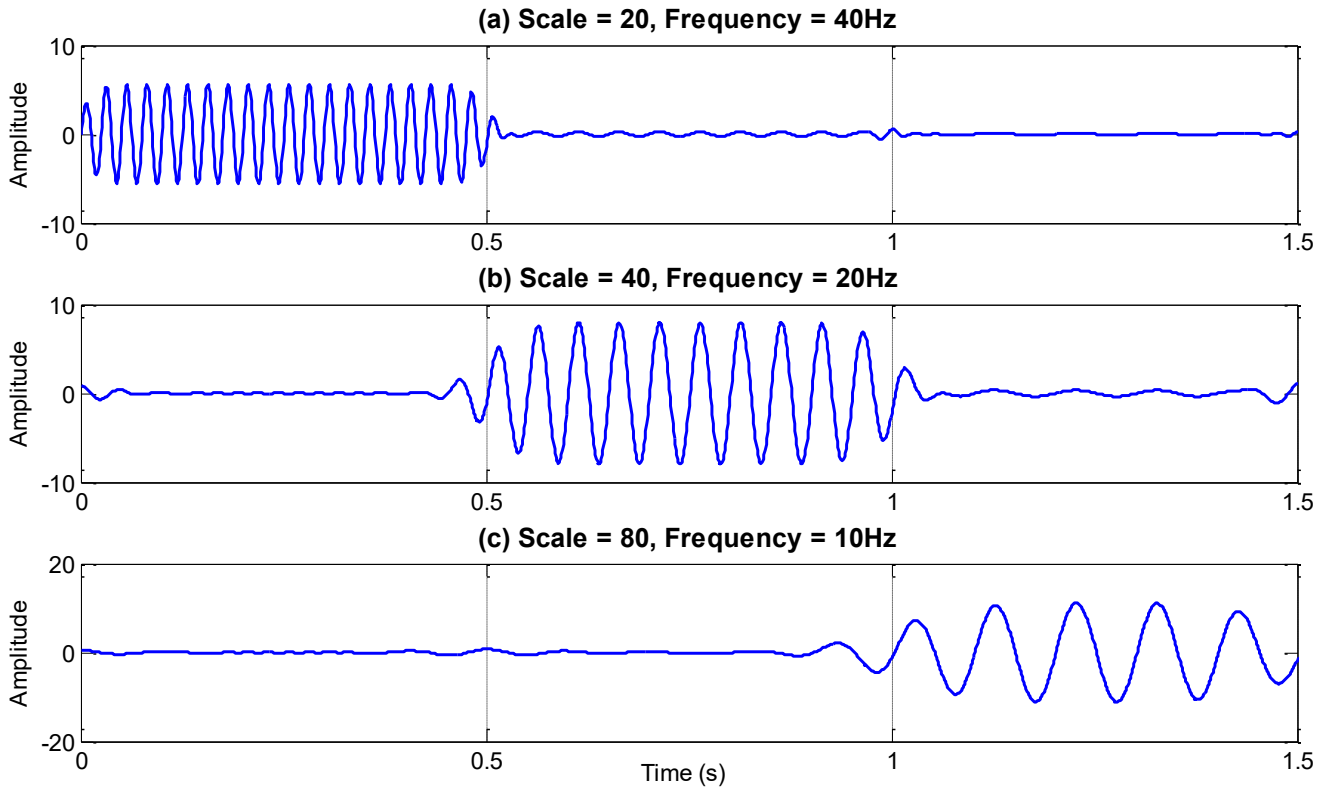


Fig 32 (a) Isolated 40Hz section (b) Isolated 20Hz section (c) Isolated 10Hz section

Frequency and scale are linked by Equation 21. [26]

$$F_a = \frac{F_c}{a \cdot \Delta} \quad [\text{Eqn 21}]$$

Where F_a is the pseudo-frequency corresponding to the scale a , in Hz; F_c is the centre frequency of the Morlet wavelet (0.8125 Hz); and Δ is the sampling period of the signal.

6.1.4.2. Real Doppler Signal

A real signal can now be introduced and the CWT can be calculated. Figure 33 shows the I-Channel signal obtained from the Doppler module and Figure 34 shows its corresponding 3D CWT.

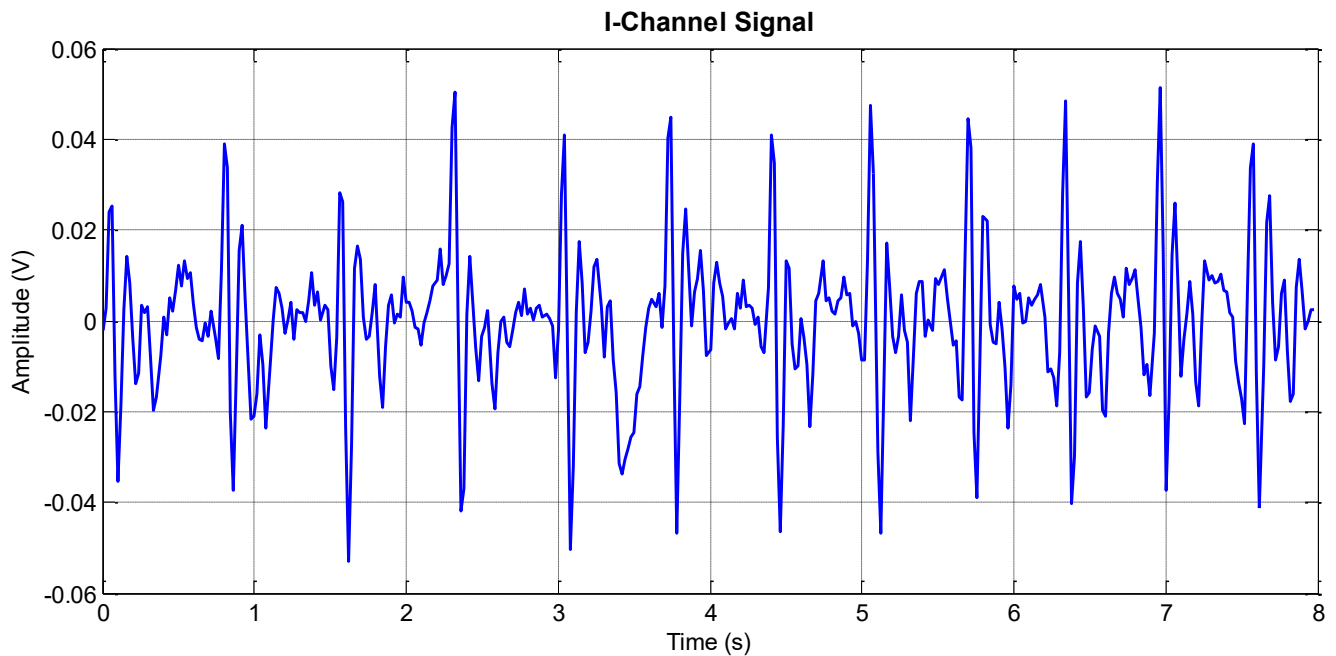


Fig 33 I-Channel signal obtained from Doppler module

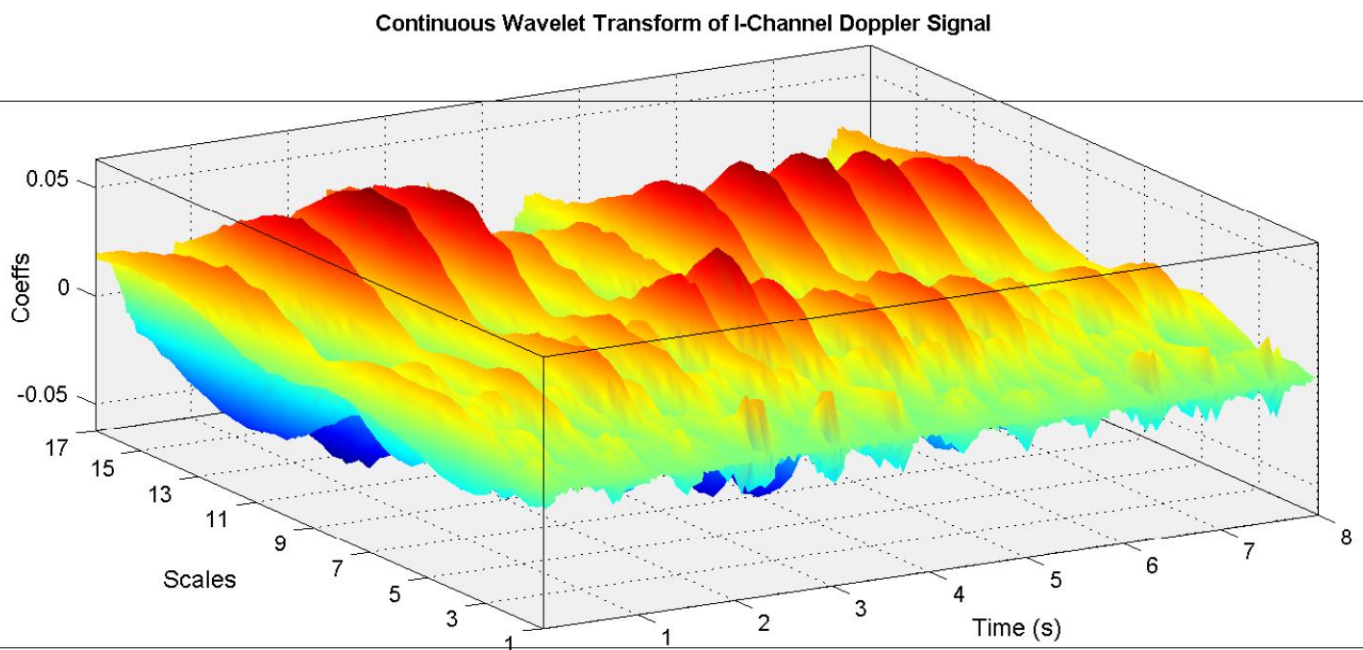


Fig 34 The CWT of the I-Channel Doppler signal (3D plot)

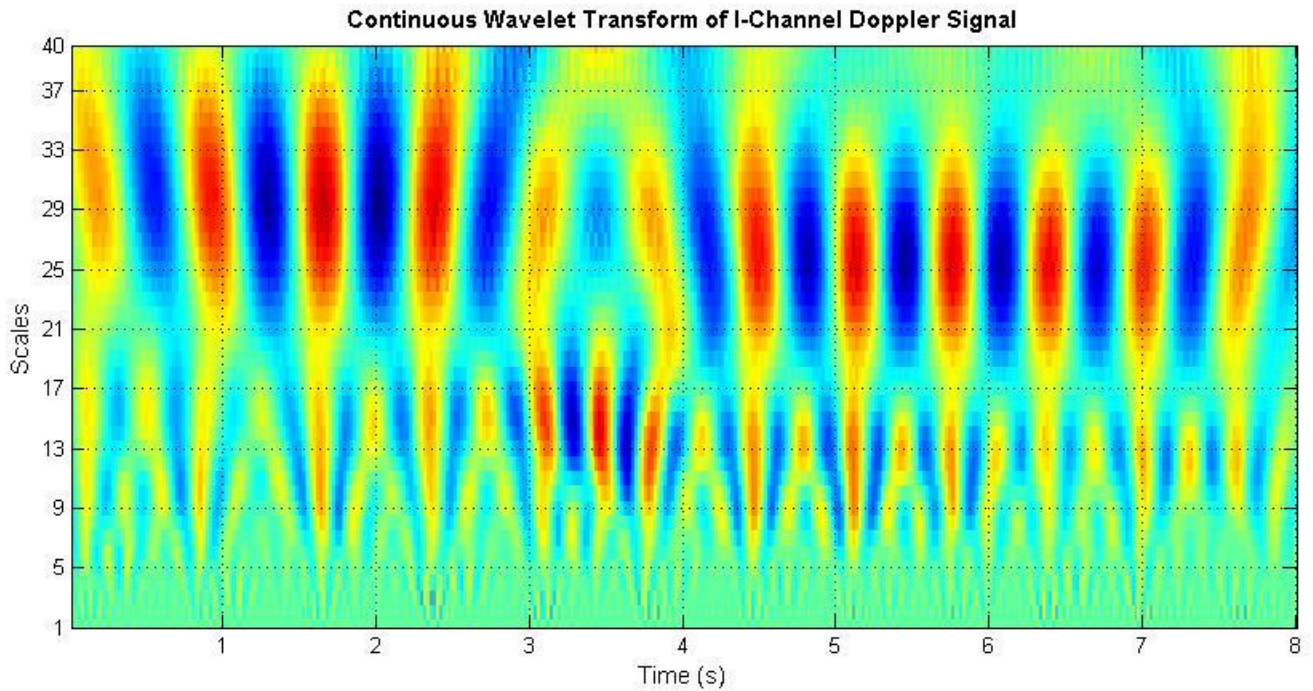


Fig 35 The CWT of the I-Channel Doppler signal (2D plot)

There are two distinct levels of frequencies in this signal; those in the scale range of 8-19 and those in the range of 20-37. By concentrating on one of these ranges, which in this case will be the more prominent range, the periodicity of the signal and subsequent heart rate can be obtained. Using the same principle as before, specific tranches of the CWT can be isolated. Figure 36 shows the least prominent CWT tranche, whereas Figure 37 shows the most prominent CWT tranche.

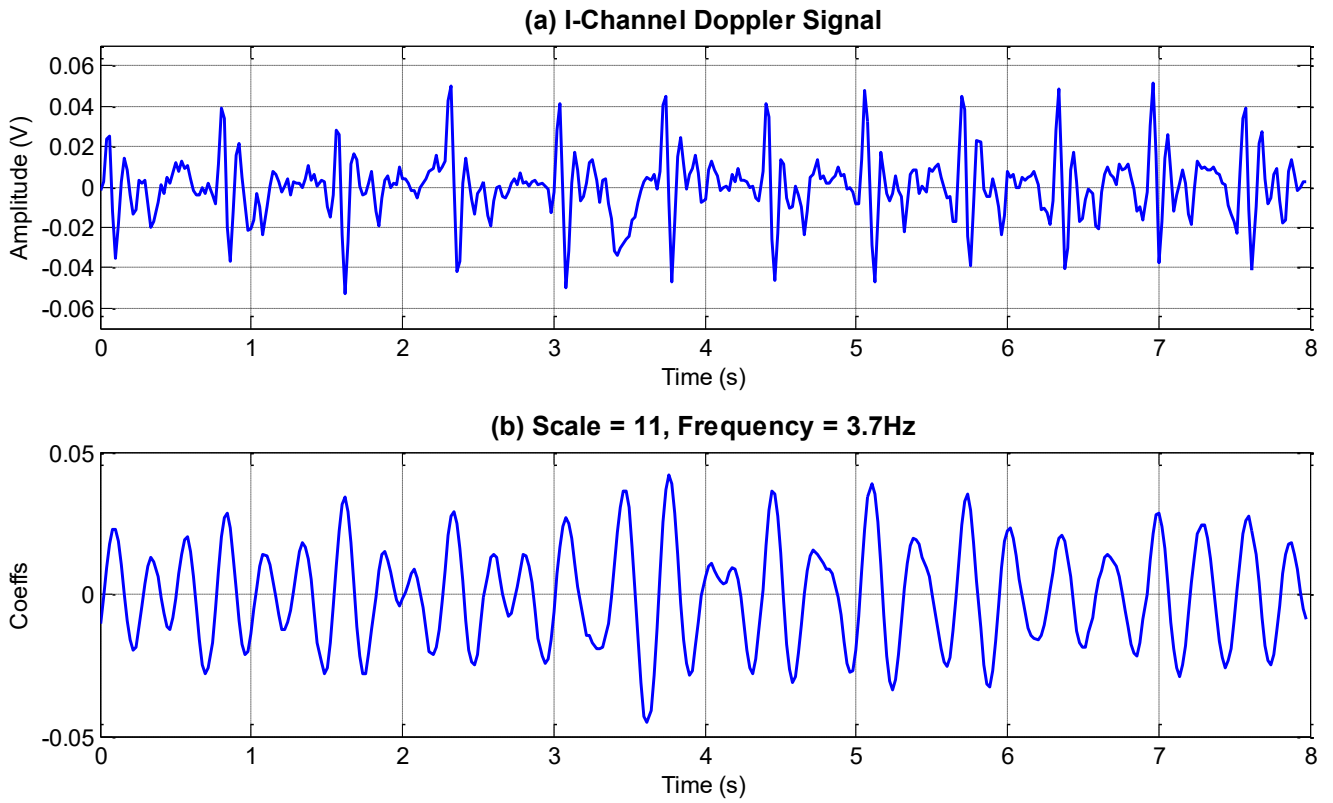


Fig 36 (a) I-Channel Doppler signal (b) Least prominent CWT tranche

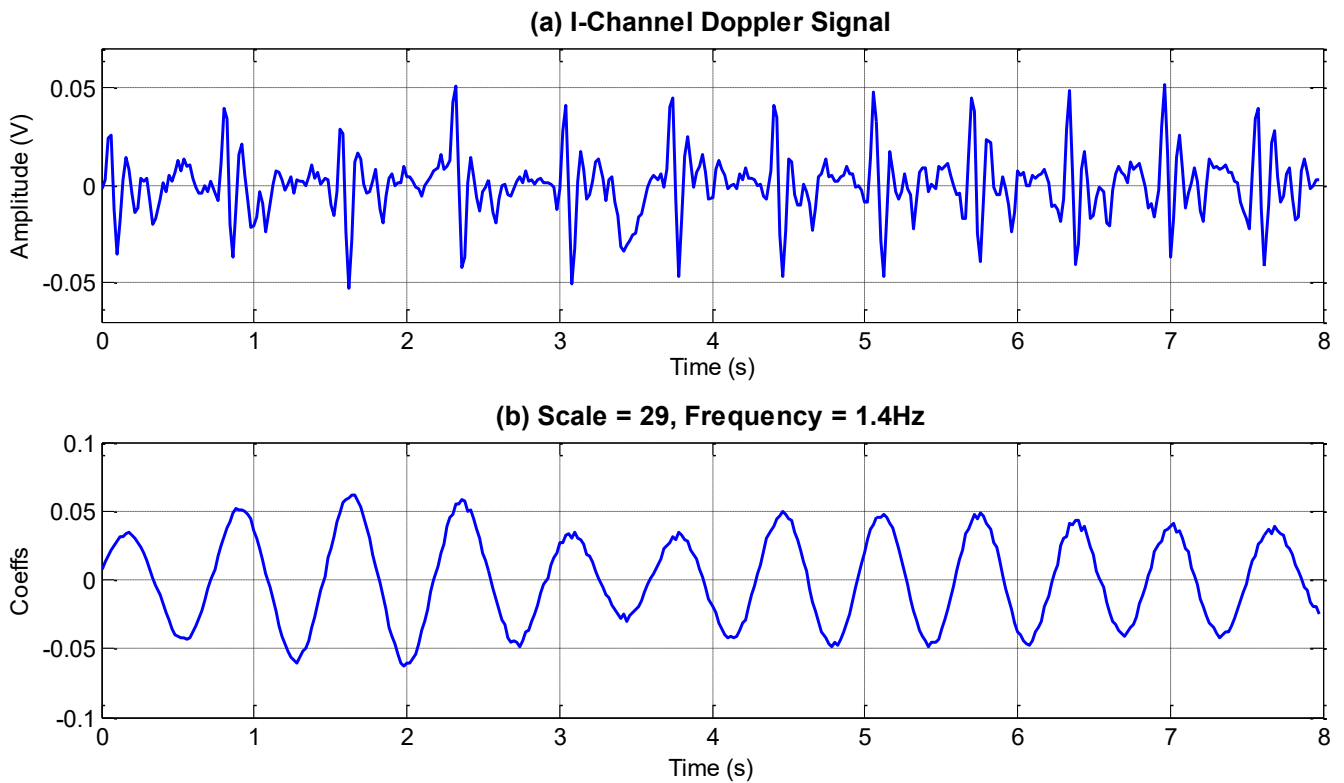


Fig 37 (a) I-Channel Doppler signal (b) Most prominent CWT tranche

By isolating certain tranches of the CWT, useful information for analysis can be obtained. Figure 36(b) shows that the higher range of frequencies, corresponding to a lower range of scales, offers little in terms of determining the periodicity of the heartbeat variations. However, in Figure 37(b) the CWT tranche shows a well-defined peak for each heartbeat variation in the time-domain signal. Therefore, this CWT tranche can be used to establish the periodicity of the heartbeats, and therefore establish the heart rate. Figure 38 shows the frequency spectrum of this tranche.

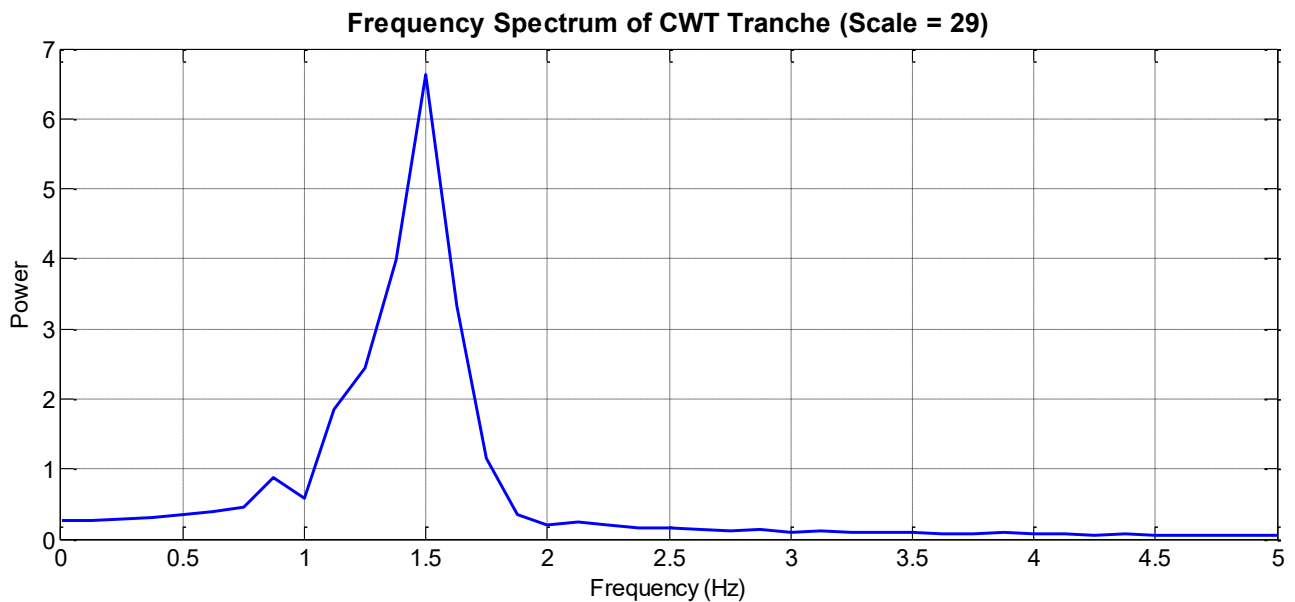


Fig 38 Frequency Spectrum of the most prominent CWT tranche (Scale = 29)

The prominent peak at 1.5 Hz tells us that the heart rate in this case is 90 beats per minute. Through the analysis of the time-domain signal, which has 12 beats per 8 seconds, this calculated heart rate is correct. Now we know that a specific element of the heartbeat variation in the time-domain signal has a frequency around 1.4 Hz, this frequency tranche can be isolated and lower frequencies induced by breathing can be ignored.

The same principle of frequency tranche isolation can also be used to determine the heart rate from the Q-channel Doppler signal. The heartbeats induce a different time-domain variation in the Q-Channel Doppler signal. Figure 39 shows this difference.

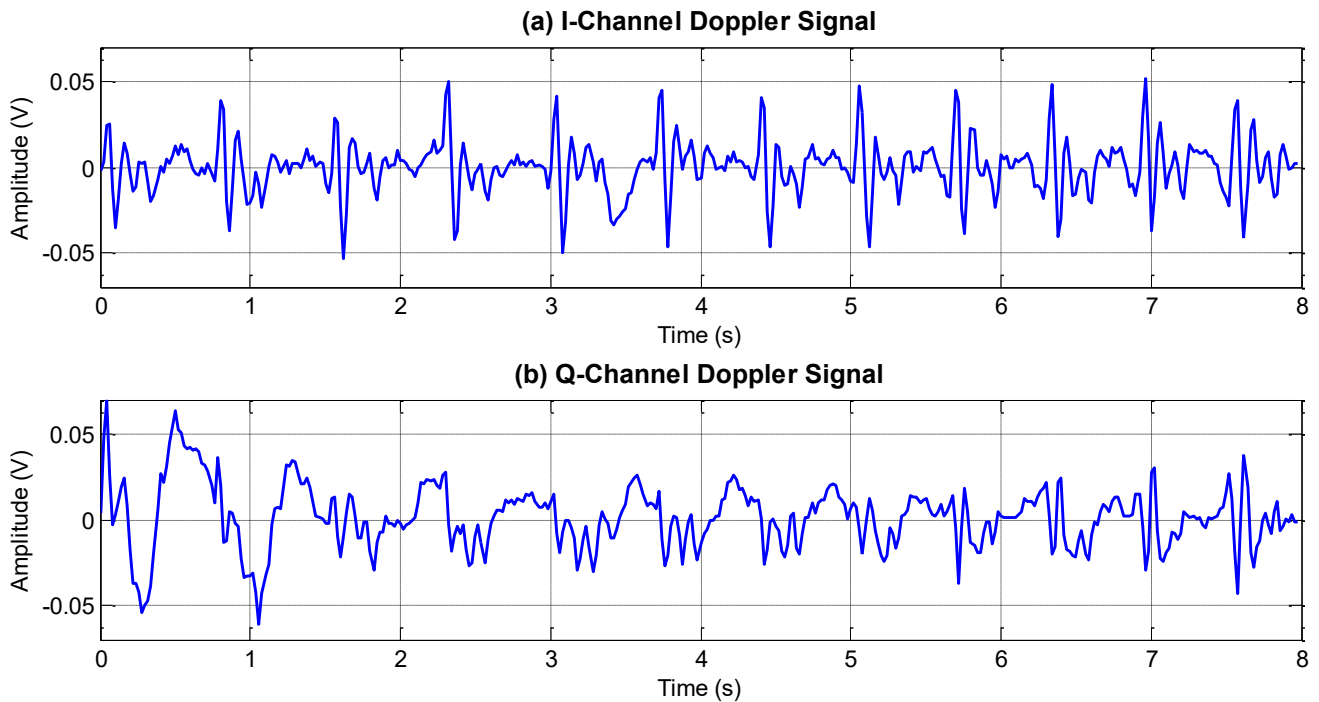


Fig 39 (a) I-Channel Doppler signal (b) Q-Channel Doppler signal

By definition the In-Phase and Quadrature channels should be different; however, in this case it is not only the phase that is different but also the time-domain signal variation induced by each heartbeat. This difference is the main reason why Spectrum Analysis & Matched Filtering techniques are not effective. [Sections 6.1.2 & 6.1.3]

This limitation is overcome when using the CWT as it is the underlying frequency of each heart beat variation which is analysed and not the time-domain signal. Figure 40 shows the 2D CWT of the Q-Channel Doppler signal and Figure 41 shows its 1.4 Hz frequency tranche (Scale = 29). Again, by looking at the frequency spectrum of this tranche the heart rate can be determined. Figure 42 shows the frequency spectrum of this CWT tranche.

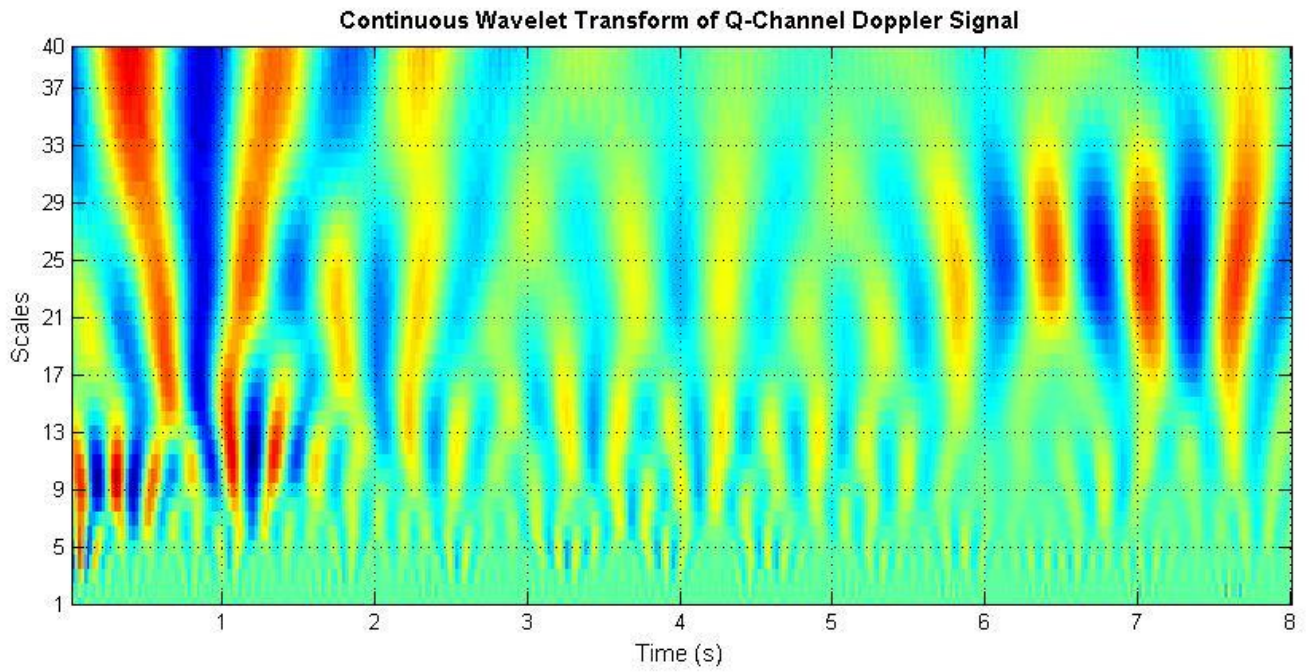


Fig 40 The CWT of the Q-Channel Doppler signal (2D plot)

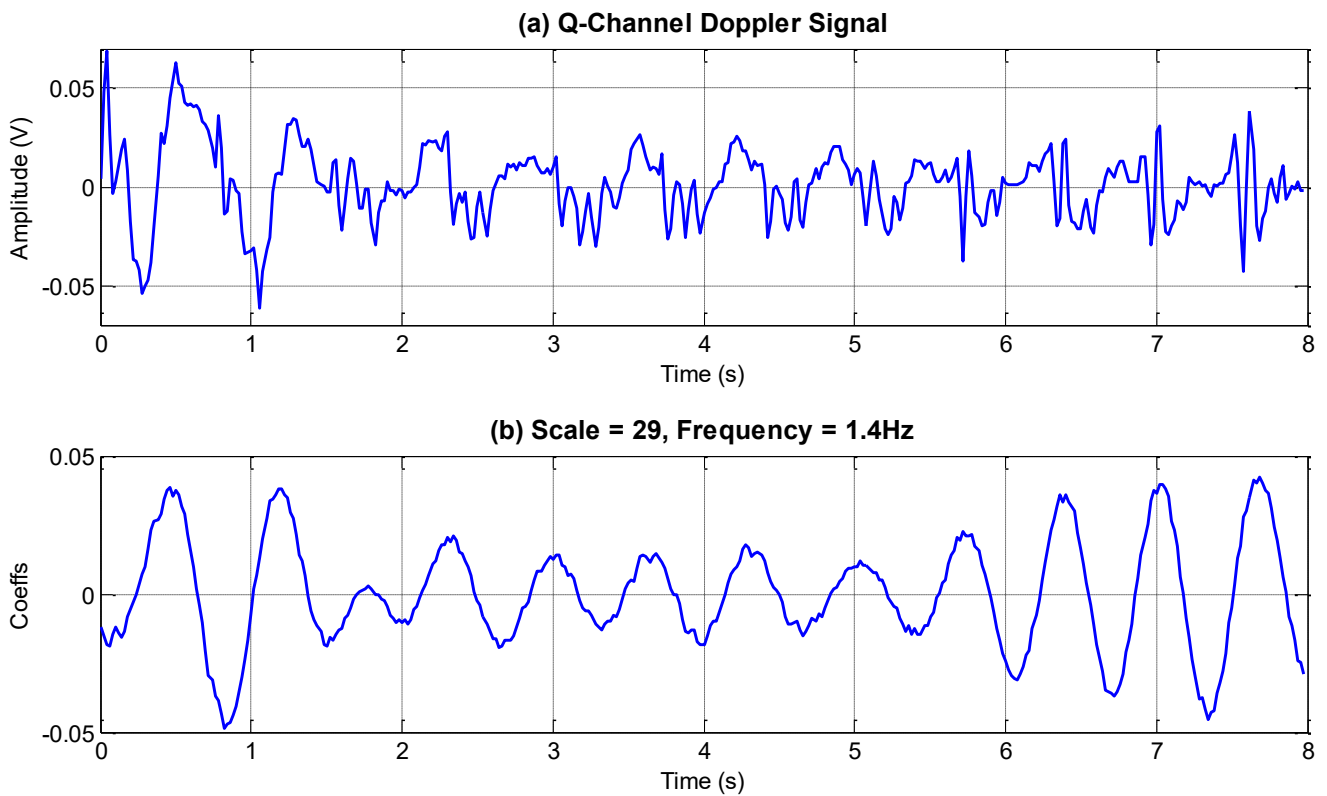


Fig 41 (a) Q-Channel Doppler signal (b) 1.4Hz frequency tranche (Scale = 29)

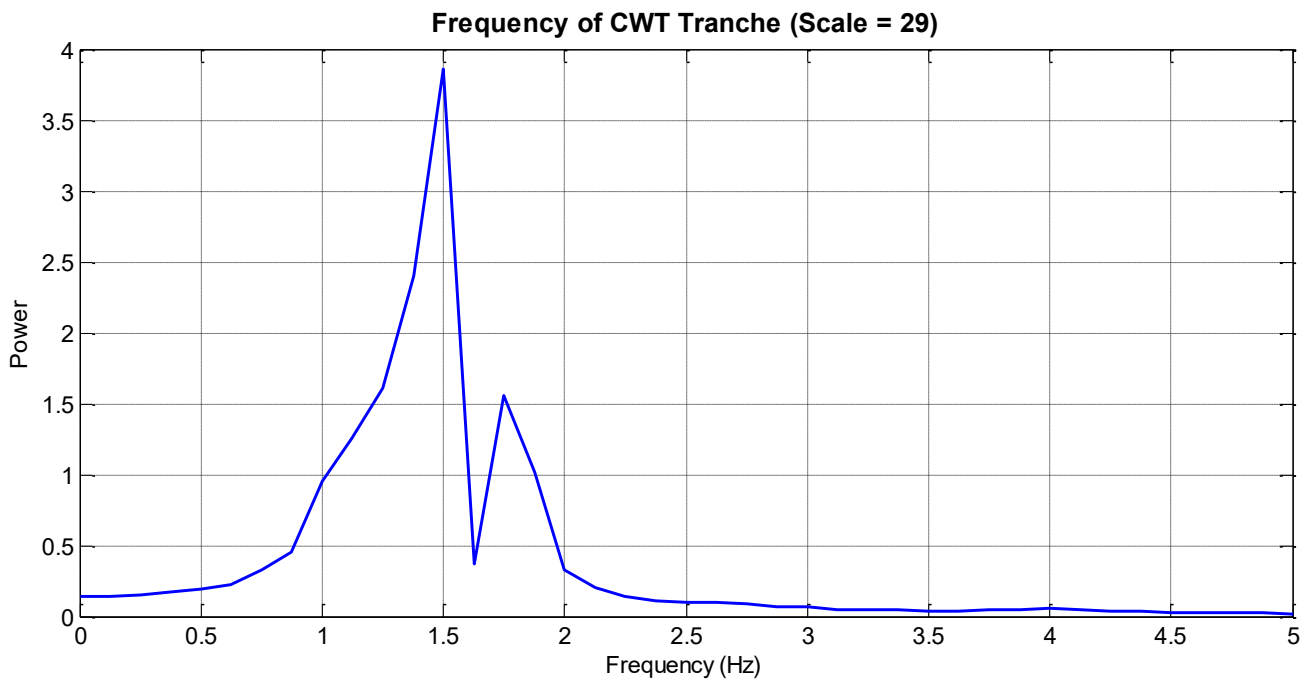


Fig 42 Frequency Spectrum of the most prominent CWT tranche (Scale = 29)

The prominent peak at 1.5 Hz tells us that the heart rate in this case is 90 beats per minute. This is exactly the same as what was calculated from the I-Channel. So this method provides a more robust approach to determining the heart rate. What needs to be established is whether it can be used when the subject is also breathing.

6.2. Development – Breathing

The noise introduced by the breathing motion of the chest presents a more complex signal processing problem. The difference in the time-domain signals can be seen in Figures 18 & 19. Section 6.1 describes the work that was carried establishing a robust algorithm that could extract the heart rate from the Doppler signal without breathing. Section 6.2 describes the implementation of these techniques on the Doppler signal with breathing.

6.2.1. Spectrum Analysis

As discussed in Section 6.1.2, the most straightforward way of finding the periodicity of a signal is through performing a FFT on it. The most prominent peak is then detected, which should correspond to the heart rate (in the case of the subject holding their breath). In practice this is quite difficult for a number of reasons, as discussed below:

- The amplitude of the signal is small in absolute terms and small compared to clutter noise. Therefore, the spectrum rarely exhibits any prominent peaks associated with the heartbeat variations.
- The exact Doppler signature of a heartbeat is not constant. It depends greatly on the person, heart rate, position, breathing pattern and many other factors which are hard to quantify and predict.

- Another problem is that depending on the absolute distance between the target and the antenna, and therefore position within the wavelength λ , the phase shift due to the heartbeats is decomposed into the I & Q channels differently. Even as the body shifts due to breathing, the position within the wavelength (12.5 mm) will change. This makes applying matched filtering practically impossible. We simply cannot anticipate what the signature will look like in each channel.
- As well as breathing the human body undergoes a lot of other movements, some of which we cannot control. These movements may be due to speaking, coughing, shifting around or any other deliberate movements. The amplitude of these movements is generally much larger than the amplitude of heart beat variations, introducing clutter noise and reducing the clarity of the received signal.

During the numerous tests that were conducted, where the objective was to find some feature of the signal which would help identify the heartbeat, it was noticed the following pattern often applied:

“The spectrum peak due to the heartbeat is likely to be contained in the channel which does not exhibit the highest spectrum peak in the region of 0.3 to 0.8 Hz”

This is illustrated in Figure 43. We could not come up with any mathematical justification for this empirical rule, and it must be noted that it does not always hold.

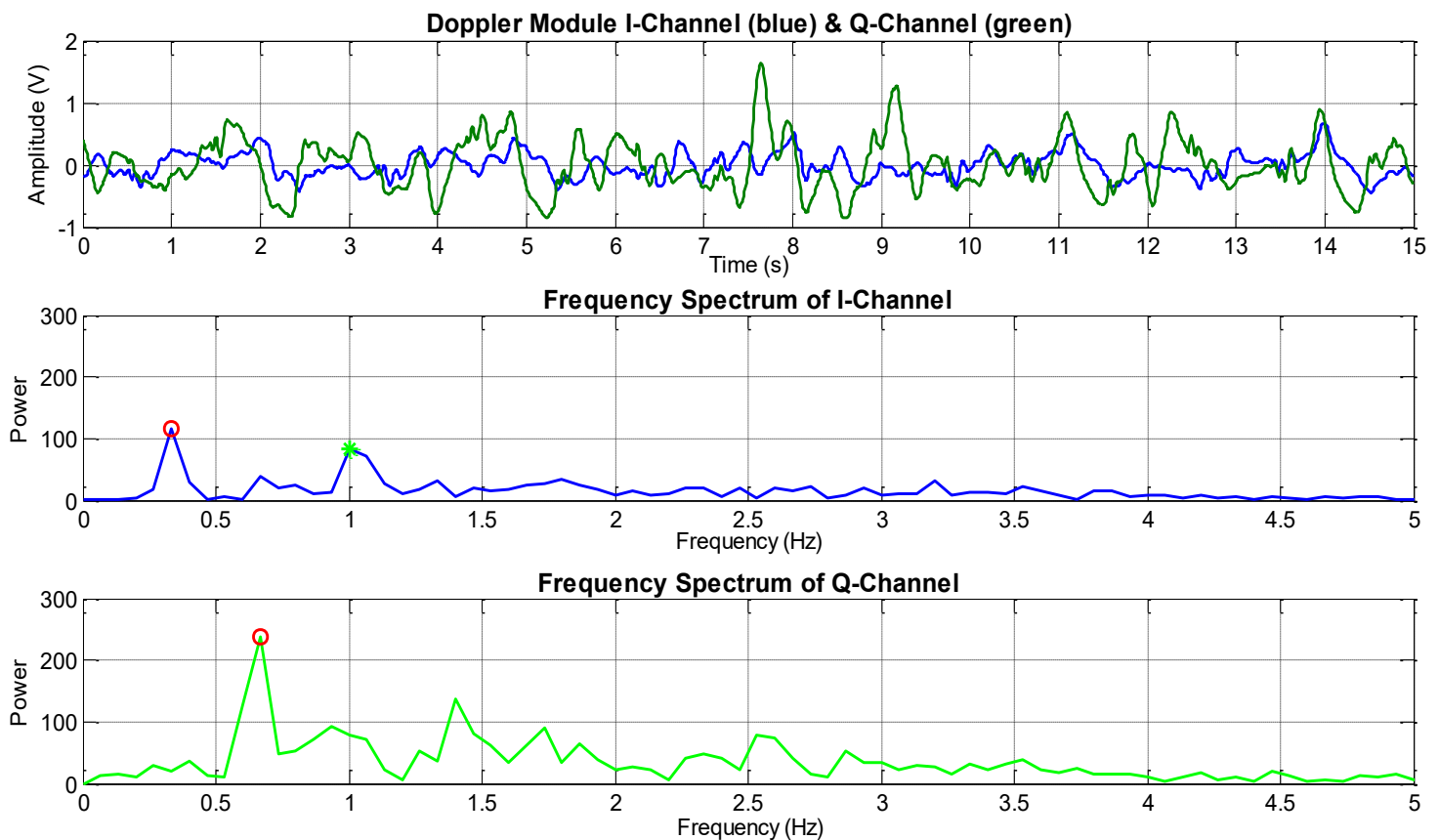


Fig 43 Rule of thumb

6.2.2. Continuous Wavelet Transform (CWT)

The CWT can be used to isolate certain frequencies of a signal, while still maintaining its variations through time. By isolating the frequency corresponding to the heartbeat variations, the lower frequency variations introduced by breathing can be ignored. Figure 44 shows the I-Channel Doppler signal with breathing and Figure 45 shows its respective 2D CWT.

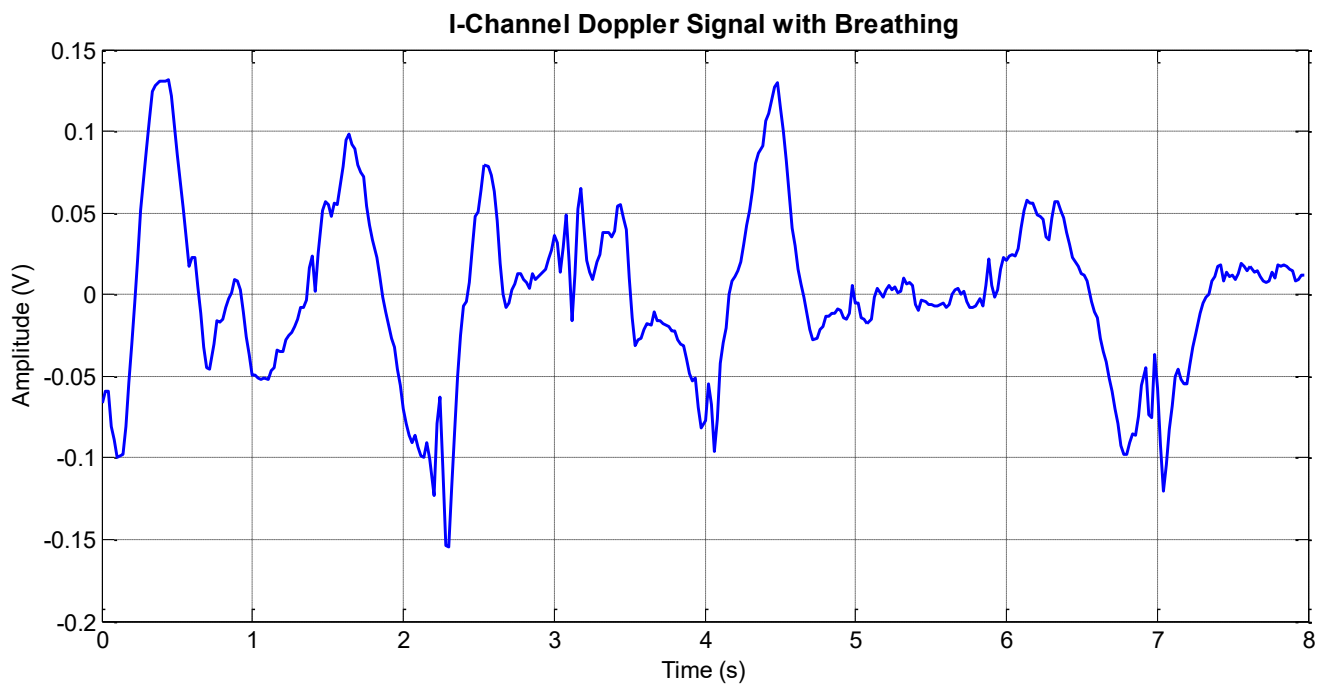


Fig 44 I-Channel Doppler signal with breathing

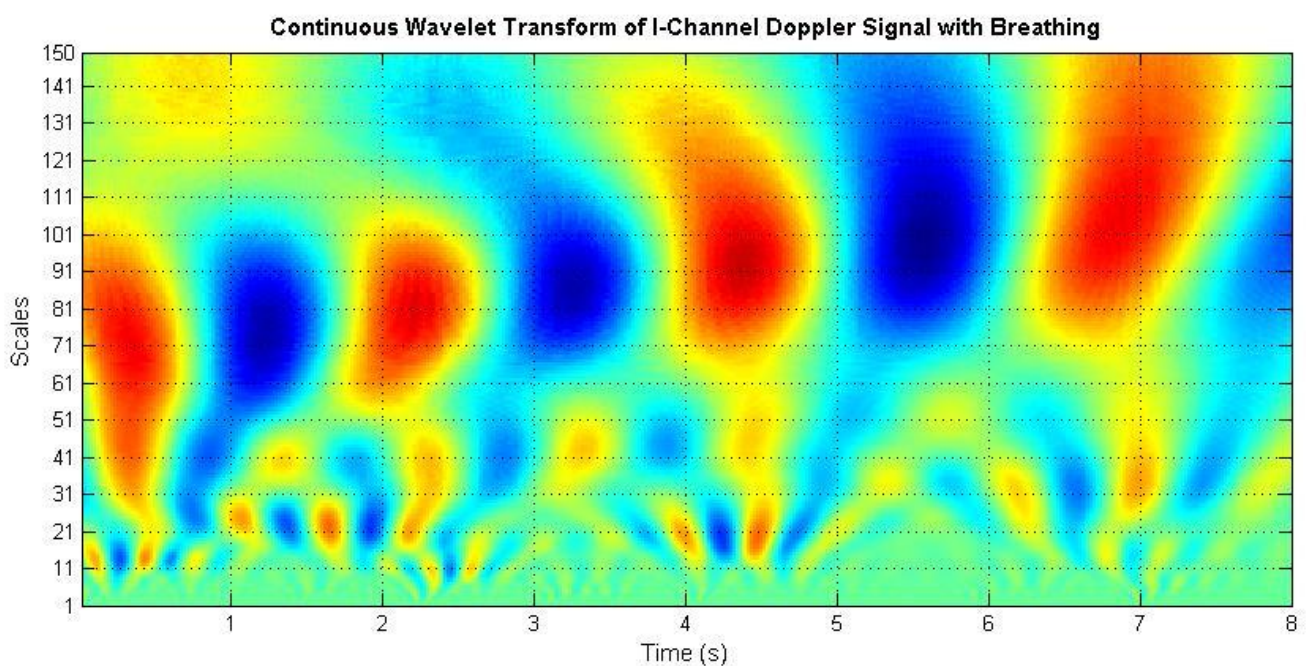


Fig 45 The CWT of the I-Channel Doppler signal with breathing (Scale Range: 1-150)

In Figure 45 it is clear to see the frequency variations which arise from the breathing motion. These variations occur at scale 90, whereas the heartbeat variations occur around scale 29, equating to a frequency of 1.4Hz. If the range of scales covered by the CWT is made smaller, the heartbeat variations can be identified. This is shown in Figure 46.

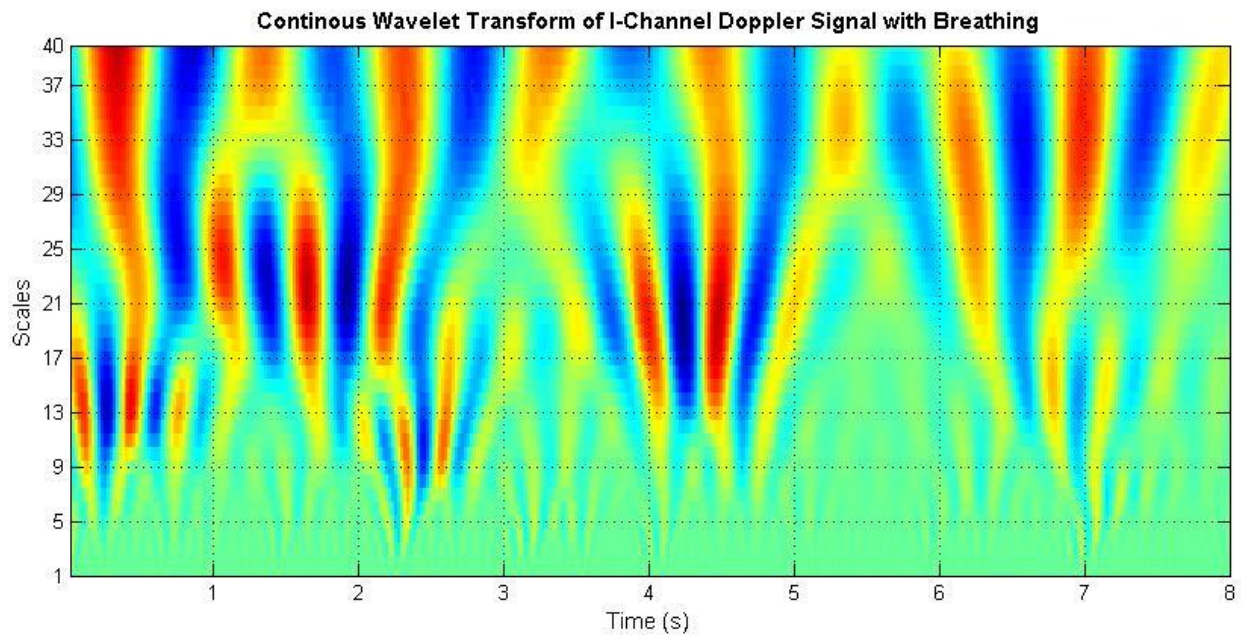


Fig 46 The CWT of the I-Channel Doppler signal with breathing (Scale Range: 1-40)

Figure 47 shows the CWT tranches for a scale of 90, showing the breathing frequency variation, and for a scale of 29, showing the heartbeat frequency variation. Figure 48 shows their respective frequency spectrums.

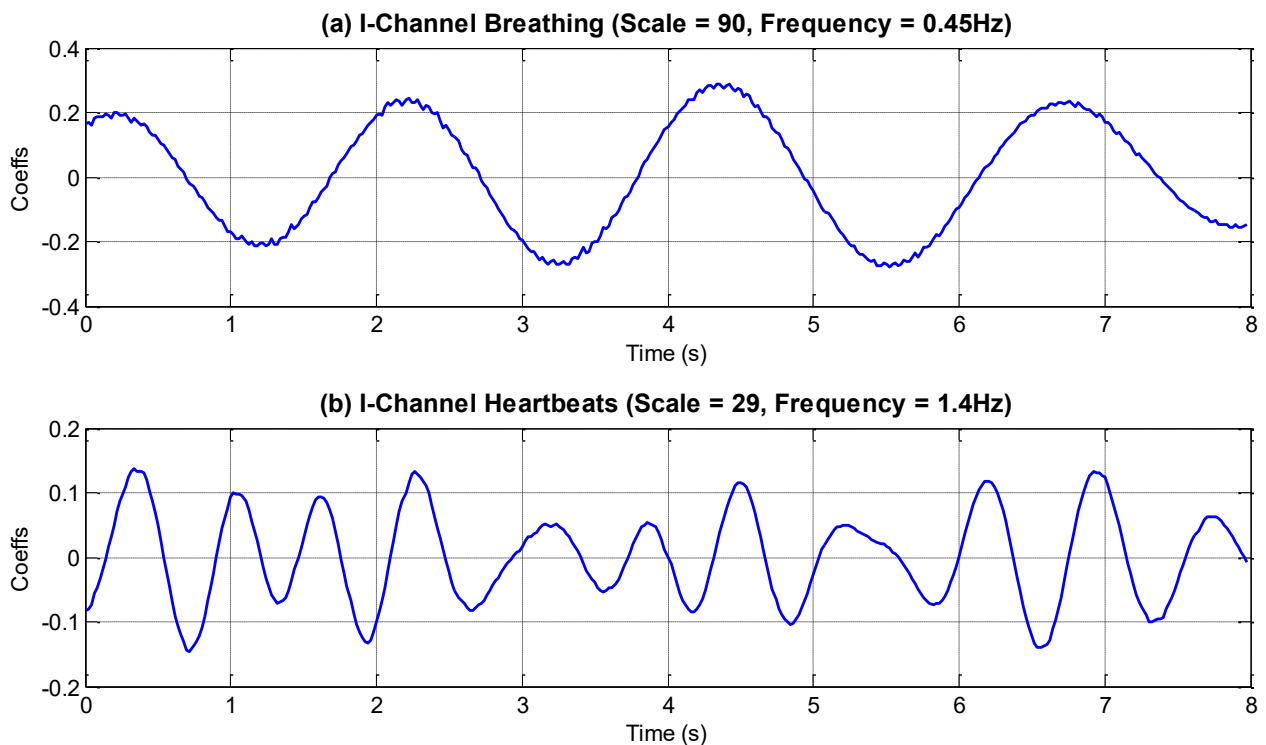


Fig 47 (a) I-Channel breathing (b) I-Channel heartbeats

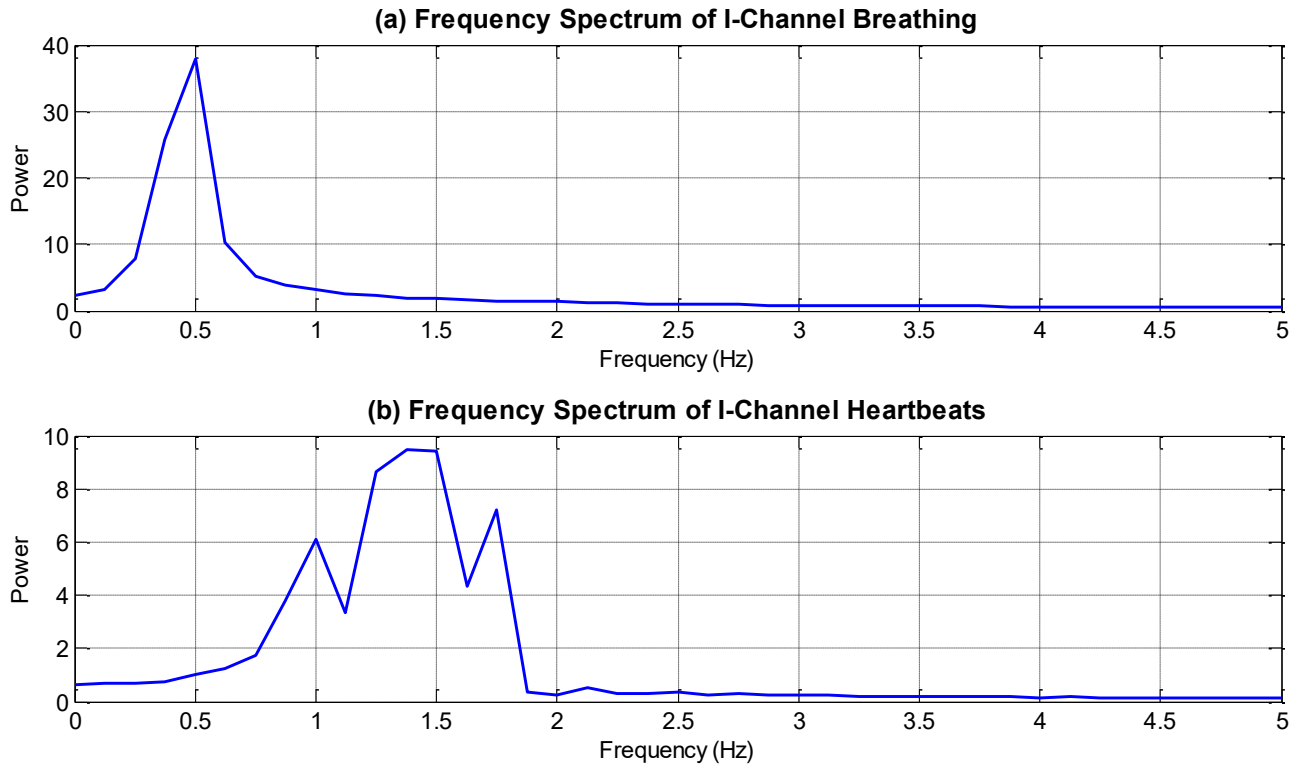


Fig 48 (a) Breathing frequency spectrum (b) Heartbeats frequency spectrum

The breathing rate of the subject is revealed by the peak located at 0.5 Hz in Figure 48(a), which corresponds to a breathing rate of 30 respirational cycles per minute. Similarly, the heart rate of the subject is revealed by the peak located at 1.4 Hz in Figure 48(b), corresponding to a heart rate of 84 beats per minute. The same technique can be applied to the Q-Channel Doppler signal. Figure 49 shows the breathing frequency variations, with its respective frequency spectrum. Figure 50 shows the heartbeat frequency variations, with its respective frequency spectrum.

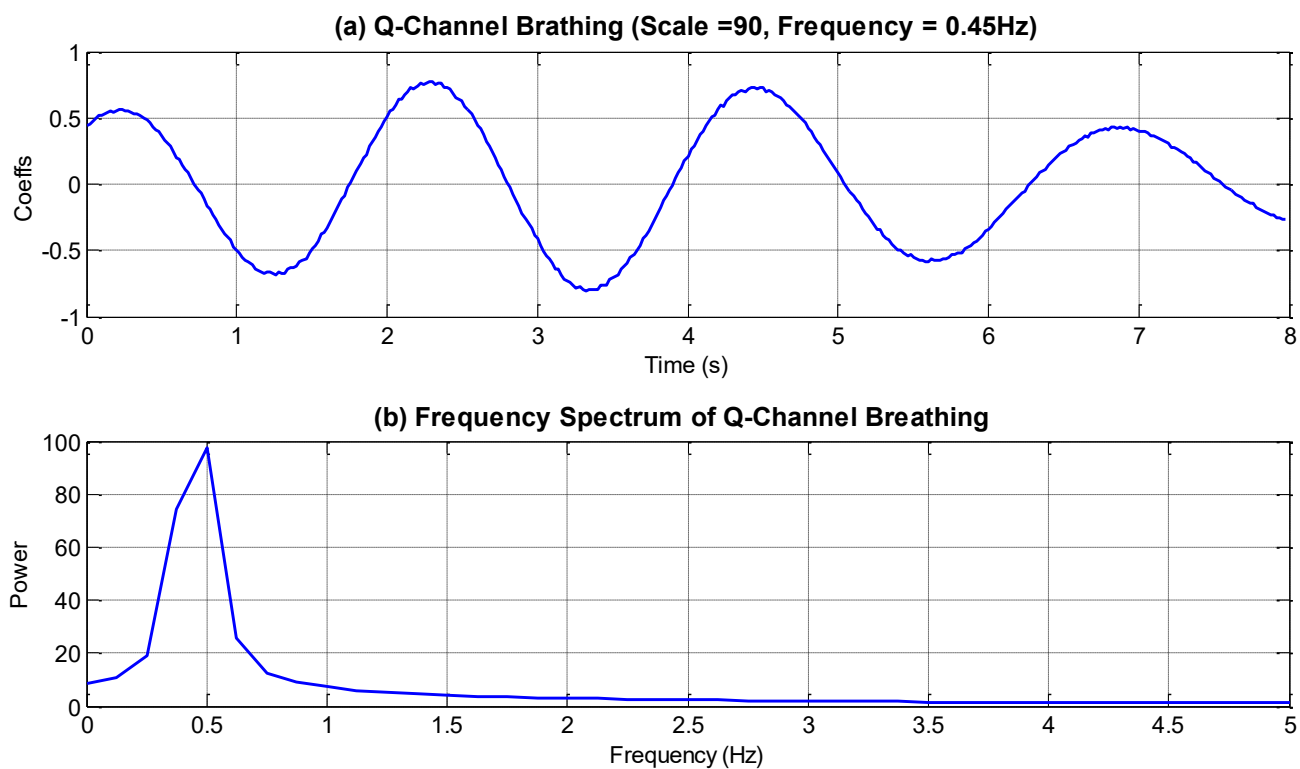


Fig 49 (a) Q-Channel breathing (b) Q-Channel breathing frequency spectrum

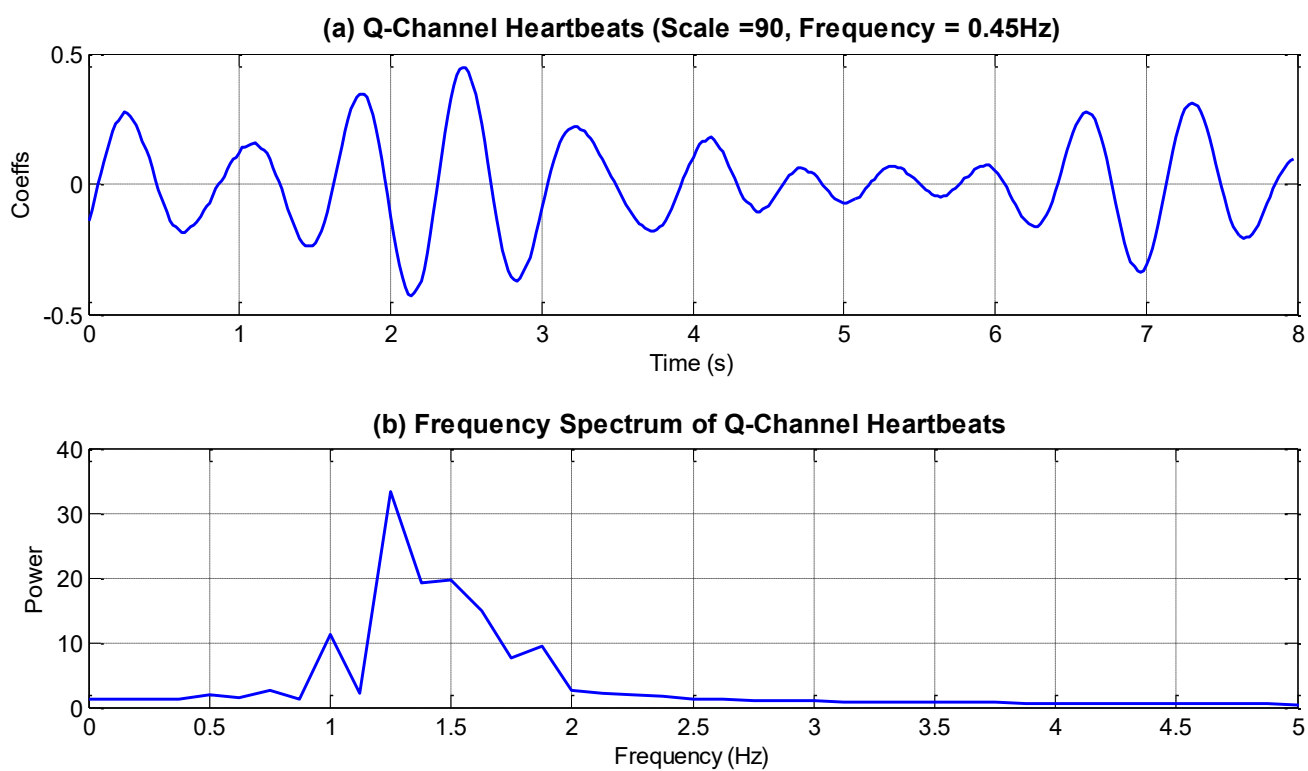


Fig 50 (a) Q-Channel heartbeats (b) Q-Channel heartbeats frequency spectrum

The Q-Channel conveys essentially the same information as the I-Channel, but there is a significant difference in the resultant heart rates. The I-Channel heart rate was calculated to be 84 beats per minute, whereas the Q-Channel is 75 beats per minute. In the pure heartbeat signals this discrepancy did not exist and both channels gave the same heart rate. The difference in calculated heart rate could have arisen from the lack of sufficient spectral resolution, which could change the location of the peaks. To increase the spectral resolution a longer sampling time may be required.

Another way this discrepancy could be circumvented is to combine both channels into a complex signal and performing the CWT on this signal. The same technique of frequency isolation could then be applied. This method would output a single heart rate value, which would be taken as the conclusive heart rate. Figure 51 shows the heartbeat frequency variations of the complex signal, along with its frequency spectrum.

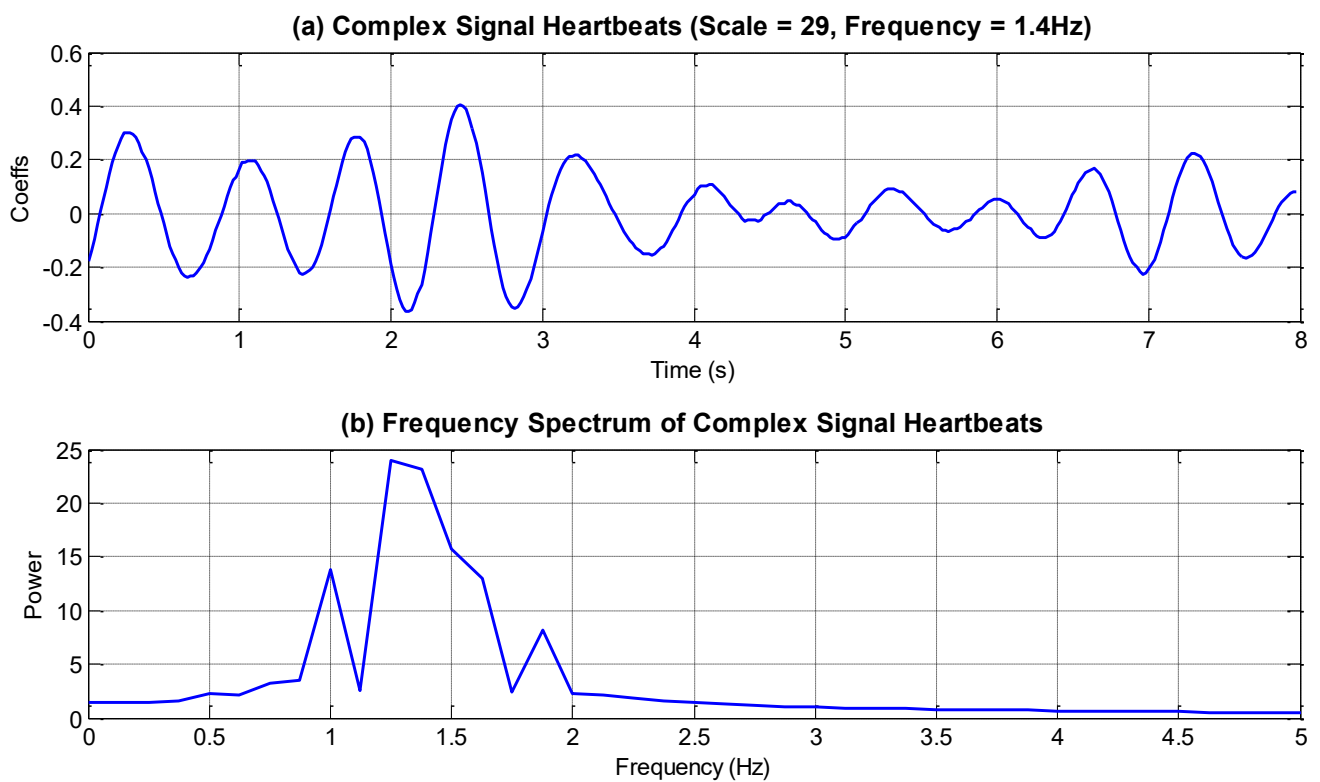


Fig 51 (a) Complex signal heartbeats (b) Complex signal heartbeats frequency spectrum

The peak in this case is located at 1.25 Hz and corresponds to a heart rate of 75, which is exactly the same as the heart rate calculated from the Q-Channel in Figure 50. The fact that there is such a large variation between the heart rates of each channel suggests that this technique may not be as robust as once thought. By increasing the overall sampling time better spectral resolution may be observed, possibly leading to more accurate results.

6.2.3. Unwrapped Phase

If the I & Q channels are combined into a complex signal, the phase of this signal is given by $\phi = \tan^{-1}\left(\frac{Q}{I}\right)$. Now if the phase is plotted against time the absolute movement of the chest is revealed, as expected. However, the phase must be unwrapped in order to overcome the frequent $\pm\pi$ jumps.

The limitation of this technique is that the absolute values can be quite large when dealing with movements associated with breathing. The movements associated with the heartbeats are very small in comparison. The spectrum of the unwrapped phase is therefore very good for determining the breathing frequency, but not effective at determining the heart rate. Figure 52 shows the unwrapped phase plot for deep and deliberate breathing. It is clear to see the movement associated with breathing.

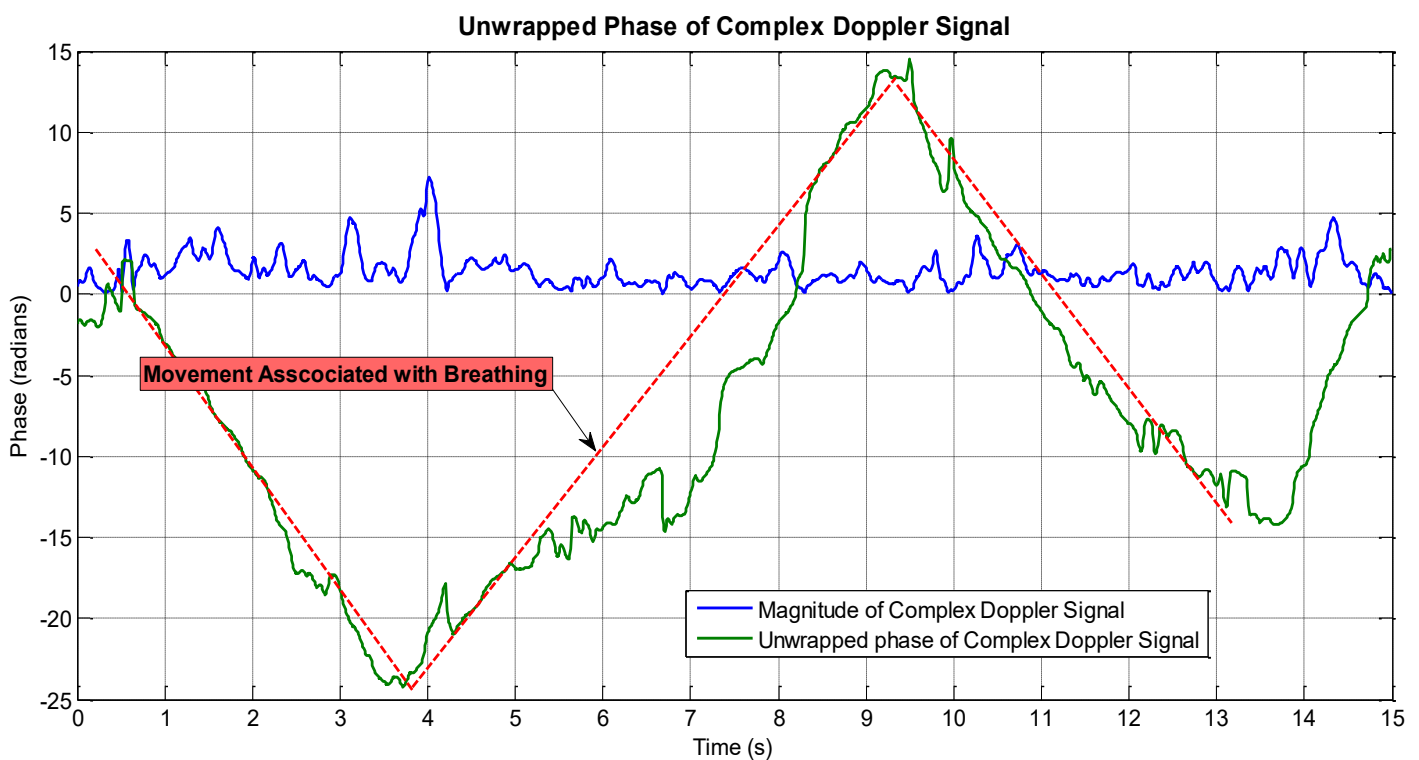


Fig 52 Unwrapped phase for deep and deliberate breathing

The unwrapped phase is particularly sensitive to signal noise due to the $\tan^{-1}$ dependency. Small variations in the I & Q channels can lead to large phase angles. From this, some plots may not have any periodicity at all and may just drift in one direction because of noise. This technique is therefore unreliable for deriving the fundamental breathing frequency.

6.3. Implementation

6.3.1. Software Structure

The general structure of the software is shown in Figure 53. The user code consists of three main logical parts: the display interface and support functions; the data sampling; and the core algorithm. Although the user code does interact with hardware directly, it uses the CMSIS library to map peripheral addresses to human-readable names (e.g. LPC_TMR0->MCR, which is Timer0 Match Control register). CMSIS also provides start-up script that which sets up system clocks, interrupts and exception handlers.

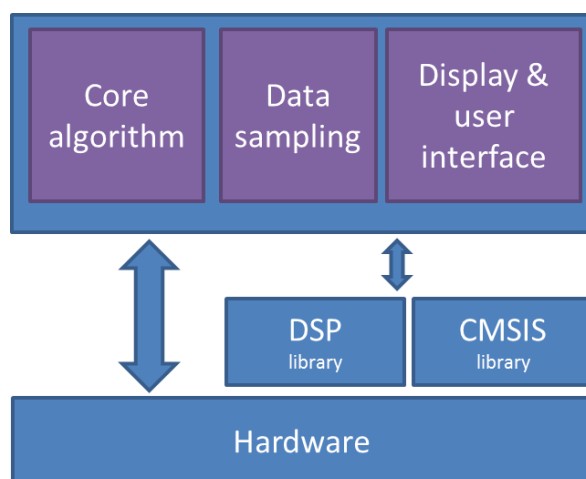


Fig 53 Software structure

The code itself is split up into files, as shown in Table 5, in order to maintain a logical structure. Detailed code documentation has been produced in a linked HTML format using the free tool Doxygen, available in the electronic supplement part of this report.

File	Operation
main.c	Contains the main application loop and interrupt handlers.
algorithm.c	Contains functions performing peak detection, maxima detection, interval filtering, etc...
utils.c	Contains support functions for LCD, SPI communication, serial (UART) and daily routines.
ADS1147.c	Functions to control and read data from ADC.

Table 5 Code files organisation

6.3.2. Serial Link

During the software programming and debugging process it became apparent that a mechanism to check intermediate calculation results, carried out by the microcontroller, was necessary. The solution was to implement a serial link comprised of a serial-to-USB converter and Matlab interface script. This link is in one direction only, from microcontroller to PC. To identify the raw data a control packet was used, shown in Table 6.

Size (bytes)	1	2	1	1	16	2	1
Offset	0	1	3	4	5	21	23
Value	START	N	Type	Command (Reserved)	Description String	Sample Rate (Hz)	END
Explanation	= 23	0...2 ¹⁶ Total size of data in bytes in upcoming data packet	0 = unsigned char (8 bit) 1 = signed char (8 bit) 2 = unsigned int (16 bit) 3 = signed int (16 bit) 4 = 32-bit float		Used to identify the data (must be set correctly)	Used for plotting correct frequency or time axis for the data	= 13
Total size = 24 bytes (constant). All data is transmitted LSB-first.							

Table 6 Control packet

A Matlab script runs a loop which reads the control packet and parses it. Based on the description string and other parameters the proceeding data is saved in the appropriate format for further processing.

6.3.3. Data Sampling

The choice of length of FFTs in the DSP library is limited to 128, 512 and 2048 points. To obtain the maximum achievable frequency resolution, it is prudent to go with the largest length, 2048 samples. This length requires an integer number of seconds of the sampled signal to make calculations convenient. A suitable value is 16 seconds, which gives an effective sampling rate of $(2048/16) = 128$ SPS. For an oversampled signal, the raw sampling rate of the ADC will be equal to $128 \text{ SPS} \times 2 \text{ (channels)} \times M$, where $M=1, 2, 4, 8, \dots$

As it was mentioned earlier, the maximum sampling rate of the ADS1147, that maintains 16 bit resolution, is 320 SPS for a single channel. This equates to 160SPS for two channels. From Section 7.1.1 it was found that oversampling the signals gave us better SNR. It is clear however that 160 SPS is not enough to oversample the signal even by a factor of 2. Therefore it was decided to sample the I & Q channels with both the built-in ADC and the ADS1147 chip. This would give us the benefit of high resolution from the ADS1147 and high speed from the built-in ADC.

The built-in ADC can be triggered by the 32-bit timer to make sure that the sampling intervals are constant. If the microcontroller runs at 40 MHz, the timer clock is 10 MHz. If the timer starts counting from zero and the match value is 78125, it gives the required interrupt rate of 128 Hz. We can only sample one channel at a time, so to sample both channels at this rate a timer interrupt rate of 256 Hz is required. Multiplying this rate by

powers of 2 will make the built-in ADC run faster to enable oversampling. The increased sampling rates are not possible to achieve with integer timer match values, and a very small sample rate error will be introduced. Table 7 shows the final sampling rates for different oversampling rates and timer match values.

Oversampling Factor, M	Timer Match Value	Effective Sampling Rate
1	39,063	127.998
2	19,531	128.001
4	9,766	127.995
8	4,883	127.995

Table 7 Oversampling rates with their corresponding real sampling rates

If the ADS1147 is used, the conversion time for a given nominal conversion rate is fixed; however, effective sampling rate can be controlled using a START pin. As long as the START pin is held high, it will continuously sample a selected channel and pull DRDY low as soon as the conversion result becomes available. The sample rates of the ADS1147 are fixed, where the two closest ones to the desired rate are 80 SPS or 160 SPS. Clearly, these are nowhere near 128 SPS. Alternatively, the START pin can be taken high at appropriate intervals to obtain the desired rate. This is illustrated in Figure 54.

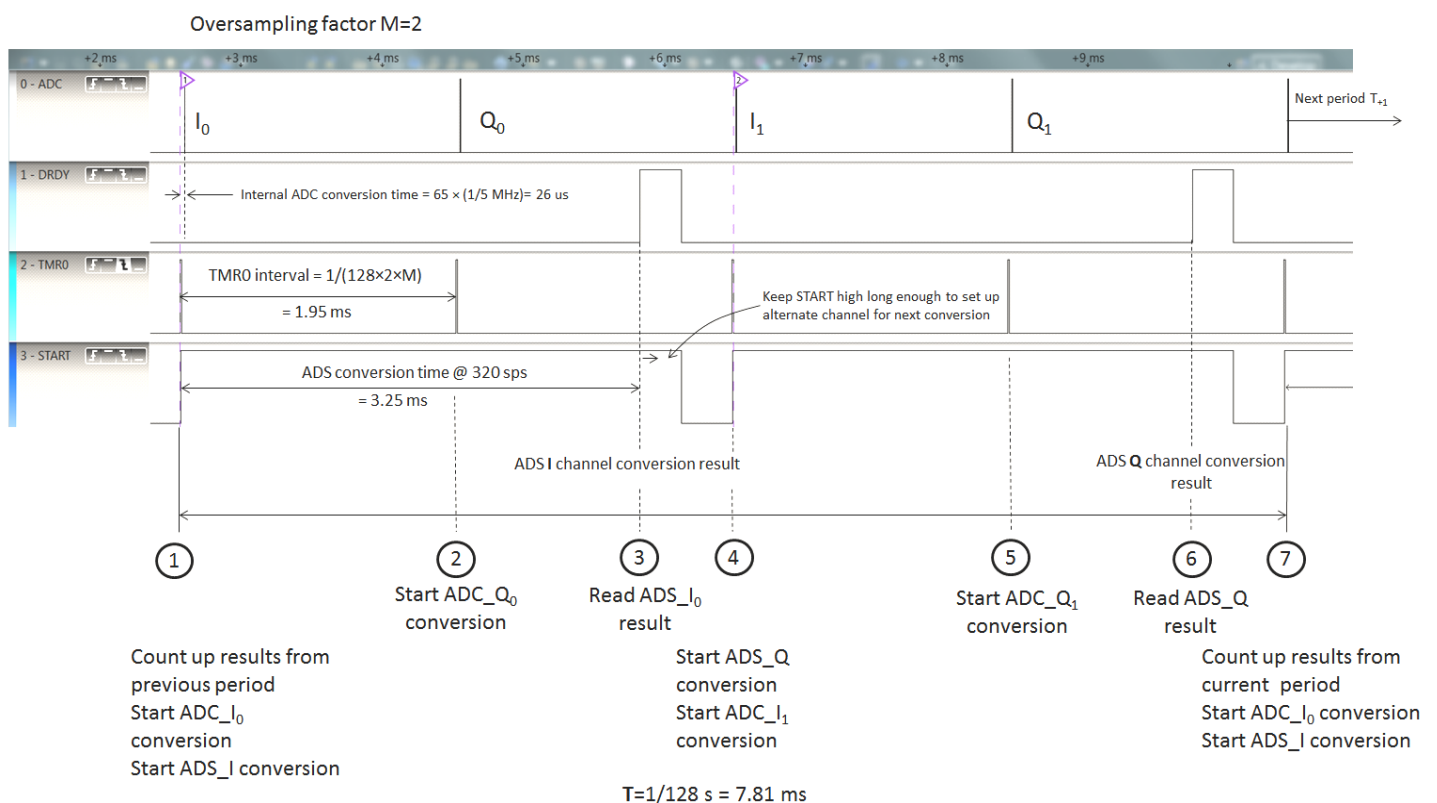


Fig 54 Timing diagram for oversampling with ADS1147

Figure 54 is rather complex, but it is important as it describes the key operations of the sampling process. The four signals shown are as follows.

- **ADC:** This is not an actual electrical signal.
 - High level indicates when ADC interrupt on conversion complete occurs.
- **DRDY:** Output signal from ADS that indicates the conversion is complete.
- **TMR0:** This is not an actual electrical signal.
 - High level indicates when Timer0 overflow interrupt occurs.
- **START:** Input signal to ADS.
 - Pulled high by the microcontroller to trigger the start of a conversion.

I channel samples within one period are designated as I_n where $n=0\dots M-1$, and the Q samples are designated in the same way. The duration of the fundamental period T is $\frac{1}{128}$ seconds, corresponding to effective sampling rate of the system for both channels. Within that period, both I & Q channels must be sampled exactly once by the ADS1147, and up to 8 times by the built-in ADC. In the diagram, the demonstrated oversampling rate M is equal to 2, so both I & Q are sampled twice by the built-in ADC.

The numbered operations shown in Figure 54 are described below:

1. The ADS_I conversion is started. The built-in ADC input is set to I-Channel and the ADC_I0 conversion is started. The accumulators for I & Q channel sample values are set to 0. Once ADC conversion completes the result is added to the corresponding accumulator. If the finger sensor is enabled, the ADC input is set to the finger sensor signal and conversion is performed. Result is stored directly in its intermediate buffer, as it is not oversampled.
2. ADC input is changed, and ADC_Q0 channel conversion is started. Once completed it is added to its accumulator.
3. DRDY signal goes high. The ADS_I conversion result is retrieved over the SPI bus and added to the I-Channel accumulator. The ADS input channel is changed, but the conversion is not initiated.
4. ADC_I1 conversion is performed and stored. ADS_Q conversion is initiated by pulling the START signal high.
5. ADC_Q1 conversion is performed and stored.
6. The ADS_Q conversion result is retrieved and stored. The ADS input is set to I-Channel, ready for next period.
7. The final sample values for the current period, T , are calculated (averaged) and stored in the intermediate buffer. The sample value conditioning and averaging process is discussed in Section 7.3.4. If the position within the intermediate buffer is 127, the flag is set which indicates that intermediate buffer is full and is ready to be copied into the main array. The main application loop detects this flag and initiates the copying and algorithm processing. Next period T_{+1} starts.

6.3.4. Sample Value Conditioning & Averaging

The analogue signal ranges and corresponding sample values are shown in Figure 55. Unlike the ADS1147 sample values, the built-in ADC values are unsigned and have 12 bit length. To

bring them to the same range and format as the ADS1147 sample values, they must be multiplied by 16 and mean-centered. This is carried out by Equation 22.

$$\text{Sample_value} = (\text{ADC}_{\text{value}} \times 16) - 32768 \quad [\text{Eqn 22}]$$

The equivalent C statement is: `Sample_value=(ADC_value<<4)-0x8000;`

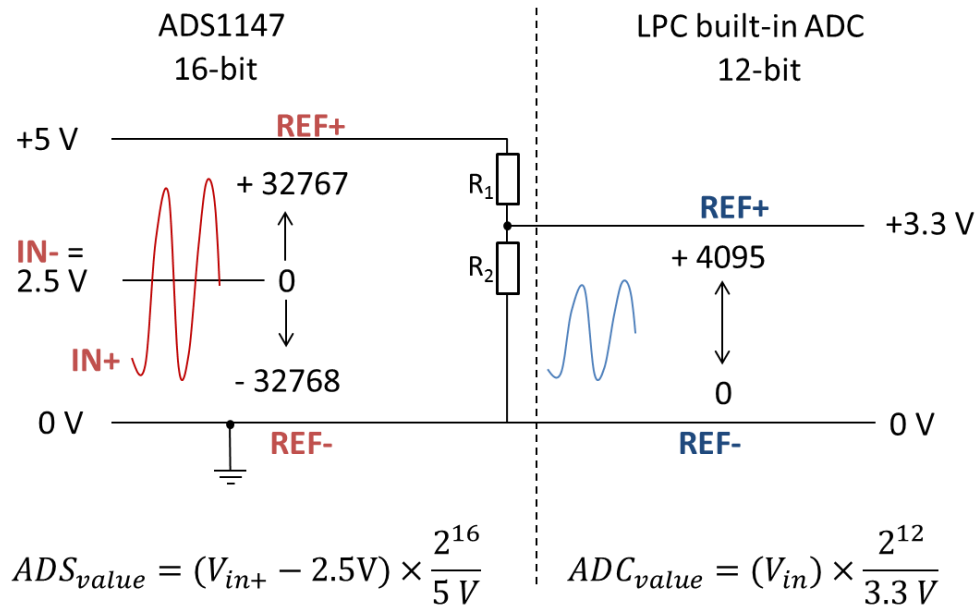


Fig 55 Normalisation

This normalisation is done before the built-in ADC values are added to the accumulator, which is 32 bits long to prevent a possible overflow. In the current implementation, ADS1147 samples are always given a weighing of 0.5 in the final sample value, and the average of the built-in ADC samples is given weighting of 0.5. This is illustrated graphically in the Table 8, and mathematically through Equation 23.

Values	ADS_I	ADC_I ₀	ADC_I ₁
Weighting	0.5	0.25	0.25

Table 8 Sample value weighting

$$\text{Sample_I} = \frac{\left(\frac{ADC_I_0 + ADC_I_1 + \dots + ADC_I_{M-1} + ADS_I}{M} \right)}{2} \quad [\text{Eqn 23}]$$

This weighting is fairly arbitrary, but simple to implement and has shown to produce accurate results with noise levels smaller than when sampling with the ADS1147 alone. Taking samples from two independent converters allows the suppression of systematic errors associated with individual converters.

6.3.5. Sampling Noise

The noise performance of the ADC process on the final PCB design was measured and compared to that of prototype PCB, which used the NI-DAQ as ADC. The noise value figures were obtained by subtracting the mean value (DC offset) from the raw-time domain samples, finding the RMS value of the 16-second period, and converting the result into Volts; as shown in Equation 24. The samples from NI-DAQ are already in volts, so the RMS value was simply computed after removing the mean value.

$$\frac{\text{RMS value of the samples over 16 seconds}}{2^{16}} \times 5 \text{ V} \quad [\text{Eqn 24}]$$

To determine the noise floor, measurements were taken when the Doppler module was both connected and disconnected. When disconnected the I & Q channel inputs were grounded. When connected it was ensured there was no movement in front of the sensor.

PCB Arrangement	Average RMS Noise (mV)	
	Doppler Module Disconnected	Doppler Module Connected
ADS1147 + built-in ADC	7.4	9.5
NI-DAQ	0.85	6.0

Table 9 Noise floor measurements

It can be seen that the ADS1147 + built-in ADC PCB has a noise floor around 10 times higher than the NI-DAQ PCB. The explanation for this is most likely to be the cross-talk between the analogue inputs and the digital IO lines, as well as the serial communication link. When the Doppler module is connected, the levels of noise are comparable.

6.3.6. Data Buffering

During the operation of the algorithm, it is clear that the array with raw time-domain samples must remain constant for all the calculations to be consistent. New samples cannot be written to it during the algorithm calculations. To facilitate that, the data samples from the built-in ADC and the ADS1147 are stored in an intermediate buffer as opposed to the main buffer, which is used for FFT and other calculations. The resultant buffering process and structure is illustrated in the Figure 56.

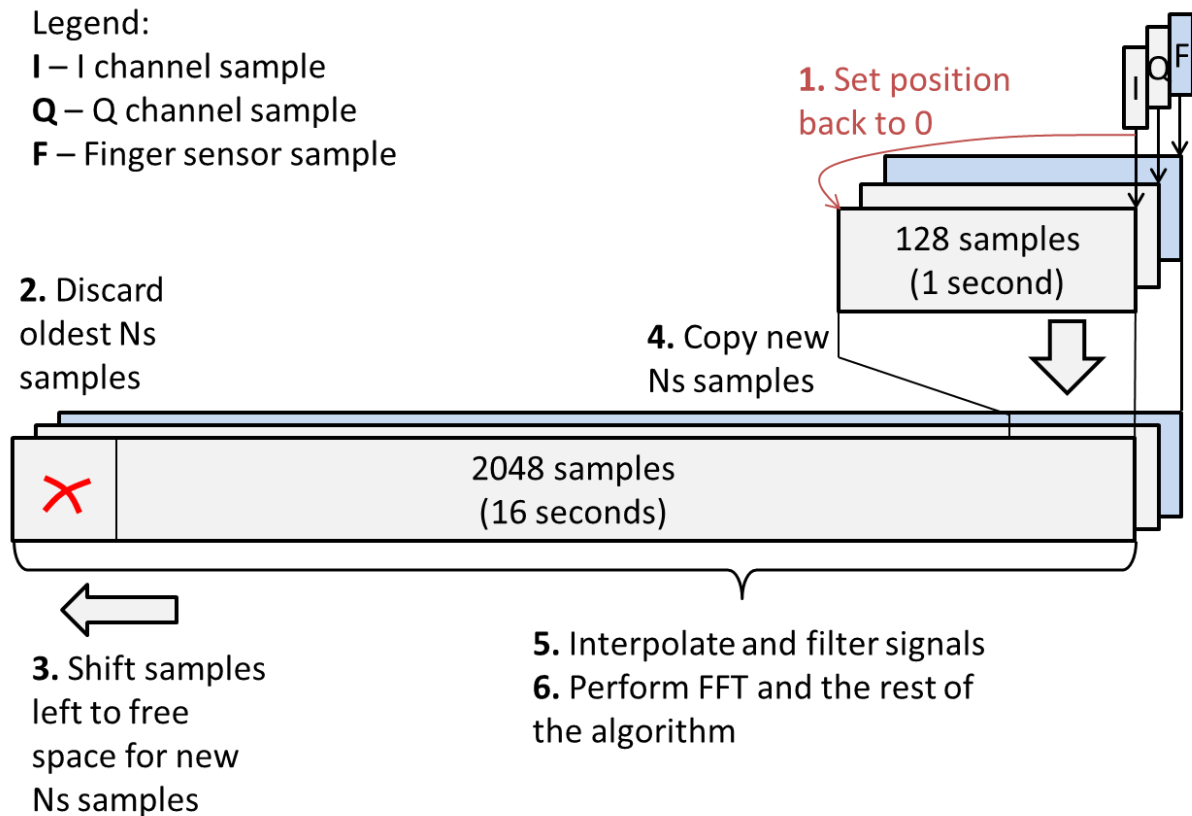


Fig 56 Buffering process and structure

After the system is powered up and initialised, samples from the I & Q channels are calculated, as described in Sections 6.3.3 & 6.3.4, and stored in their respective intermediate buffers, starting from position 0. When the intermediate buffers are full the data in the main buffers is shifted left by 128 samples, discarding it and making space for new data. Then, the intermediate buffers are copied in at the end of their respective main buffers. After that, the algorithm processing is initiated. The algorithm can run for up to a second before the data is overwritten, which allows plenty of time for quite complex calculations.

6.3.7. Floating Point Calculations

The serial link to MATLAB allows the comparison between the fixed-point calculation results and the MATLAB floating point results, helping to detect any errors before they cause obscure bugs.

The samples are stored in Q15 fixed-point format, as shown in Figure 57. This means that the number is stored in a 16 bit container (signed short int type in C), which has a virtual decimal point between bits 15 and 16. The 16th bit is the sign bit, representing a negative number in 2's complement arithmetic when it is 1. In this format, the numbers are represented as fractional values between 0.999969482421875 and -1 ($2^0-2^{-15} \dots -2^0$). It has to be noted that the decimal point only exists in context of arithmetic operations and human representation, and does not change the binary data in any way.

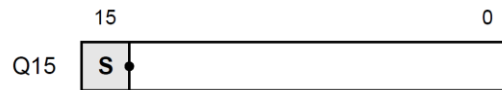


Fig 57 Q15 fixed-point format

After testing the data acquisition code and verifying the operation of the buffers, the 2048-point Q15 FFT was performed. Once the result was sent to MATLAB through the serial link, it was noticed that the magnitude of the result was 0 most of the time. However, after some observation it became clear that the spectrum magnitude was not always 0; it did show up for relatively large signal amplitudes at a particular frequency, as shown in Figure 58.

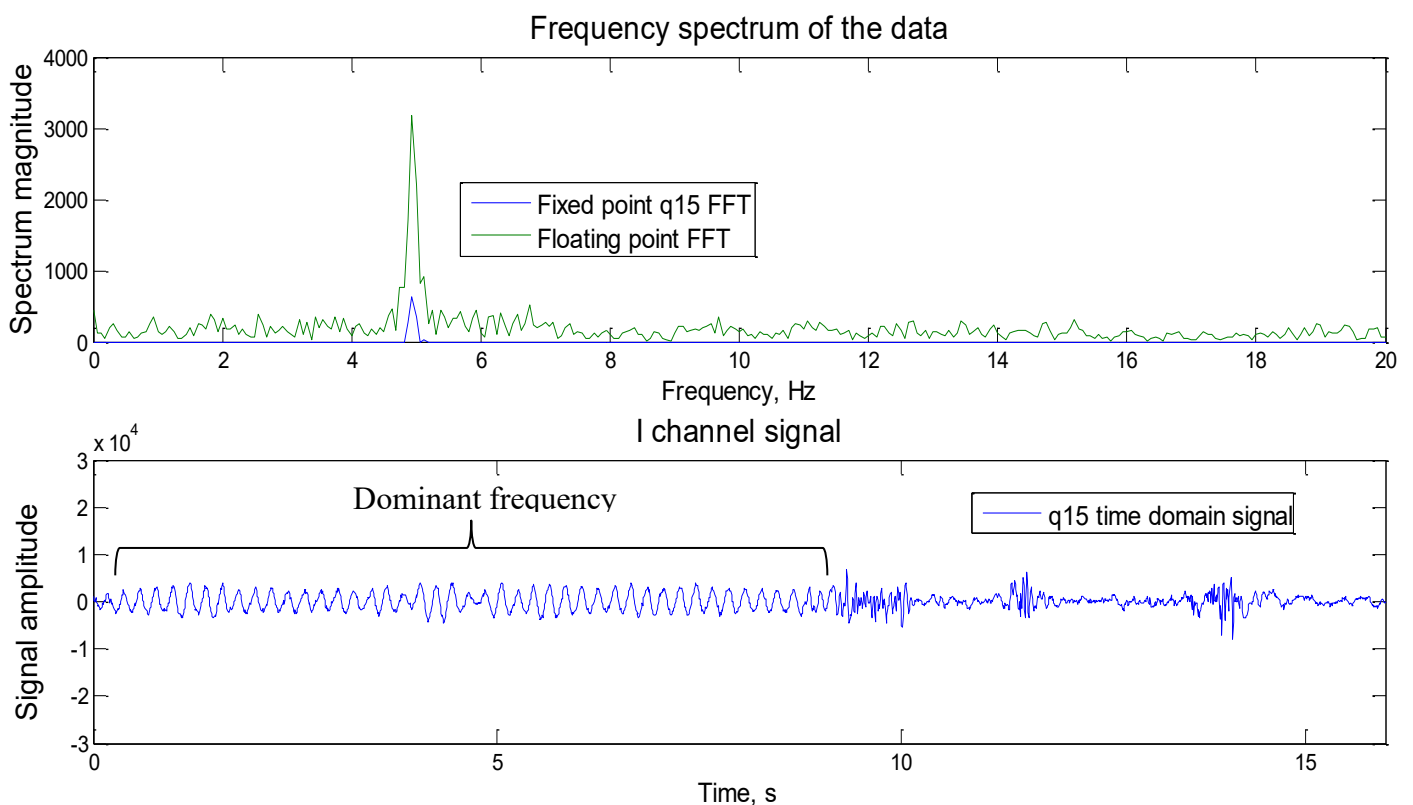


Fig 58 Fixed point and floating point FFT comparison

The raw time-domain signal was also sent to MATLAB, which performed a double-precision FFT on it. In Figure 58, both FFT outputs show the dominant frequency peak correctly; however, for all the other frequencies which are small but non-zero, the Q15 FFT output is 0.

The FFT involves a lot of multiply-and-add operations, and considering that samples are quite small and have values with different signs, calculations with round-off error result in a

zero. A few experiments with up-scaling raw samples and intermediate results did not improve the output, or introduced overflow errors. It became clear that floating point FFT was required for our application. Fortunately, the ARM DSP library provides optimised single-precision FFT functions.

After changing the code to floating-point operation, the operation was verified. This time, both embedded and MATLAB implementations produced essentially the same result. There were some minor differences, but this can be attributed to the difference between single and double-precision arithmetic.

6.3.8. Algorithm Outline

The algorithm is split into two independent parts: the operations on the Doppler sensor data; and the operations on the finger sensor data. The aim of the project is to measure the heart rate using Doppler sensor data only. The finger sensor (F-Channel) was found to be a more reliable and robust way of measuring heart rate, but it requires physical contact with the sensor. In our project it is used to verify the heart rate value obtained from Doppler sensor data.

The outline of the operation of the algorithm is as follows:

- Wait for the new one-second blocks of data to be ready (for I, Q and F channels).
- Copy the new data blocks into their respective main 16-second arrays.
- I-Channel:
 - Convert and copy time-domain sample values to a floating-point array.
 - Perform floating-point FFT.
 - Find the spectrum magnitude.
 - Find 3 largest peaks in spectrum magnitude.
 - Estimate HR.
- Q-Channel:
 - Repeat I-Channel steps.
- Consolidate information from both channels to arrive at the final estimate of heart rate
- F-Channel:
 - Copy to a scratch buffer (FFT modifies the original data).
 - Perform fixed-point Q15 FFT on the scratch buffer, and find the spectrum magnitude.
 - Find the highest peak in the region 0.9 – 2.5 Hz.
 - This corresponds to a rough estimate of the heart rate.
 - Calculate corresponding estimated interval between the heartbeats.
 - Find the heartbeats peaks in the time-domain signal.
 - Compute the differences between the peak positions and intervals.
 - Filter the intervals to eliminate erroneous and missing peaks.
 - Find the mean value of filtered intervals.
 - Derive a precise estimate of the heart rate.

The I & Q Channel operations and F-Channel operations are described in more detail in Section 6.3.8.1 and Section 6.3.8.2 respectively.

6.3.8.1. I & Q Channel Operations

Figure 59 shows an illustration of the I & Q Channel algorithm operations. It should be noted that all the calculations are carried out on the microcontroller, and the data is only passed to MATLAB for displaying. As such, the portable device is completely stand-alone as specified.

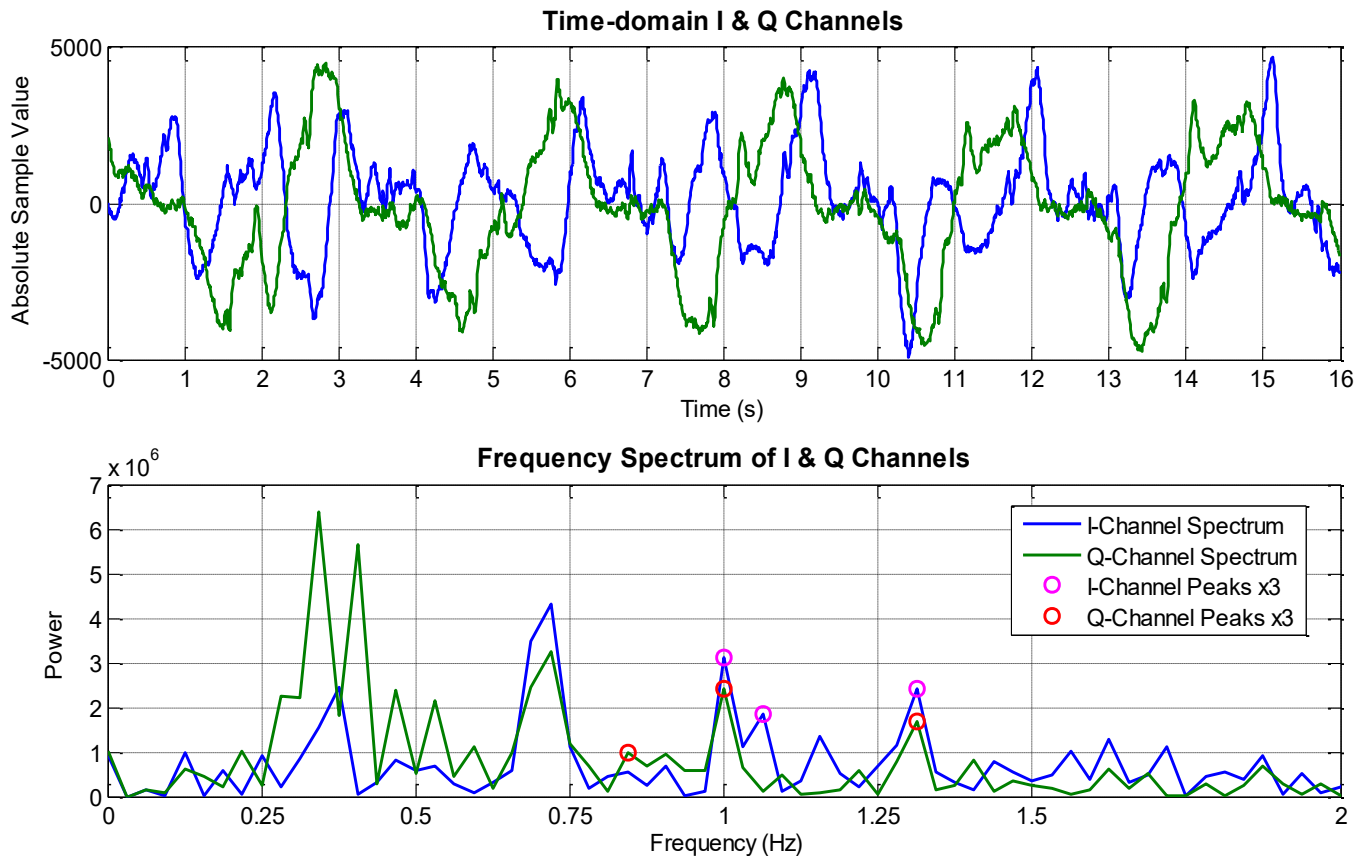


Fig 59 I & Q Channel algorithm operations

The new value of the heart rate in each channel is found by weighting the positions of three highest spectrum values ($x_1 \dots x_3$) by their height ($val_1 \dots val_3$), and inversely weighting them by the difference from the last heart rate ($d_1 \dots d_3$). This way, maxima that are significantly different from the last estimate are weighted down. The maxima are only found within the interval from 0.8 Hz to 1.6 Hz, corresponding to heart rates between 48 and 96 bpm. Below that maxima are likely to represent breathing, above that they can be attributed to harmonics.

The final estimate is found by taking the average of the I & Q channel estimates. Looking at only the highest peak in the spectrum is not suitable, as the highest peak may shift around from cycle to cycle, somewhat randomly. This is a rather simplistic way of estimating the heart rate, but no other estimation method was found during the project that achieved better performance. An illustration of this estimation process is shown in Figure 60.

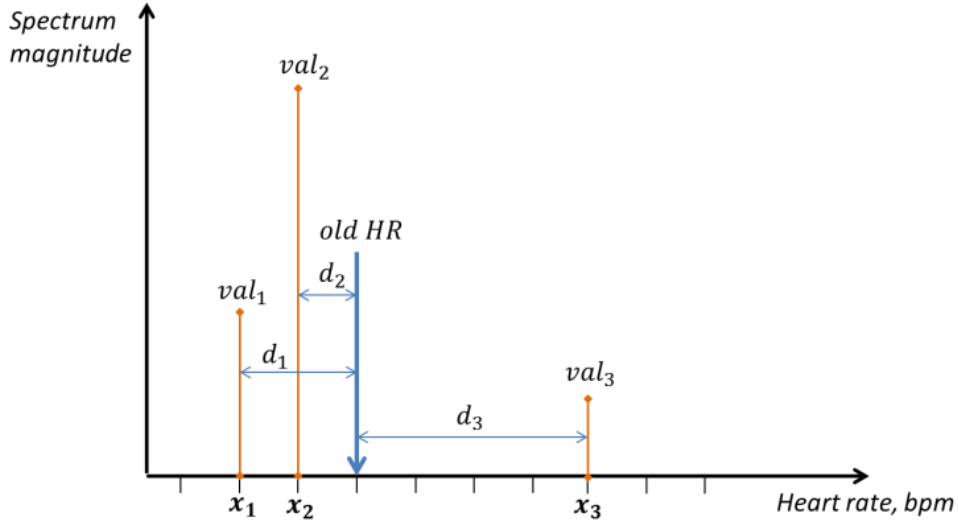


Fig 60 Heart rate estimation process

The formula which governs this estimation process is given by Equation 25.

$$\text{new HR} = \frac{\left(\frac{x_1 \times val_1}{d_1} + \frac{x_2 \times val_2}{d_2} + \frac{x_3 \times val_3}{d_3} \right)}{\left(\frac{val_1}{d_1} + \frac{val_2}{d_2} + \frac{val_3}{d_3} \right)} \quad [\text{Eqn 25}]$$

6.3.8.2. F-Channel Operations

Figure 61 is a convenient representation of the F-Channel algorithm operations and illustrates how the heart rate is calculated from the finger sensor signal.

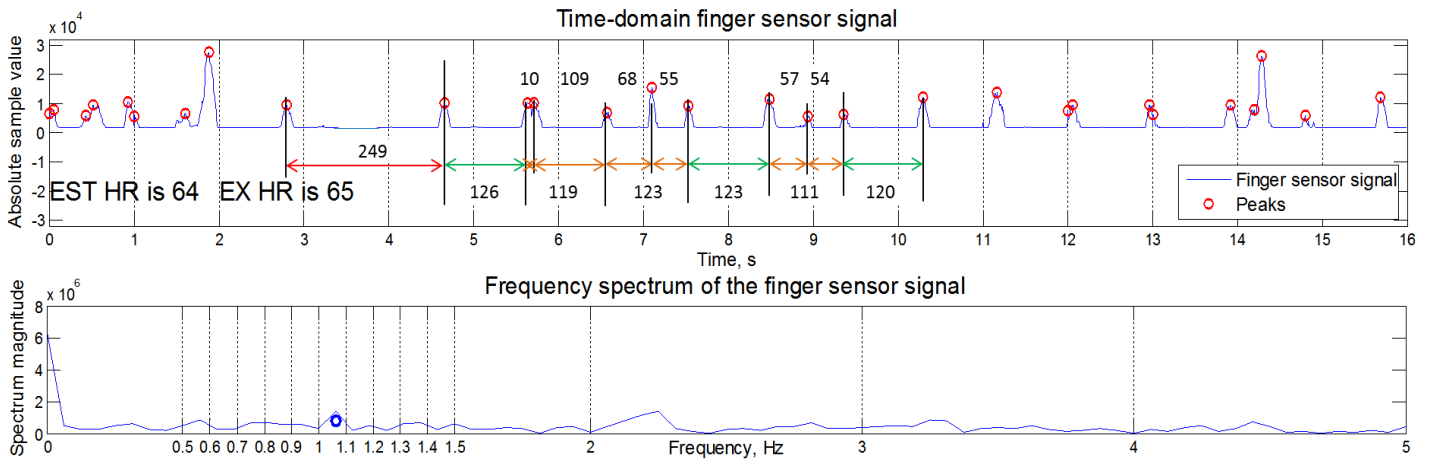


Fig 61 F-Channel algorithm operations

Firstly, the spectrum magnitude is calculated from the raw finger sensor signal data, shown in Figure 61. Fixed-point FFT is sufficient for this, because the signal amplitude is large, and spectrum peaks are not lost due to round-off errors. The highest peak is identified in the region 0.9 – 2.5 Hz, corresponding to heart rates in the range of 54 – 150 bpm. Special

measures are taken to make sure the second harmonic is not detected instead of the fundamental. The peak is marked with a blue circle. Its position is 136 in the spectrum array, corresponding to a frequency of $136 \times dF = 1.0625$ Hz and an estimated heart rate of 64 bpm. This heart rate corresponds to an expected peak spacing of:

$$\frac{60 \text{ bpm/Hz}}{64 \text{ bpm}} \times 128 \text{ samples} \cdot \text{Hz} = 120 \text{ samples}$$

The interval tolerance to account for deviation in individual heart beat intervals is set to 10%, which equals to ± 12 samples. In other words, the intervals that lie within the range of 108 – 132 samples are accepted.

Next, the time-domain peaks are detected using the valley-trough method, shown with red circles in Figure 61. The intervals are computed as differences between the peak positions. The raw data peaks may include some spurious peaks due to finger movement, and some missed peaks too. Including all of them in heart rate estimation would lead to inaccurate results. Therefore the raw interval values are filtered as illustrated in Figure 61.

The filtering process can be explained by three cases:

- Interval *lies within* the expected interval tolerance: accept it and move to next one.
 - This is shown with green arrow lines.
- Interval is *larger* than the expected interval tolerance: discard it and move to next one.
 - This is shown with red arrow lines.
- Interval is *smaller* than the expected interval tolerance: start the cumulative loop:
 - If the sum is *less* than interval tolerance:
 - Continue adding up.
 - If the sum is *within* interval tolerance:
 - Accept the sum value and go to following interval.
 - If the sum is *larger* than interval tolerance:
 - Discard and restart from next interval after sum counting beginning.

Referring to Figure 6 and following the outlined algorithm, from left to right the annotated values are:

- 249 → too large → discarded.
- 126 → within tolerance → accepted.
- $10 + 109 = 119$ → within tolerance → accepted.
- $68 + 55 = 123$ → within tolerance → accepted

After the intervals are filtered, the average interval is calculated. The heart rate is then calculated using Equation 26.

$$\text{HR} = \frac{128}{\text{mean}(\text{filtered_intervals})} \times 60 \quad [\text{Eqn 26}]$$

The value calculated is the average heart rate over the past 16 seconds, which is the length of the data. For the case discussed above the heart rate is 65 bpm, which is very close to the original estimate of 64 bpm. However, it can be seen that even the intervals between

individual beats may vary by up to around 10% in a completely healthy individual. We can obtain the value of the instantaneous heart beat by just using the last valid interval, or we can average the intervals for any number of seconds to obtain a longer-term average.

The variation in individual heartbeat intervals puts under question possibility of detecting heart dysrhythmia using the finger sensor, let alone the Doppler signal. From a heart rate detection point of view, this spread in interval values results in spread of the spectrum peak, making it harder to detect and track.

7. Results

The most obvious way to establish the accuracy of the Doppler module results is by plotting them along with the finger sensor results over a certain length of time. This is shown in Figure 62.

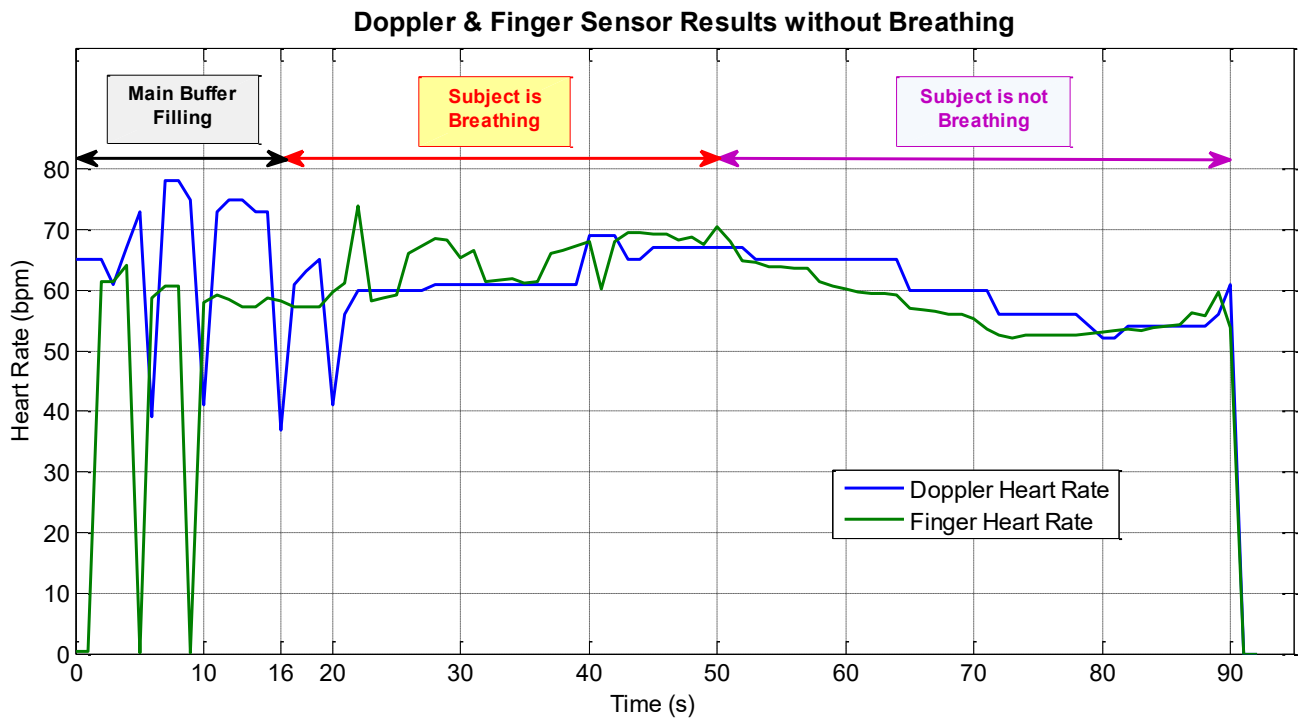


Fig 62 Doppler & Finger sensor results without breathing

During the first 16 seconds the main buffer is still being filled, so the results are completely inaccurate. After that, the finger sensor locks onto the heart rate but the Doppler sensor still takes some time to display accurate results. In Figure 62, the subject is holding their breath from 50 seconds onwards. Figure 63 shows the same plot for a subject who is breathing throughout the measurements.

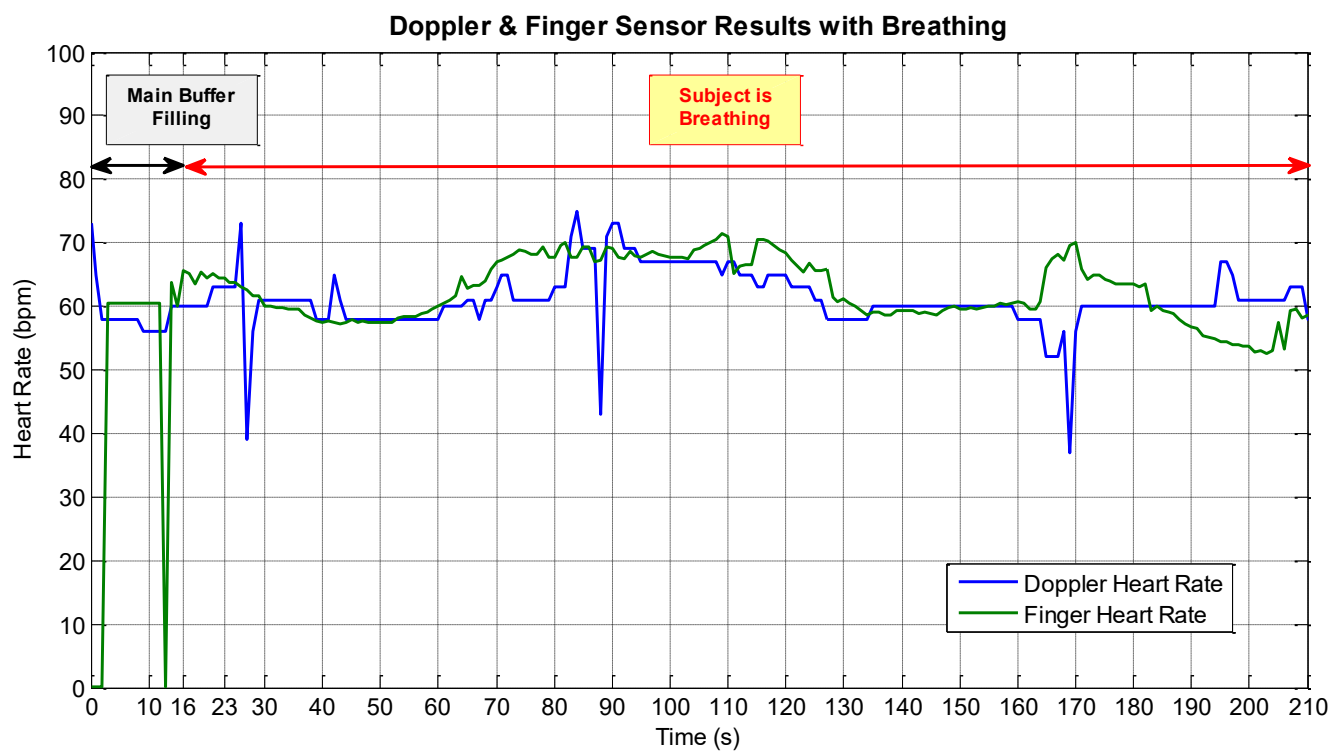


Fig 63 Doppler & Finger sensor results with breathing

8. Conclusion

To conclude, it is correct to say that the revised aim of developing a portable device which could accurately determine the heart rate of a subject through contactless means was achieved. Although this aim was achieved, there is a severe limitation to the accuracy of the device, relating to the magnitude of clutter motion that is acceptable before inaccurate results are obtained.

In addition to the contactless Doppler sensor, a blood oximetry-based finger was designed and built in order to provide an accurate reference measure of the heart rate for development purposes. In the results section, it is demonstrated that the results derived from the Doppler sensor do correlate with those derived from the finger sensor, albeit under a limited set of conditions.

As mentioned, non-cardio-respiratory motion poses a severe problem to the accuracy of the device. Periodic movement, such as tapping your foot, can be misinterpreted as the heartbeat. In most cases, the amplitude of such movement will be larger than that of the heartbeats and its frequency may well lie in the expected heart rate range. These interfering signals are therefore very difficult to filter out. This would inevitably cause a problem when trying to use the device for medical and security applications, which require a high level of reliability and accuracy. For the developed device, even small movements demonstrate this limiting factor.

In light of this, the developed system can be used successfully for a number of applications, although some improvement may be required. The main application would be monitoring subjects that can be expected to be fairly still, such as those which are sleeping or unconscious. Furthermore, the device could be used to study sleep-related disorders or to monitor hibernating animals.

At this point in time, it is not feasible for the device to be used in sports applications as the movement associated with exercise produces a significant amount of interference, which renders the results useless. In order for the use of this device in sports applications a substantial amount of further work must be carried out.

9. Further Work

There are two main directions of further work: firstly, the device hardware must be improved; secondly, one should seek to improve the detection algorithm. In terms of hardware improvement, the main parts to look at are increasing the operating frequency from 24 GHz and reducing the system noise. With adequate expertise and funding a higher frequency lower noise device could be easily achieved.

On the other hand, improving the detection algorithm is a complex problem. An improvement could be made by taking into account range information as well as Doppler information. This would aid in increasing the robustness of the algorithm and aid in increasing clutter suppression. The absolute distance between the device and the chest of the subject could be measured using one of the following methods: pulsed radar; laser range finding; or ultrasound range finding. It is important that any such range-finding solution measures the average distance to the chest with high accuracy.

The range information can be used to improve the detection algorithm in a number of ways. If the average range information is normalised to the unwrapped phase information and their difference is found, the phase variations associated with each heartbeat will be left. The range information can be used to find the location of the subject within the wavelength interval, and aid in predicting how the demodulated signal will be presented in the I & Q channels. If the range and absolute volume of movement of a subject is known, it is possible to estimate their level of physical activity. This information can then be used to determine their probable heart rate range, which can assist the algorithm in pinpointing the heart rate.

The methods discussed involve changing the current system; however, some improvements can be made without doing so. Focus should be turned towards analysis of the demodulated signal in the time-domain, especially on the complex channel and unwrapped phase information. Various combinations of autocorrelation and phase variation recognition should be investigated. Perhaps even a neural-network algorithm can be trained; provided information about the true position of the heartbeats is available from a different sensor, such as the finger sensor.

10. Project Management

Month	Week	Stage
September	1	Background research
	2	Submission of Project proposal
October	1	Initial work carried out on URSP N210
	2	
	3	
	4	
November	1	Doppler module acquired
	2	Breadboard prototype designed/built
	3	
	4	Breadboard prototype tested
December	1	Initial algorithm development (no breathing)
	2	Prototype PCB designed/built
	3	
	4	Prototype PCB tested
January	1	Further algorithm development (breathing) Final product designed
	2	
	3	
	4	
February	1	Intensive Modules
	2	
	3	
	4	
March	1	Final product PCB built Algorithm implementation Final Product Testing & Debugging
	2	
	3	
	4	
April	1	Preparation of final report
	2	
	3	
	4	

Table 10 Project timeline

Team Member	Section	Description
William Ashman	3.2 3.3 4.1 6.1 6.2	Doppler-Effect Doppler Radar Universal Radio Software Peripheral (URSP) Development – No Breathing Development – Breathing
Mohamed El Hawary	3.1 4.2 4.3 5.5	Heart Breadboard Design Prototype PCB design Final PCB design
Dmitry Ignatyev	5.1-5.4 6.2.3 6.3 7	Final Product Design Unwrapped Phase Implementation Results
Joint work	1 2 8	Introduction Project Aim Conclusion
Angel Noorbakhsh	3.4	Antenna

Table 11 Work distribution

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Appendix A

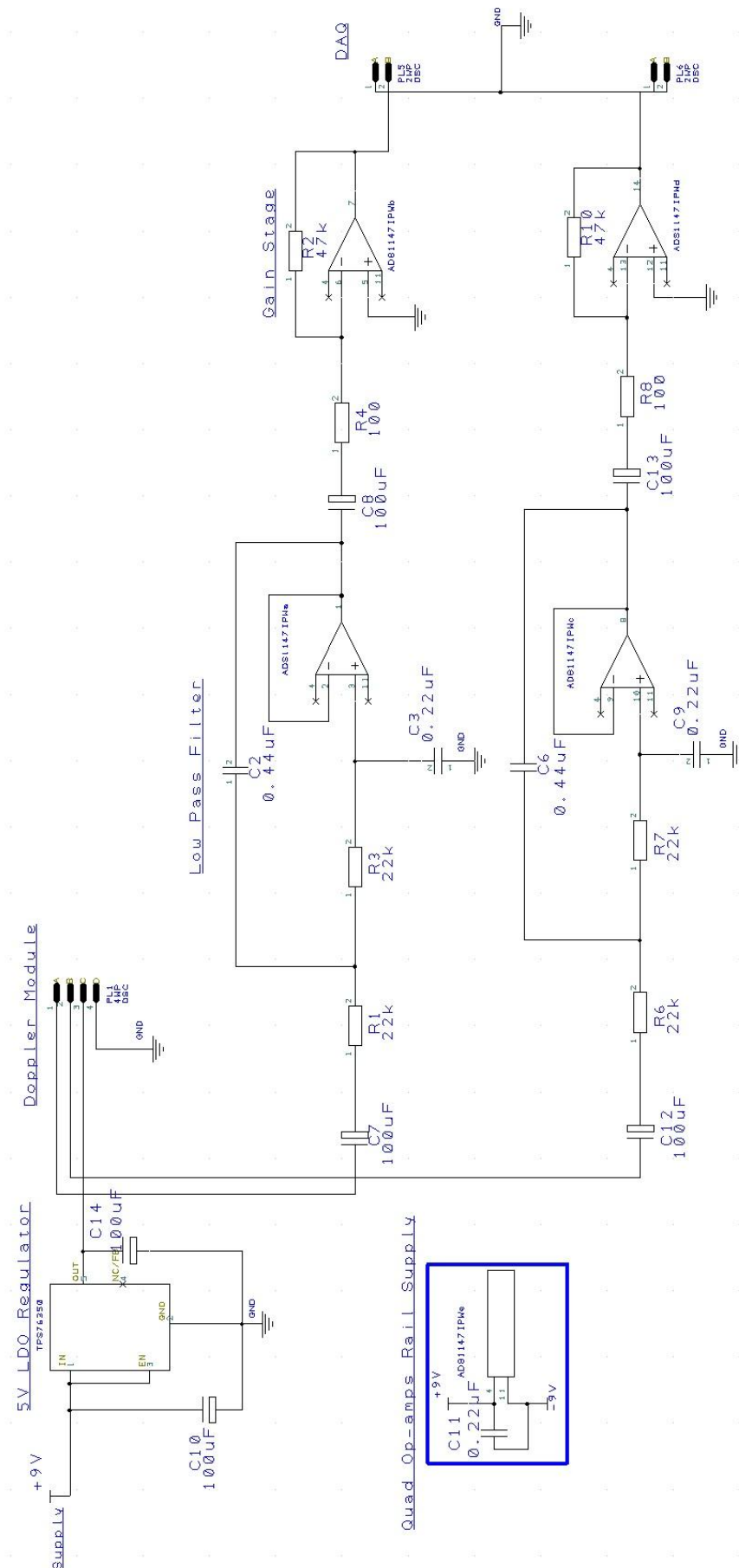


Fig A1 Prototype PCB circuit schematic

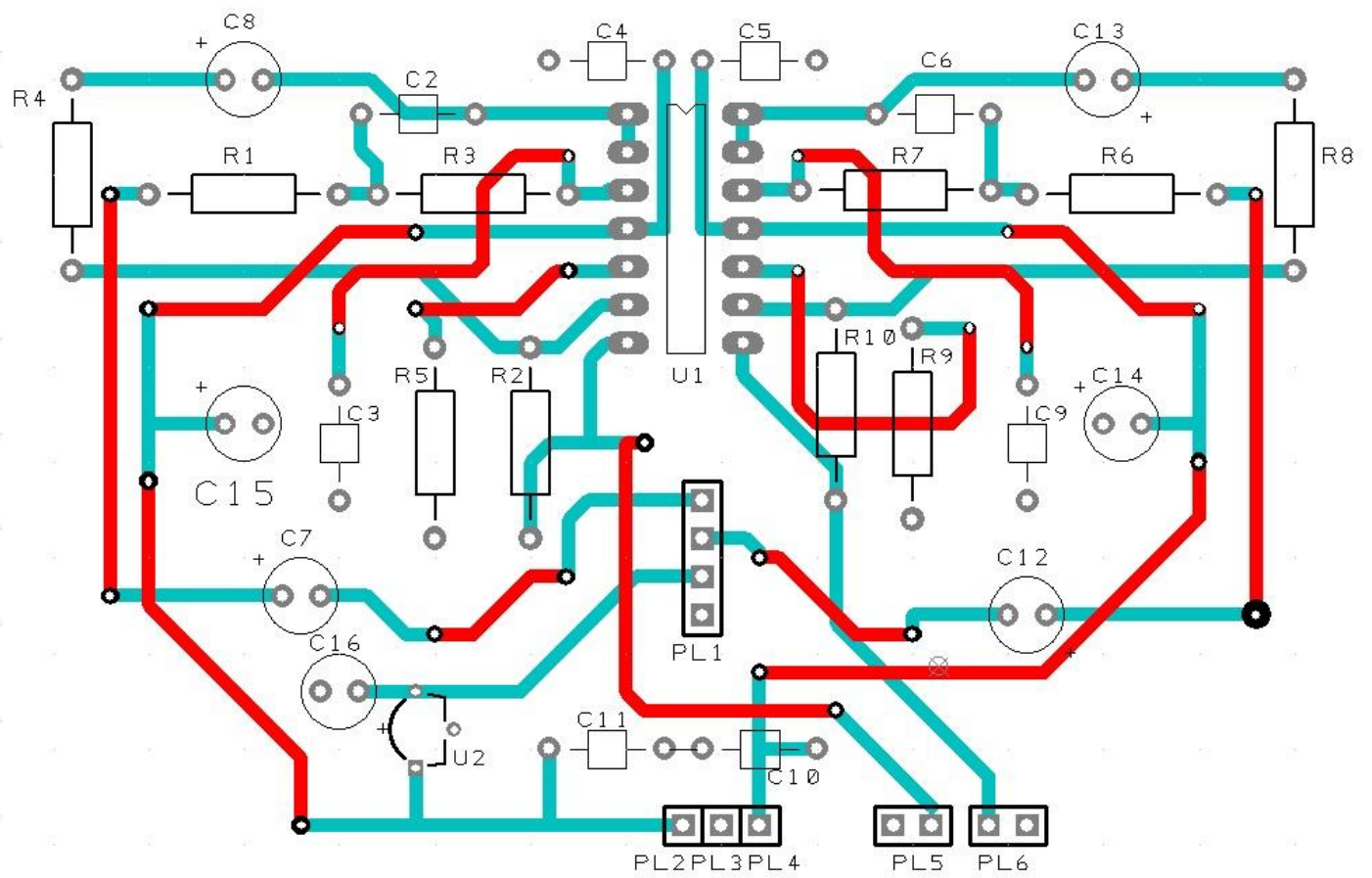


Fig A2 Prototype PCB layout

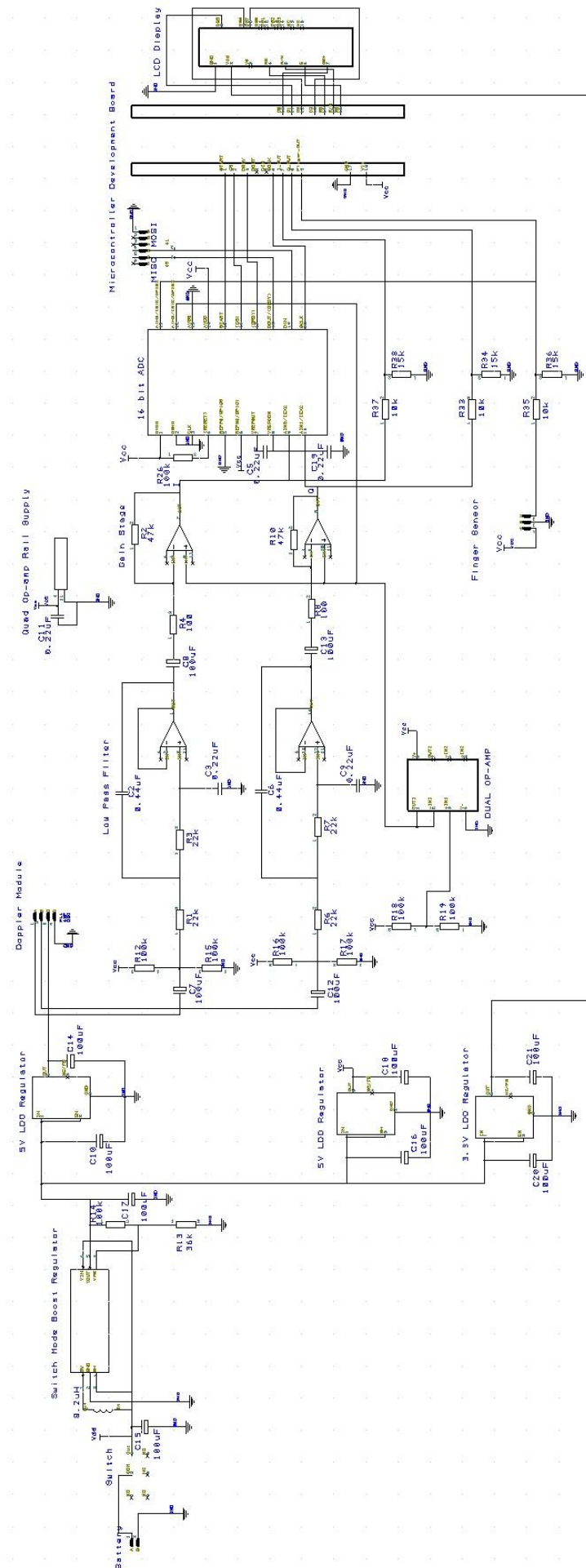


Fig A3 Final PCB circuit schematic

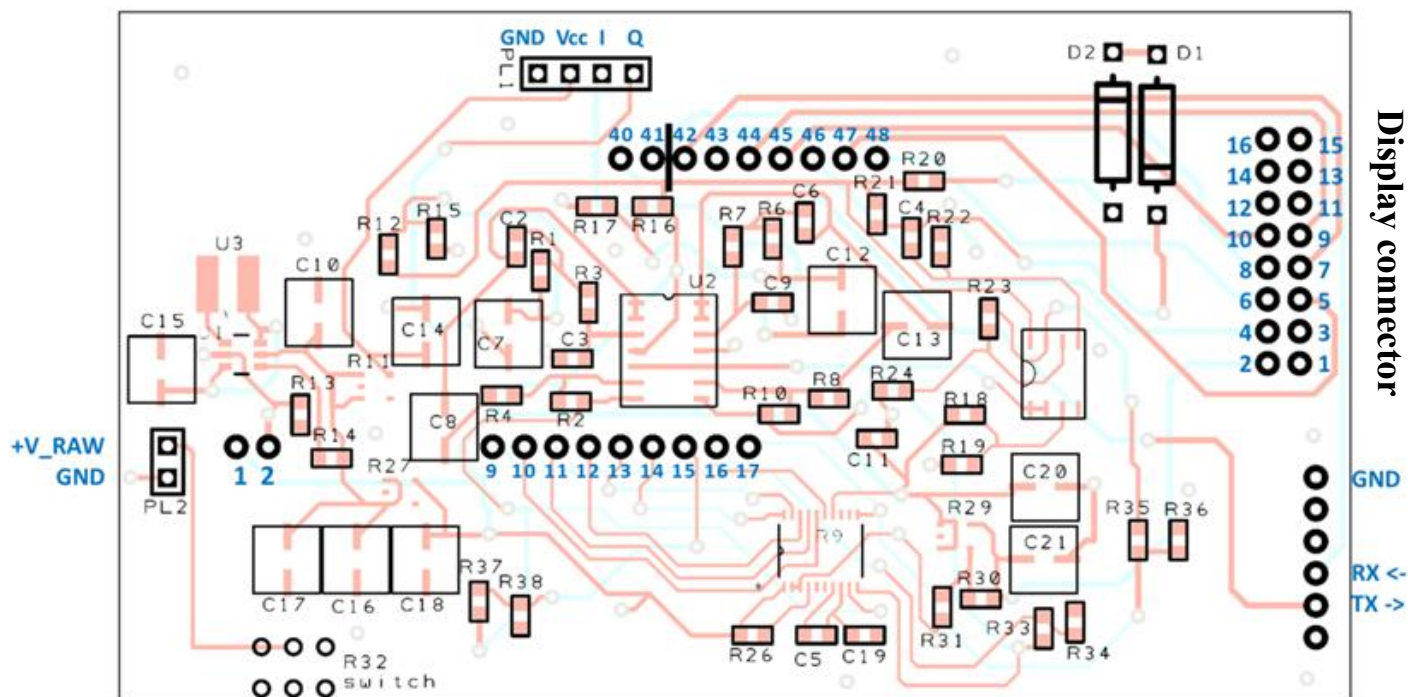


Fig A4 Final PCB layout

Pin (on LPC board)	Board connection	LPC port
1	GND (ground)	
2	Vcc (5V, not 3.3V)	
9	To ADC, pin 15 (START)	P0.0 ADC_START
10	to ADC, pin 18 (DRDY)	P0.1 ADC_DRDY
11	To ADC, pin 19 (DIN)	P0.18 ADC_DI (MOSI)
12	To ADC, pin 18 (DOUT)	P0.17 ADC_DO (MISO)
13	To ADC, pin 20 (SCLK)	P0.15 ADC_SCK
14	To ADC, pin 16 (CS)	P0.16 ADC_CS
15	Channel I	ADC0
16	Channel Q	ADC1
17	Finger sensor	ADC2
Pin (on LPC board)	Board connection	LPC port
40	TX (MISO) (RS232)	P0.10
41	RX (MOSI) (RS232)	P0.11
	Display connector (16 pin)	
42	pin 11 (D4)	P2.0
43	pin 12 (D5)	P2.1
44	pin 13 (D6)	P2.2
45	pin 14 (D7)	P2.3
46	pin 4 (RS)	P2.4
47	pin 5 (R/W)	P2.5
48	pin 6 (E)	P2.6

Table A1 Final PCB pin-out guide

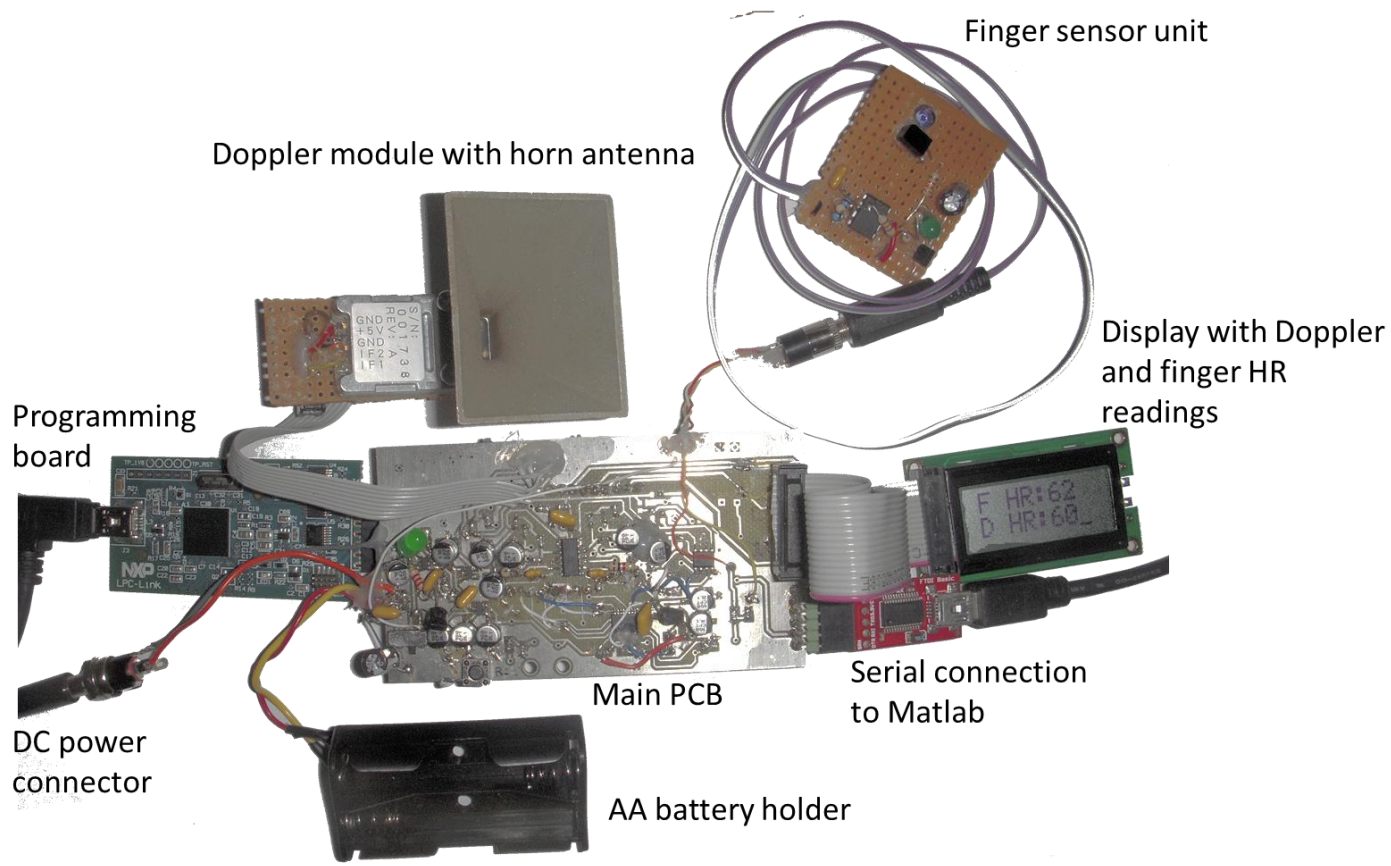


Fig A5 Operating prototype

Appendix B

All order numbers are Farnell stock codes					
Order number	Part number	Quantity	Price each	Cost	Comment
Semiconductors					
1863917	MCP1623	1	£0.35	£0.35	switch mode boost regulator
8461538	TPS76933DBVT	2	£0.33	£0.66	3.3 V LDO regulator
8461384	TPS76350DBVT	2	£0.33	£0.66	5 V LDO regulator
1823860	ADS1147IPW	1	£8.06	£8.06	16 bit, 4 input ADC
1750145	LM324ADT	1	£0.16	£0.16	quad op-amp
1750158	TS27L2IDT	1	£0.24	£0.24	dual op-amp
Connectors and switches					
1216726	JR1819-ROHS	1	£0.71	£0.71	DC 2.1 mm socket
1737256	MP-136	1	£0.37	£0.37	DC 2.1 mm plug
1099238	MC6FD016-30P1	2	£0.47	£0.94	IDC cable connector
1702630	KEYSTONE - 2462	1	£0.85	£0.85	battery holder
3418546	AMP - 826632-8	2	£0.64	£1.28	16 way 2x8 pin header (2.54 pitch)
1280662	PSG01484	1	£0.30	£0.30	3.5mm 3 way audio jack
1267374	MJ-074N	1	£0.78	£0.78	3.5mm audio socket (chassi mount)
1248126	1-826631-6	1	£1.08	£1.08	16 way right angle LPC programming connector
1668303	SSQ-108-02-G-S-RA	1	£1.10	£1.10	8 way right angle socket (LPC programming)
1522046	MCLSS22	1	£0.30	£0.30	slide switch
Passive components					
1864102	MCSD43-8R2MU	5	£0.09	£0.45	inductor, 8.2 uH
1828891	GRM188R71C224KA01D	10	£0.04	£0.42	0.22 uF X7R capacitor
1469811	CRCW060347K0FKEA	50	£0.01	£0.65	47K resistor 1% 0603
9330615	MC 0.063W 0603 1% 15K	50	£0.01	£0.65	15K resistor 1% 0603
1469779	CRCW060322K0FKEA	50	£0.01	£0.65	22K resistor 1% 0603
9330364	MC 0.063W 0603 1% 100R	50	£0.01	£0.30	100 R resistor 1% 0603
2065969	EEEEFT1C101AR	12	£0.22	£2.64	SM electrolyth 100u caps
2008366	CR0603-JW-104GLF	50	£0.01	£0.45	100k 0603 resistor
Display					
2063123	MC20805A6W-GPTLY	1	£6.04	£6.04	LCD display
Microcontroller board					
1825878	LPCXPRESSO, LPC1768	1	£22.28	£22.28	Microcontroller development board
Microwave components					
http://freaklabs.org/	NJR4261	1	\$85 inc. shipping	£73 inc. duties	Doppler sensor module 24 GHz
http://www.microsemi.com/	MDT86552	1	Free sample	£0.00	Horn antenna (normally \$65 dollars each)
PCB					
UCL electronics lab	100 x 55 mm FR4 2-layer board	1	Free	£0.00	Estimated commercial prototype cost - £30
			Total	£125.37	

Table B1 BOM used in final PCB

PCB	
Mistakes:	
Holes diameter is too small for all the IDC connectors	
LCD: the 4 data bits (P2.[0..3]) should be connected to 4 MSB (D7..D4) and not the LSB of the LCD data butts.	
ADS references should be connected: REF0N to ground and REF0P to op-amp Vcc/2.	
3.3 V regulator for the microcontroller should be powered directly from a switching converter, to minimise common-mode power tracks with sensitive circuitry.	
SPI interface to ADS is connected wrong.	
Improvements:	
Ground thermals are too thick and make soldering problematic.	
Larger footprint for capacitors and SOT23-5 footprints for ease of soldering.	
Add ceramic decoupling capacitors at each IC.	
Add power indicator LED.	
Use common-cathode Schottky diode or dedicated chip to manage power from DC power supply.	
Add ferrite bead/inductor after switching converted to minimise ripple noise.	
Add transistor enable switches for both LCD backlight and IR LED, controlled by the μ C.	

Figure B2 Known mistakes and suggested improvements for final PCB